



Motivation, Computation and Applications of Quasi-Periodic Orbits (QPOs) in Space Trajectory Design

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What are QPOs?

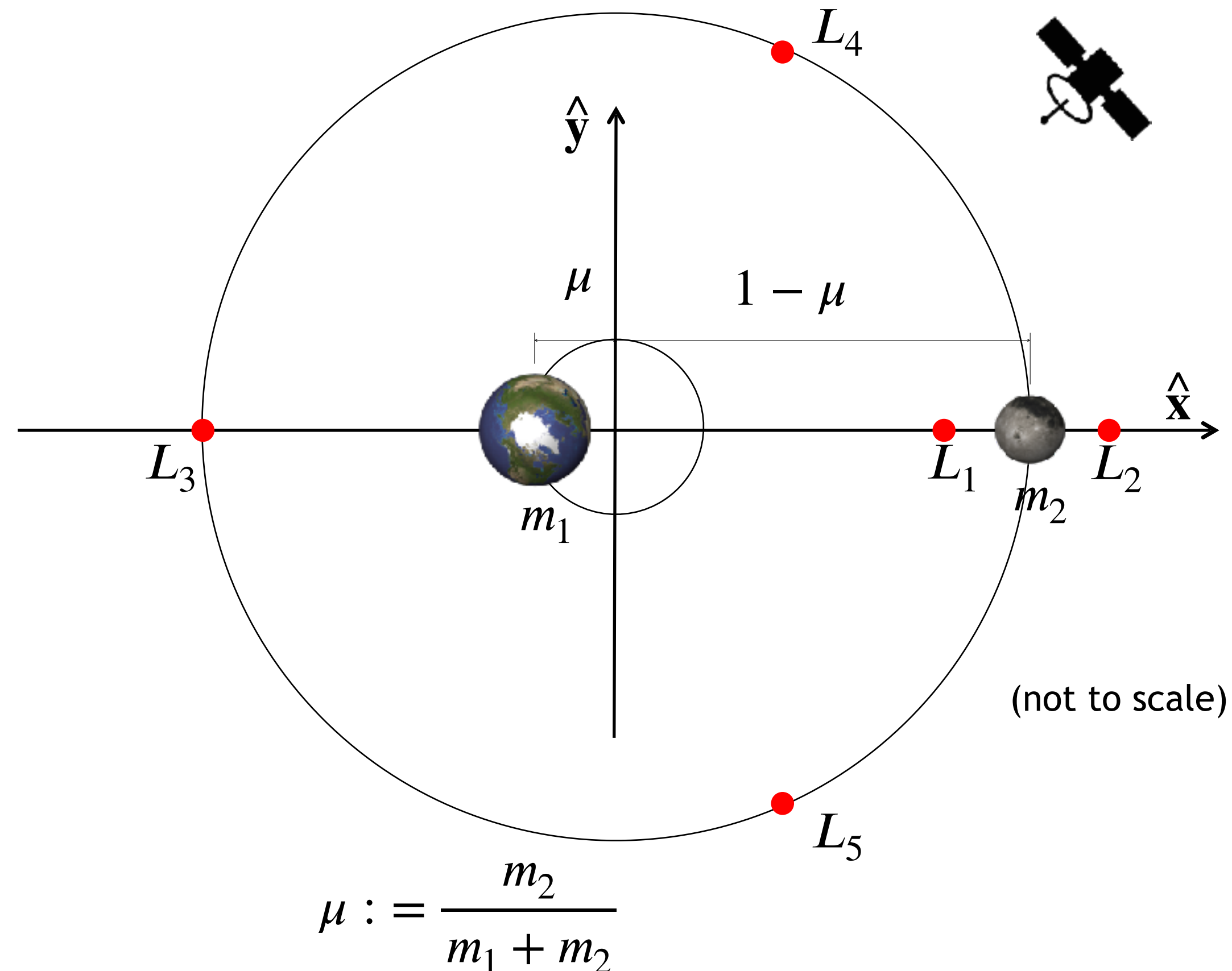
The Fundamental Problem in Spacecraft Dynamics is...



- ... the Circular-Restricted 3-Body Problem
 - Motion of a point mass attracted by 2 massive bodies in a circular orbit
 - Coordinate frame rotates with the bodies, centripetal accelerations that can be balanced with gravity

While the CR3BP cannot be *solved* in general, it has *many* known or knowable solutions:

- Equilibrium points (co-orbiting circular orbits)
- Periodic orbits
- Quasi-periodic orbits
- Stable / Unstable manifolds

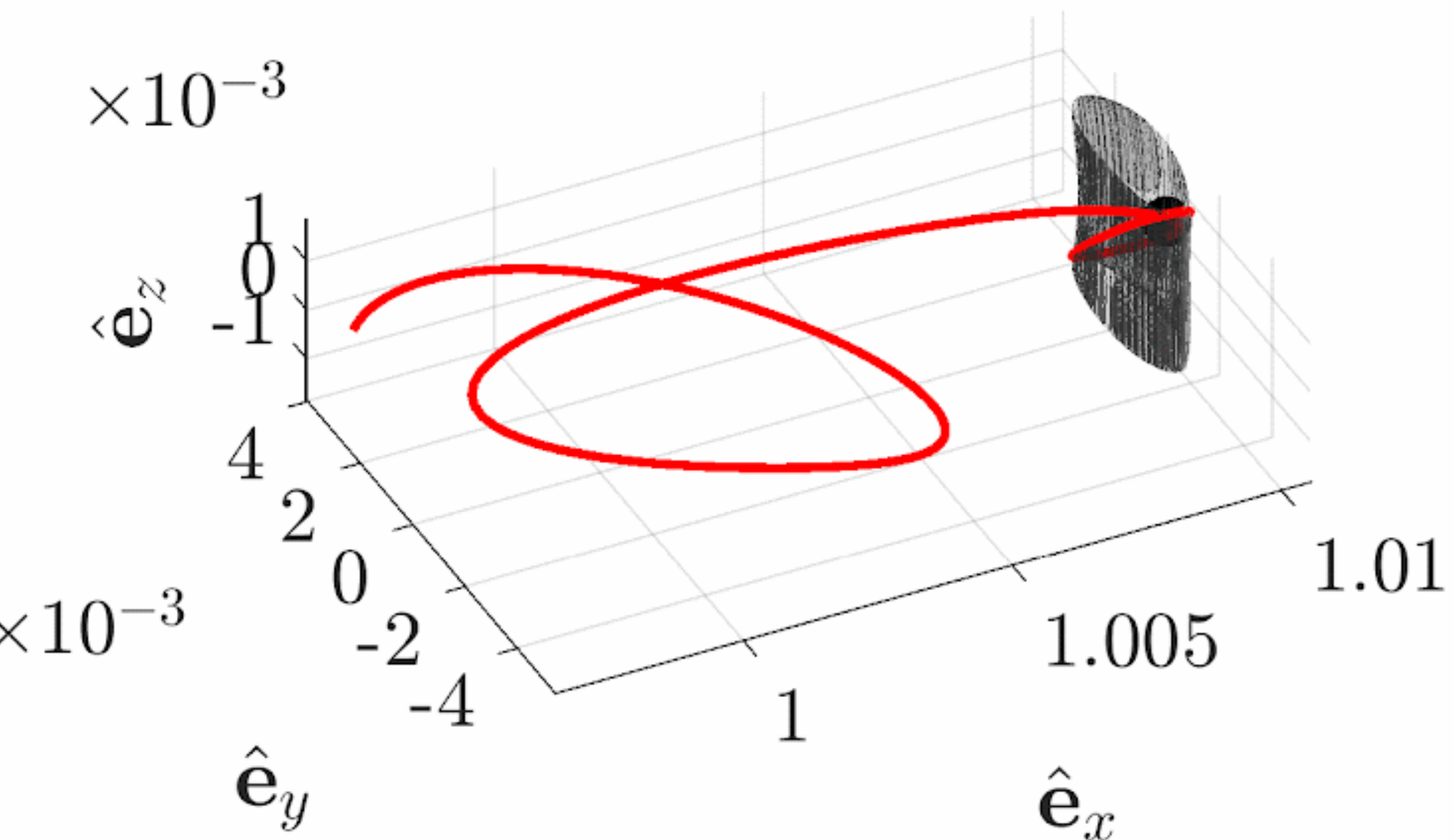




Solution Types

- Examples of these different types of solutions can be found relative to

Unstable Manifold from a Quasi-Periodic Orbit



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Equilibrium Point

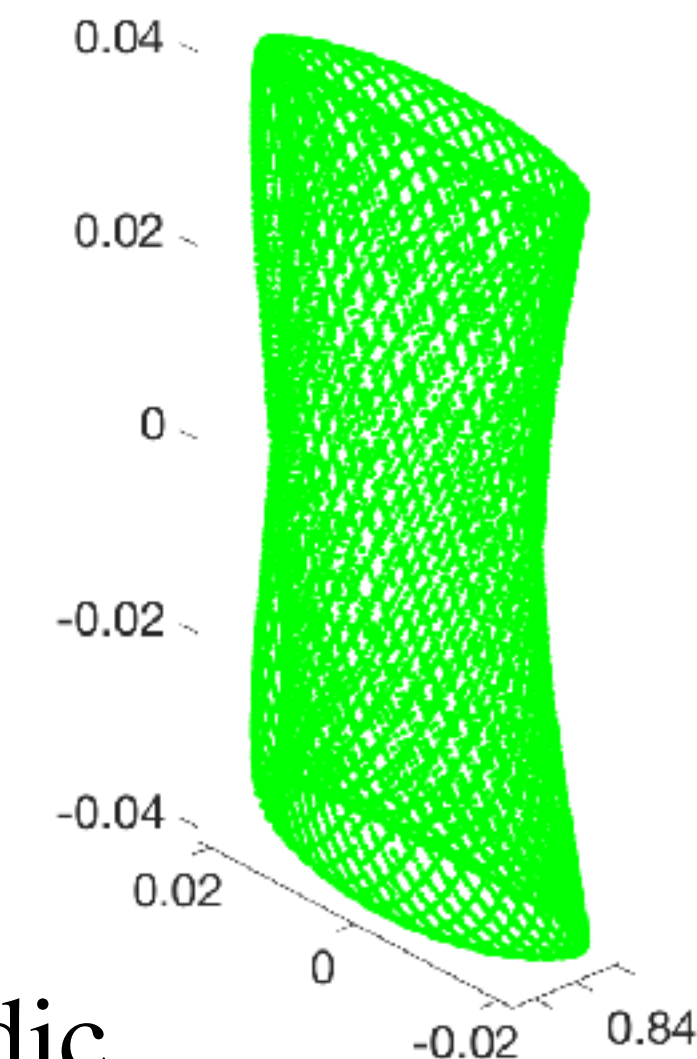
Unstable
Manifold

Periodic Orbit

Stable
Manifold

Center Manifold

Quasi-Periodic





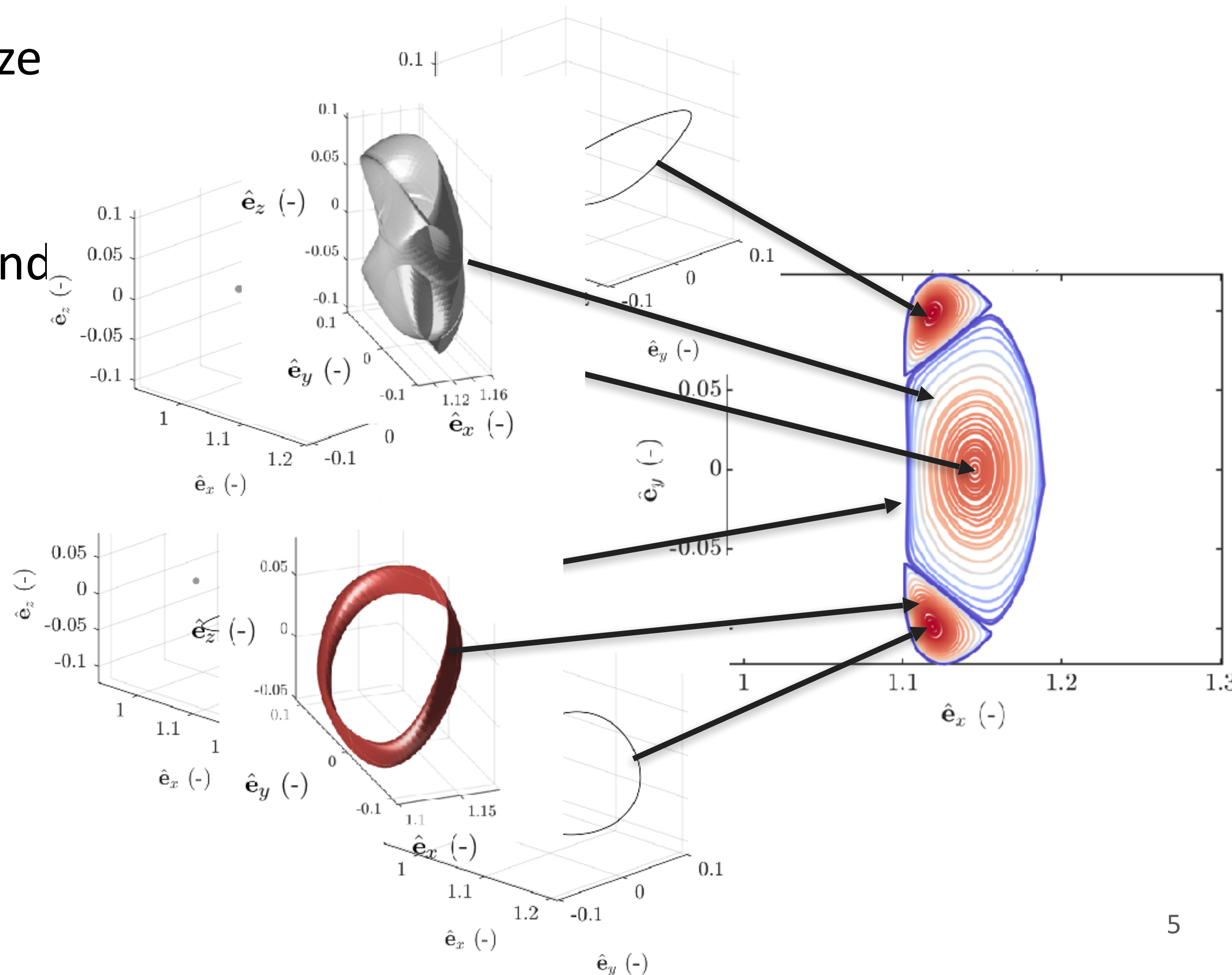
Orbits in the Circular Restricted Problem



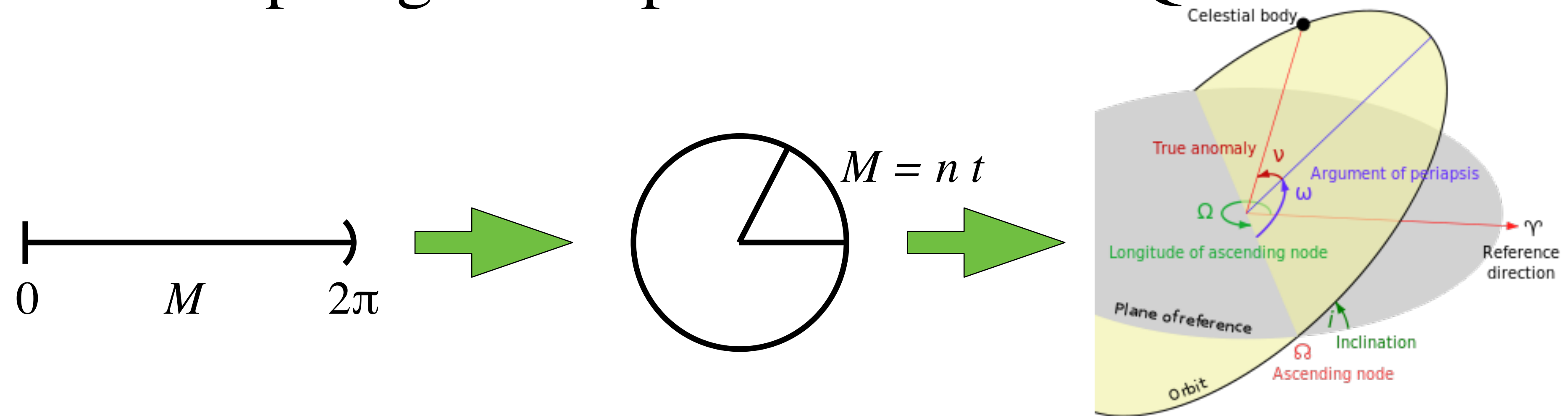
We can use a Poincare map to summarize the orbits at a given energy level

At our chosen energy level, around L1 and L2 there are:

- A discrete set of periodic orbits
 - Vertical Lyapunov
 - Southern and northern halos
 - Planar Lyapunov
- One-parameter families of quasi-periodic orbits
 - Lissajous
 - Southern and northern quasi-halos

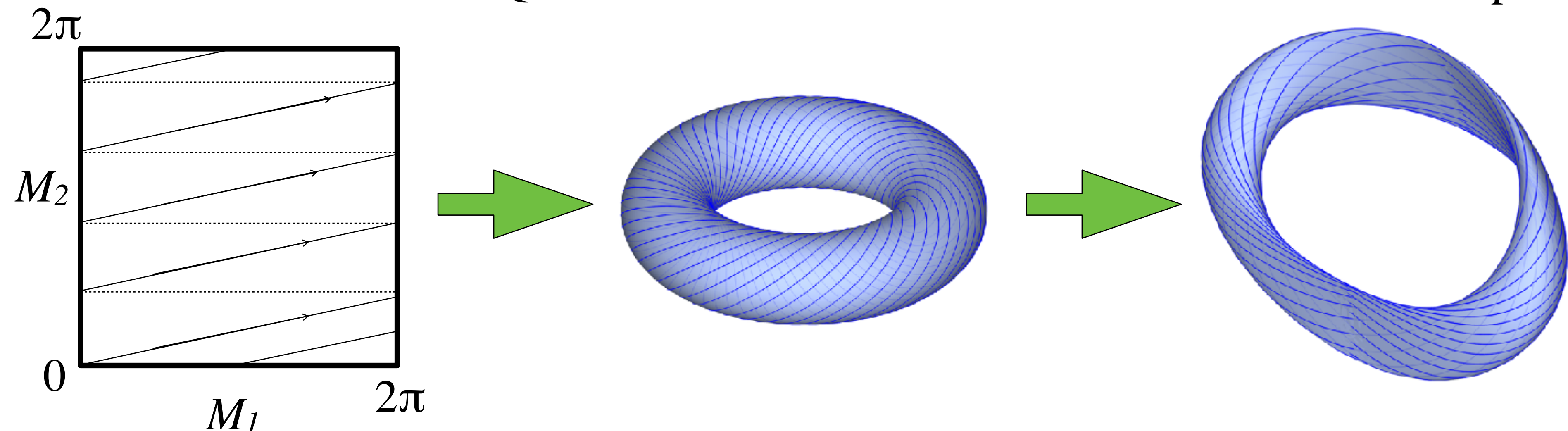


Topological Representations of QPO



Periodic Orbits are 1-D Circles embedded in 3-d space

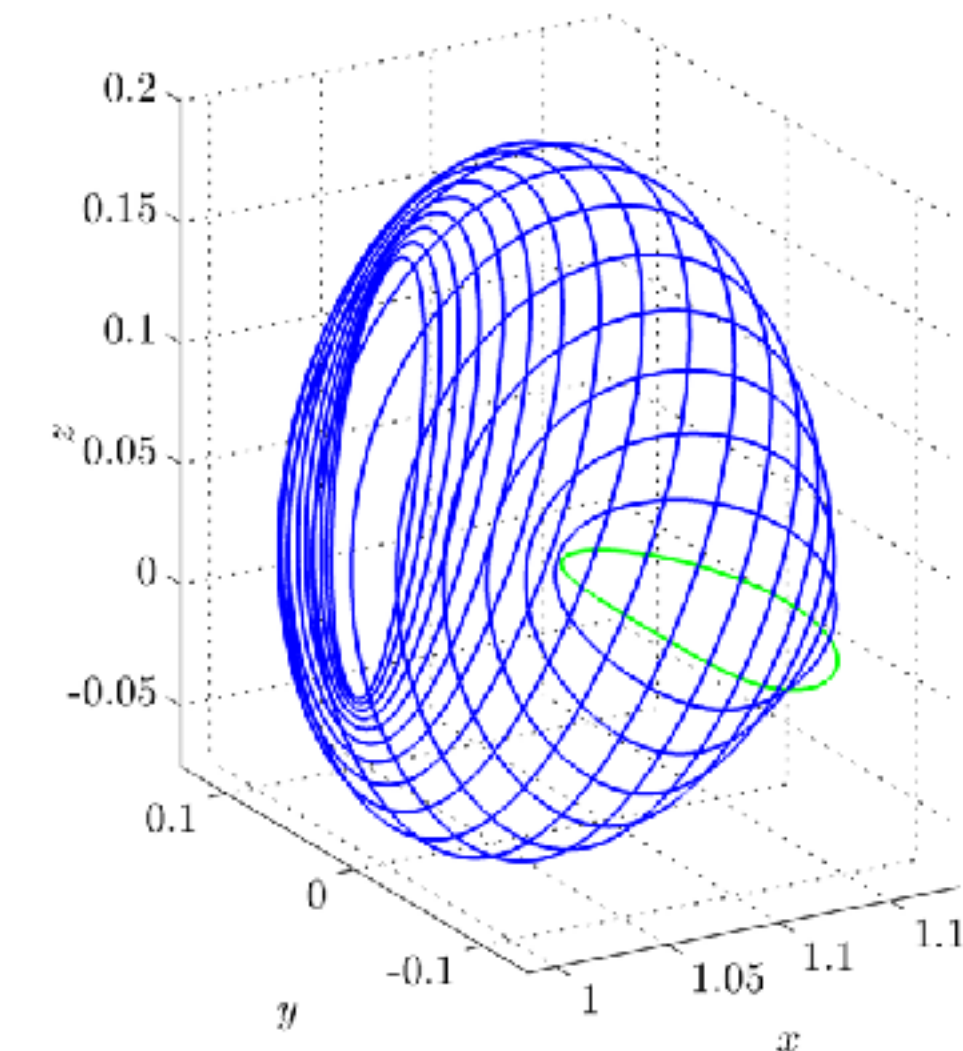
Quasi-Periodic Orbits are 2-D Tori embedded in 3-d space



Families of Solutions

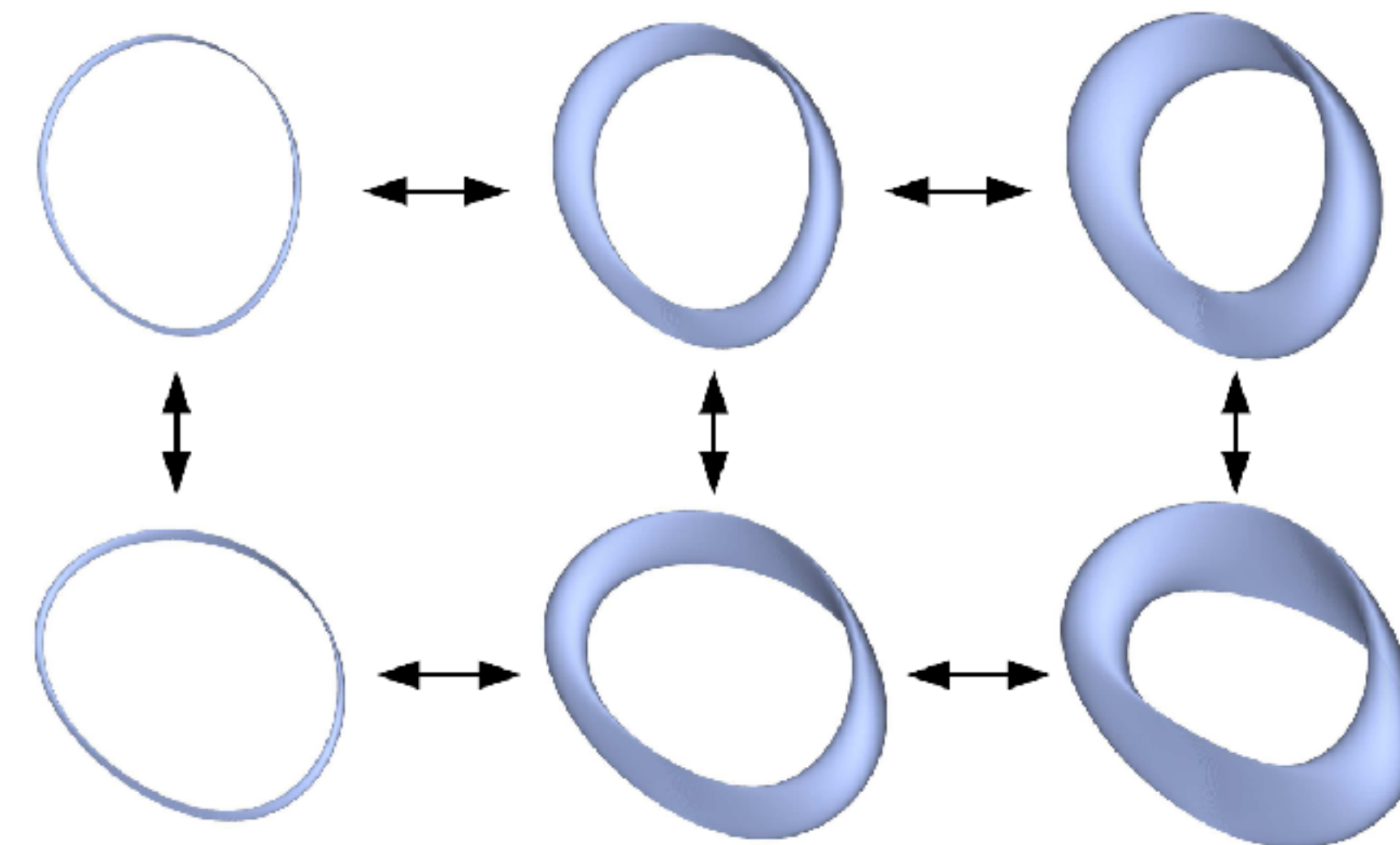


- Equilibrium points exist as singular solutions — isolated with no parameter
- Periodic orbits exist as 1-parameter families — Amplitude changes with orbit “Energy” or orbit frequency (period)

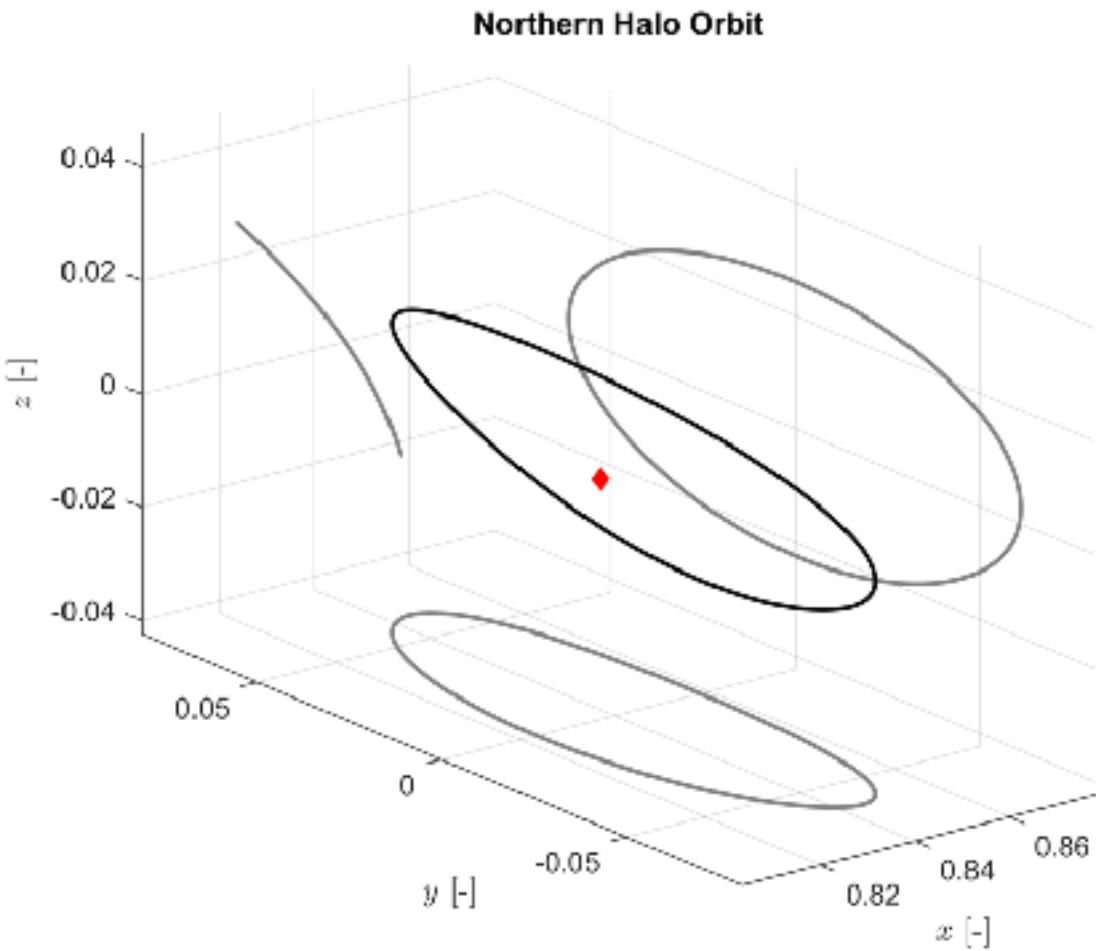


Halo Periodic Orbits, each at a different energy level

- 2-D Quasi-periodic orbits exist as 2-parameter families
 - Energy + Rotation Number or 2 frequencies
- 3-D Quasi-periodic orbits exist as 3-parameter families
 - Energy + 2 Rotation Numbers or 3 frequencies

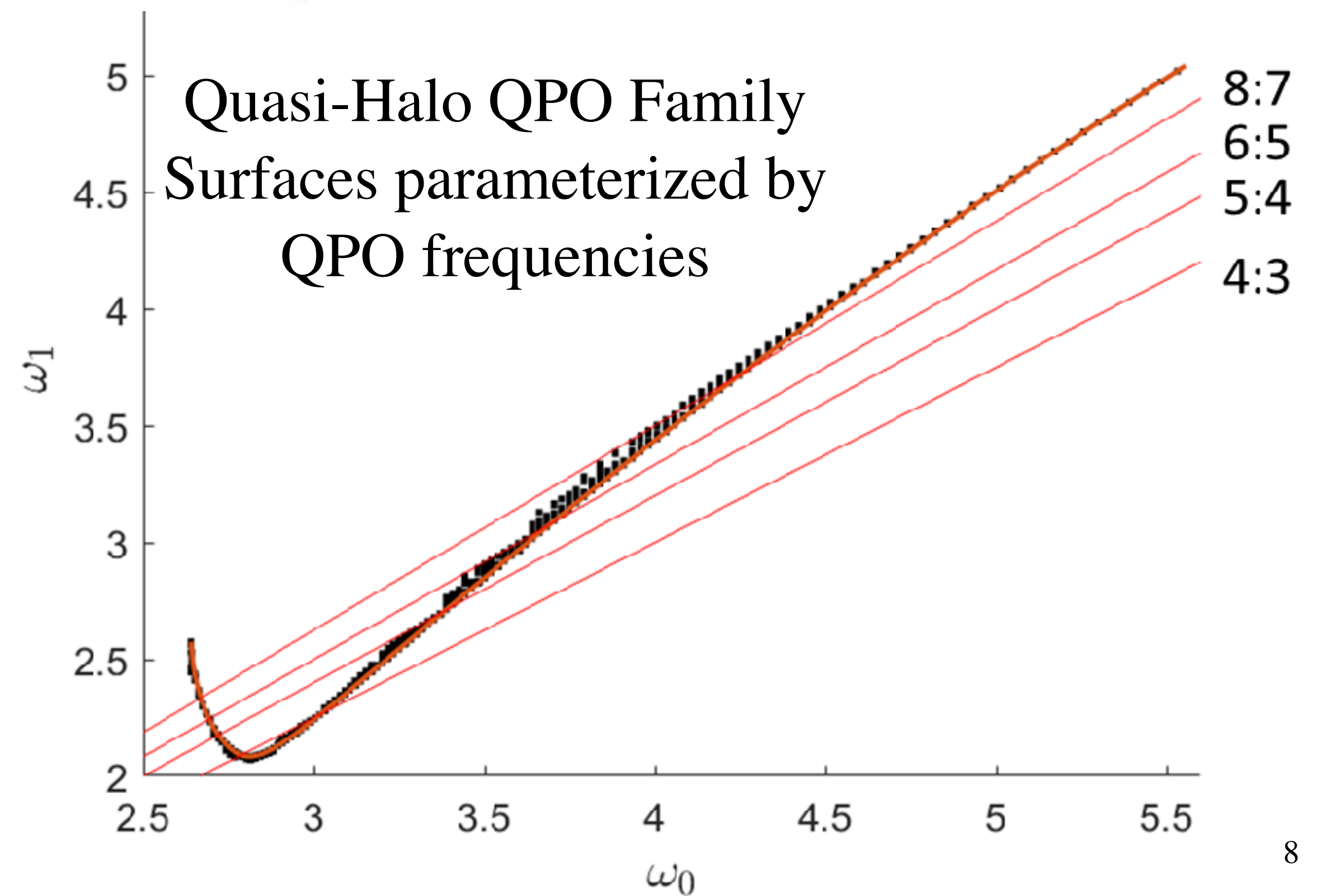
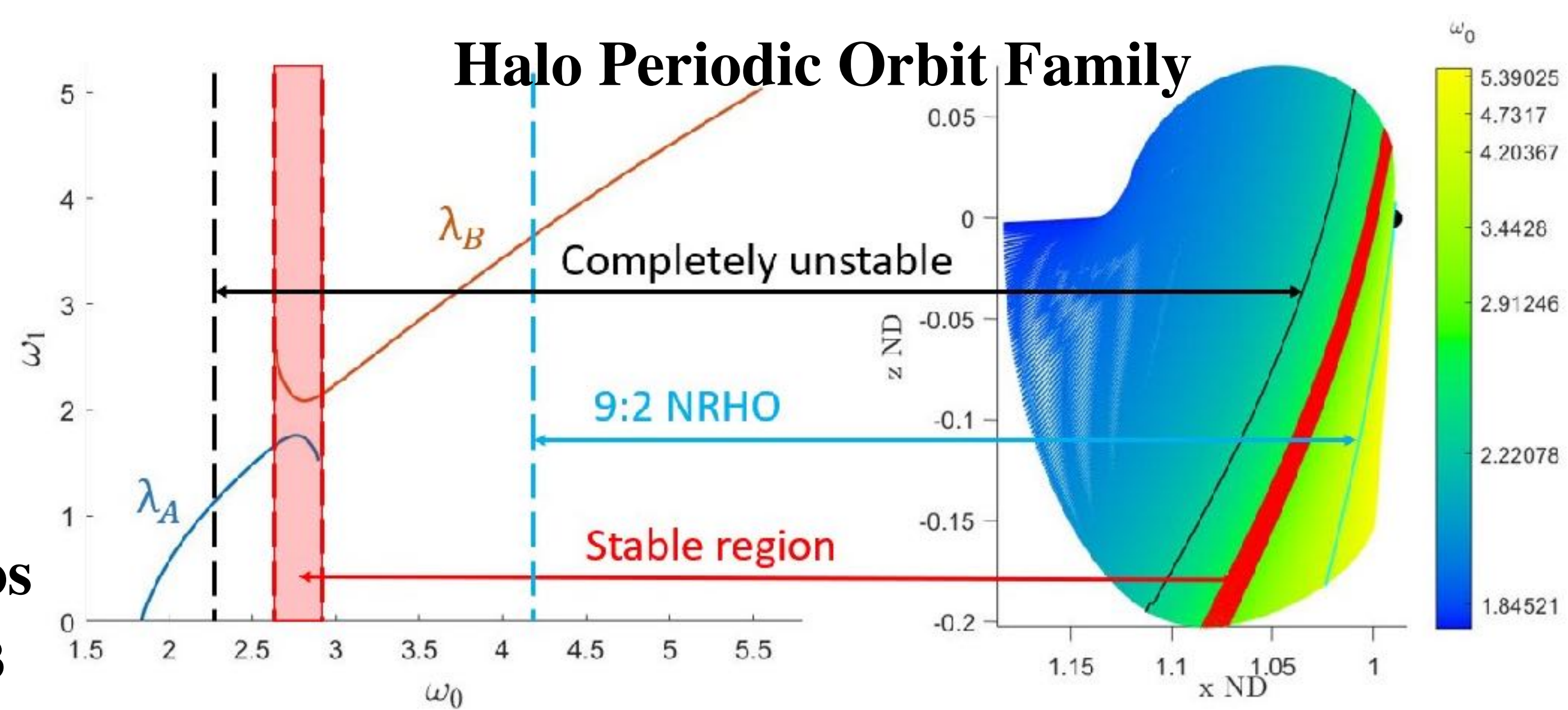
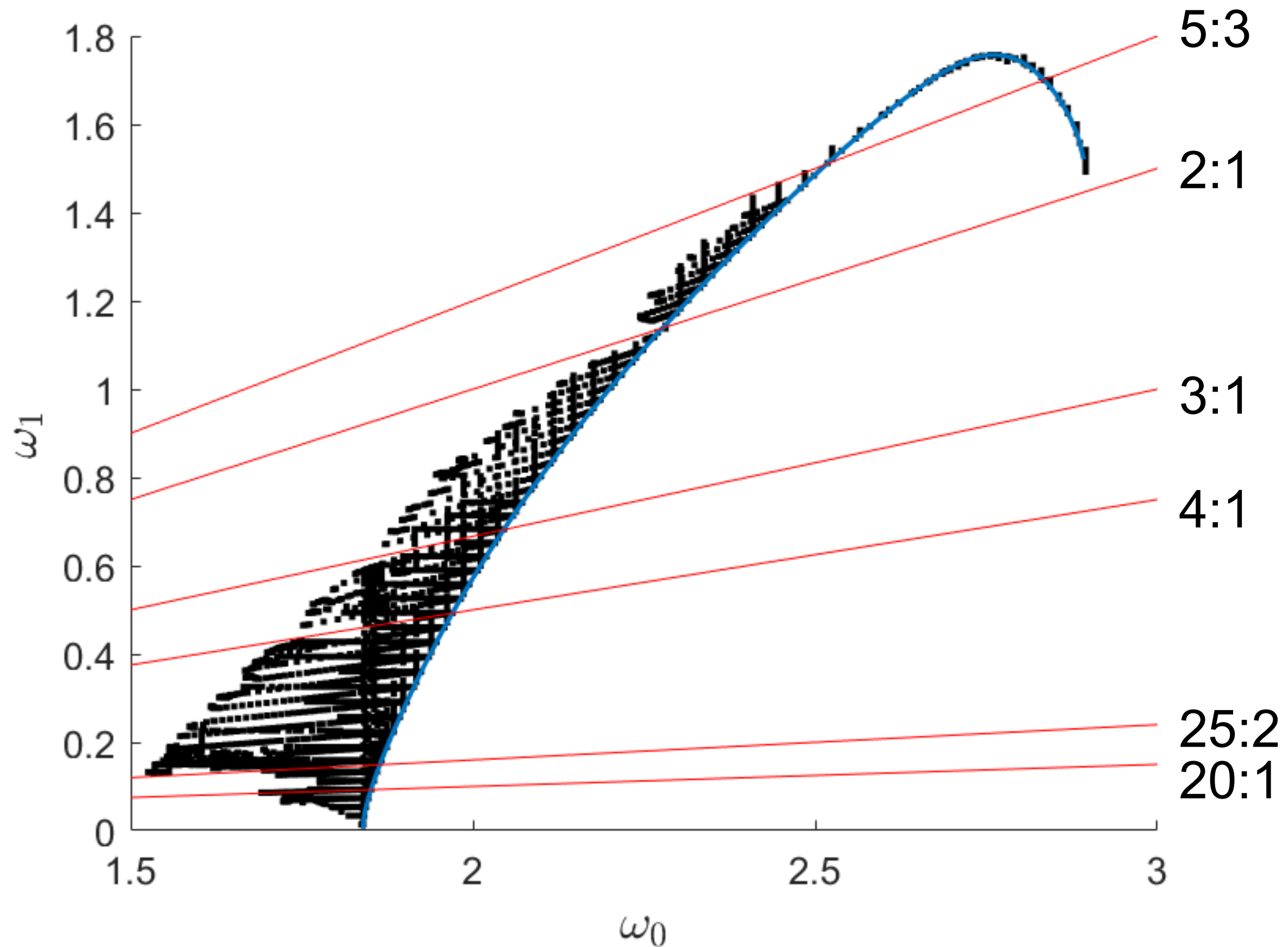


Quasi-Halo
Quasi-Periodic
Orbits



Halo orbit family:
Period $2\pi/\omega_0$
Center manifold
frequency ω_1

Quasi-Periodic Orbit Family — Quasi-Halos

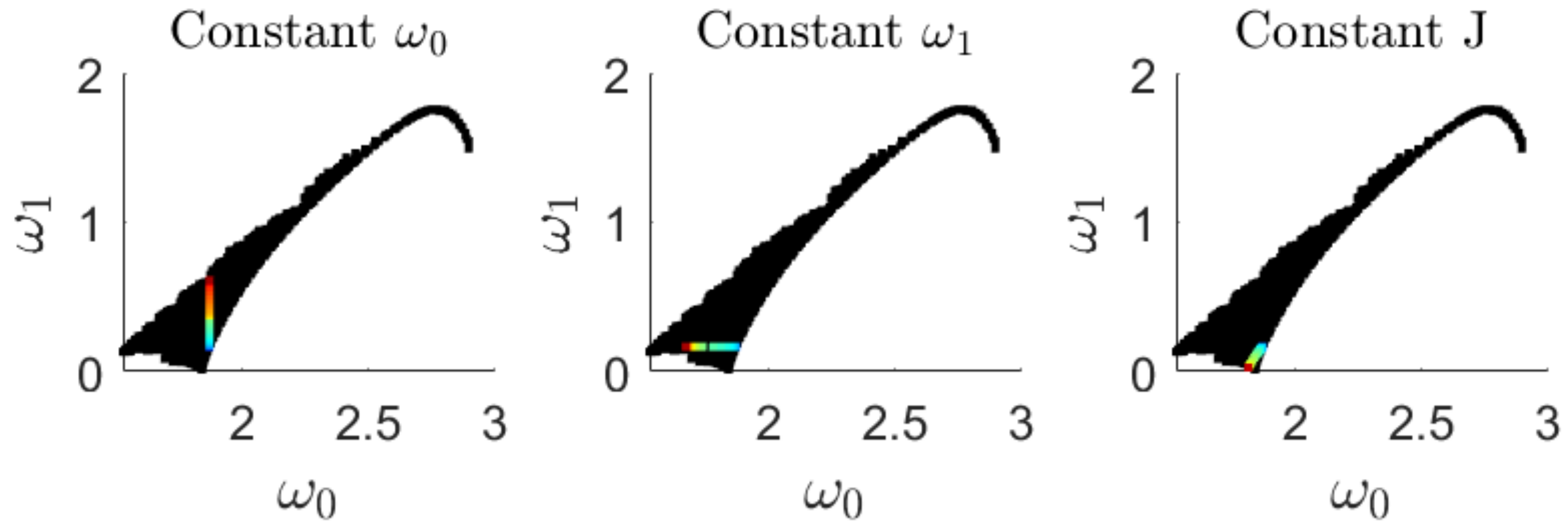




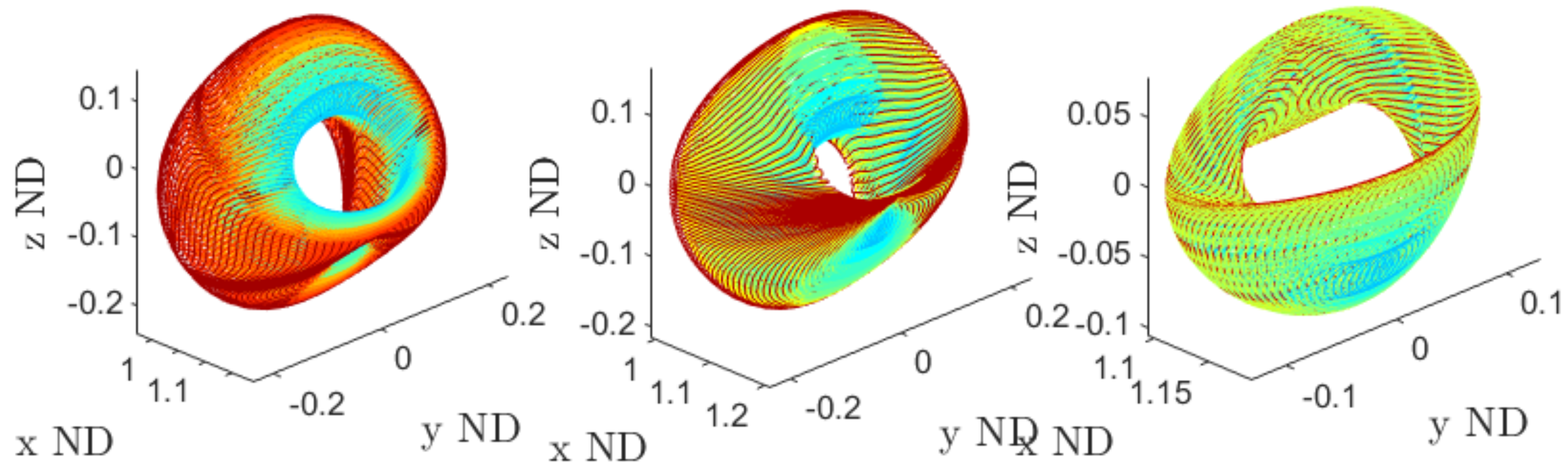
QPO Variation Across Frequency

$$\omega_0 = 1.868404 \text{ — } \omega_1 = 0.166293 \text{ — } J = 3.126576$$

Family Surface



Computed Orbits

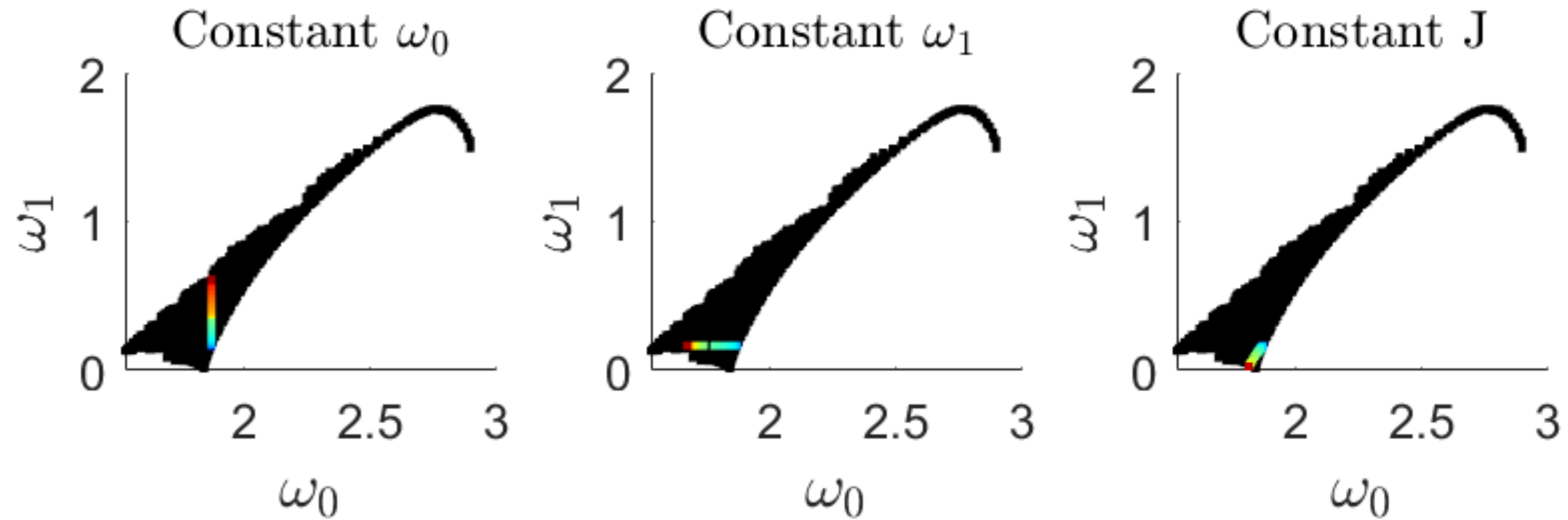




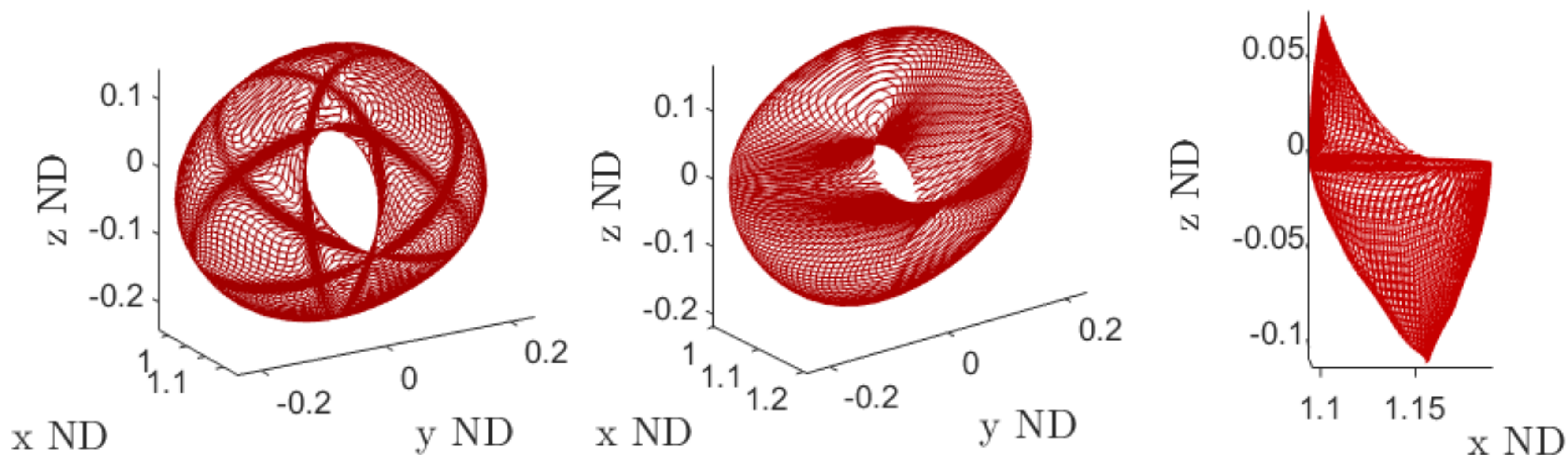
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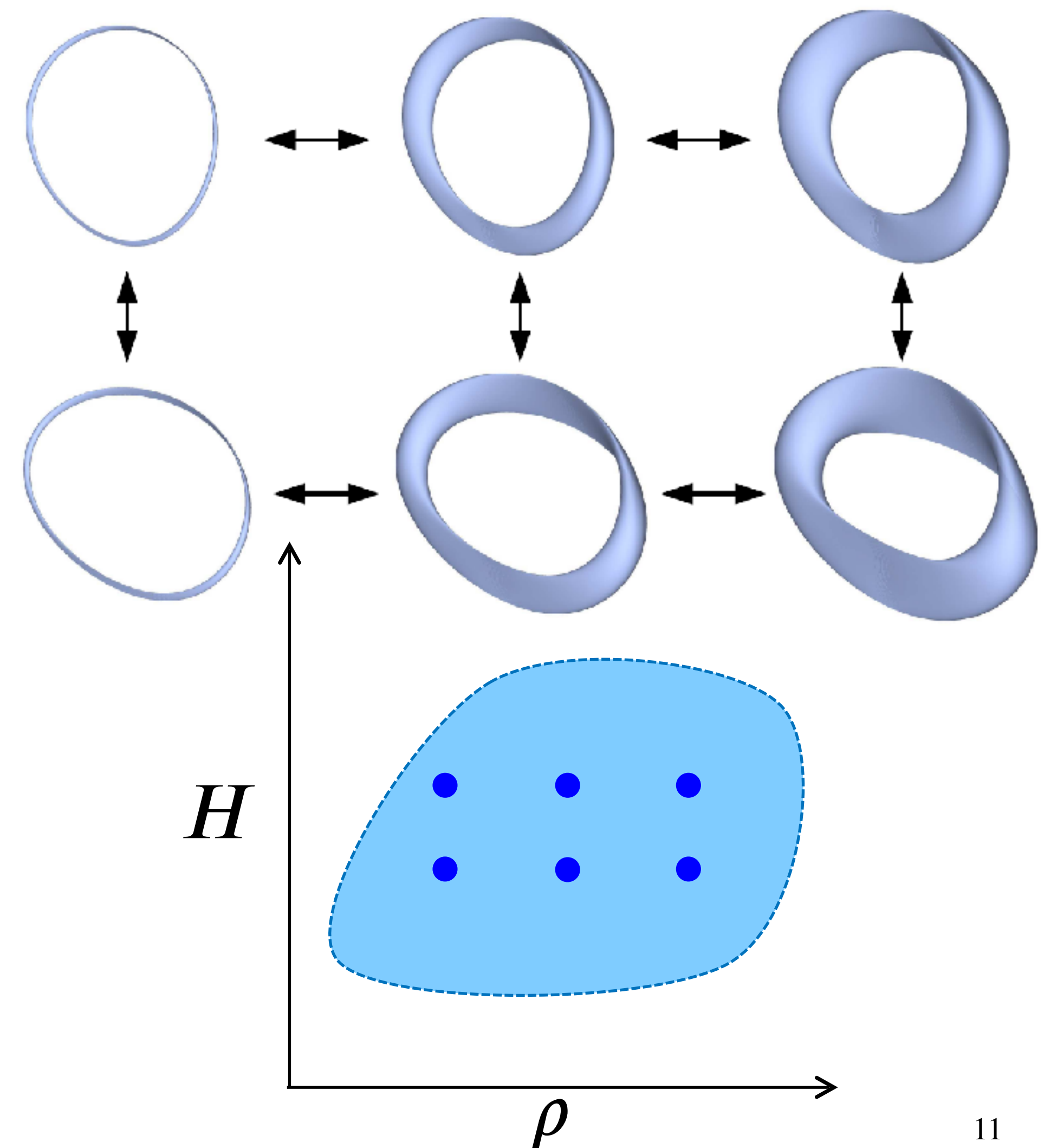


Last members
of the QPO
sequence

Continuity of Quasi-periodic orbits



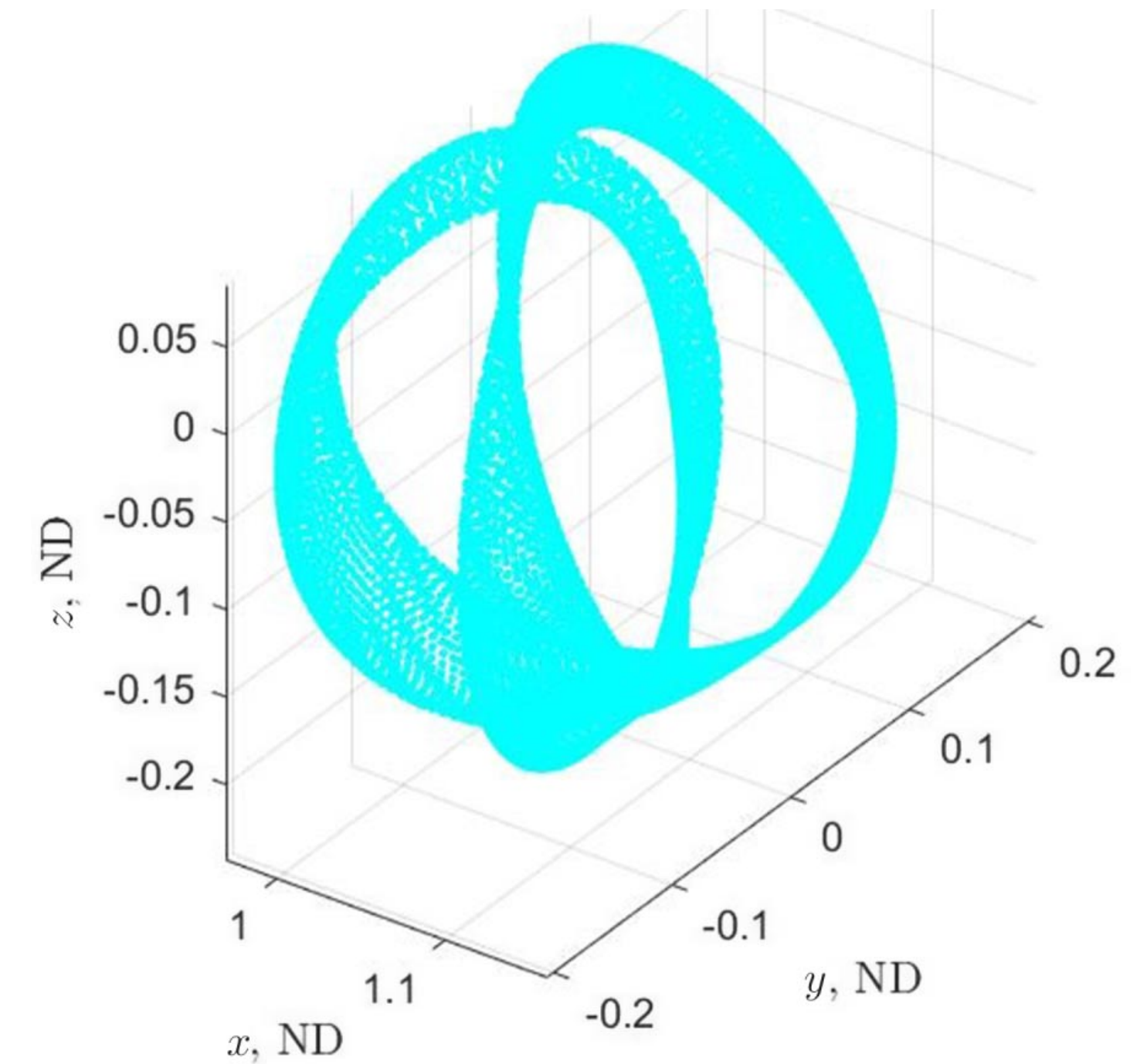
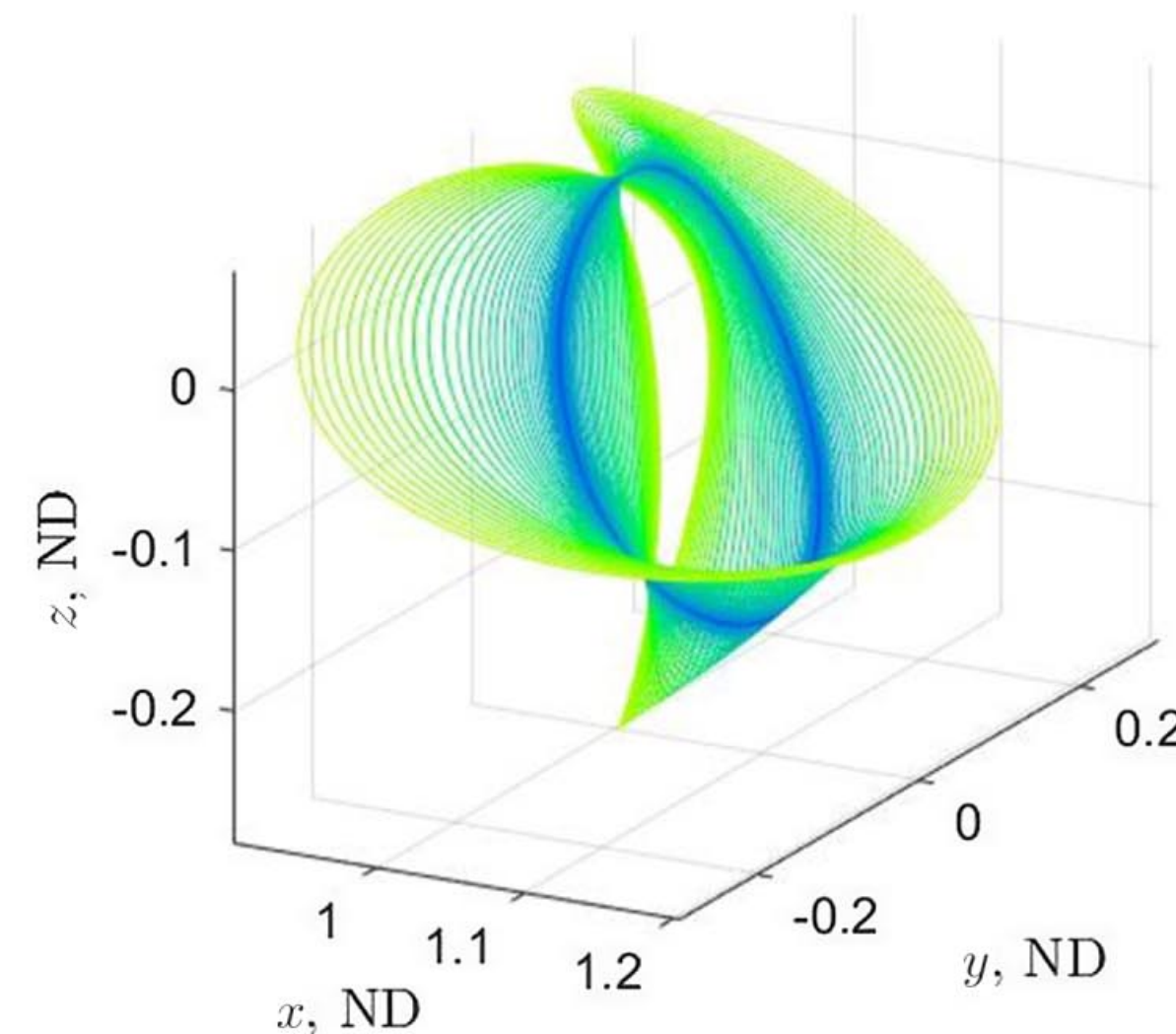
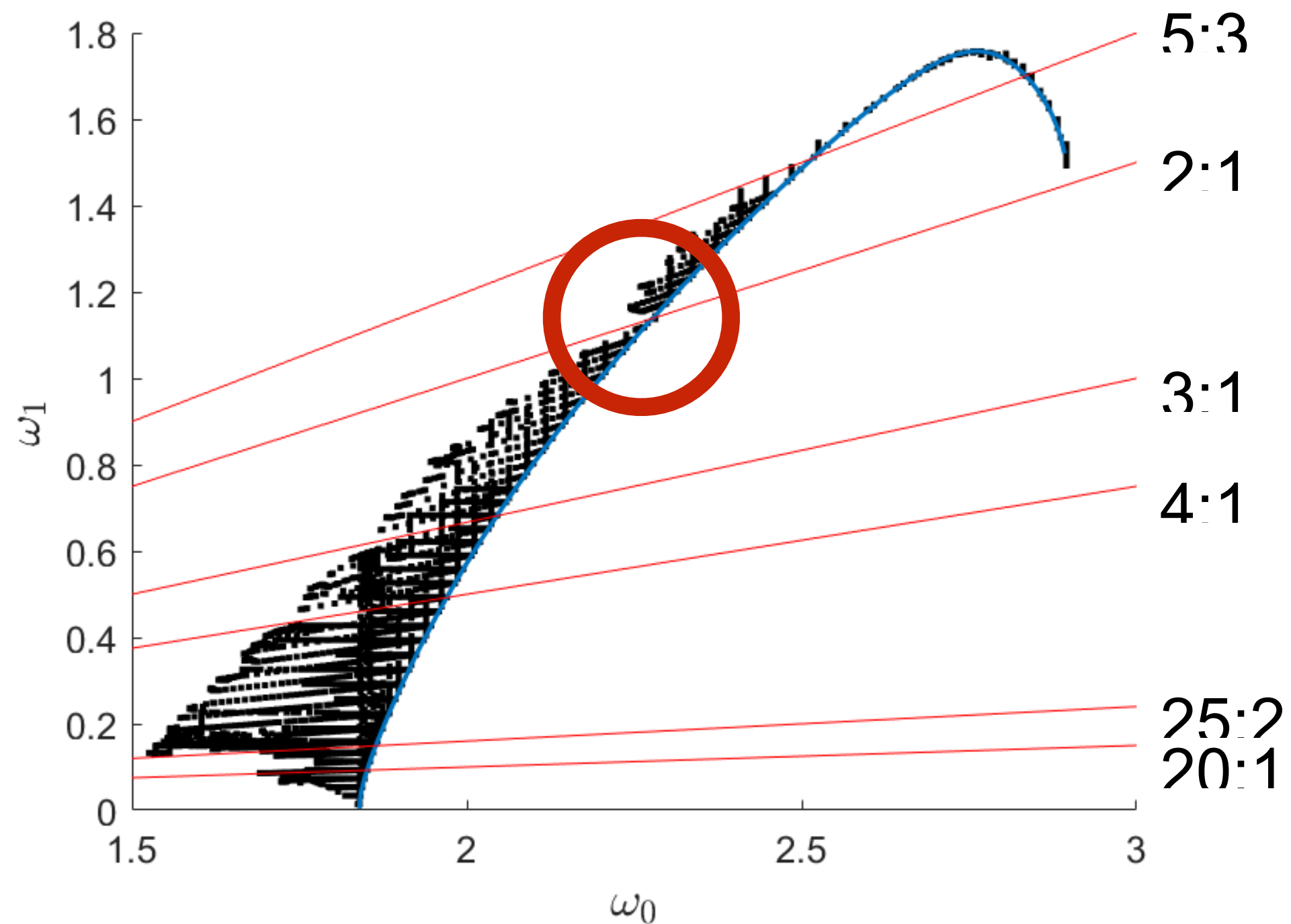
- We treat the families as “continuous”
- But similar to periodic orbit families, there should be small gaps in these families when they encounter resonances
 - Such resonances form a Cantor set
 - They generally have “exponentially small” measure [Jorba & Villanueva, 1997]
- At low order resonances we do find larger gaps, such as at the 2:1 resonance
 - Can still compute double periodic orbits at these resonances
 - These orbits have their own center manifolds and hence QPO families



2:1 QPO Families



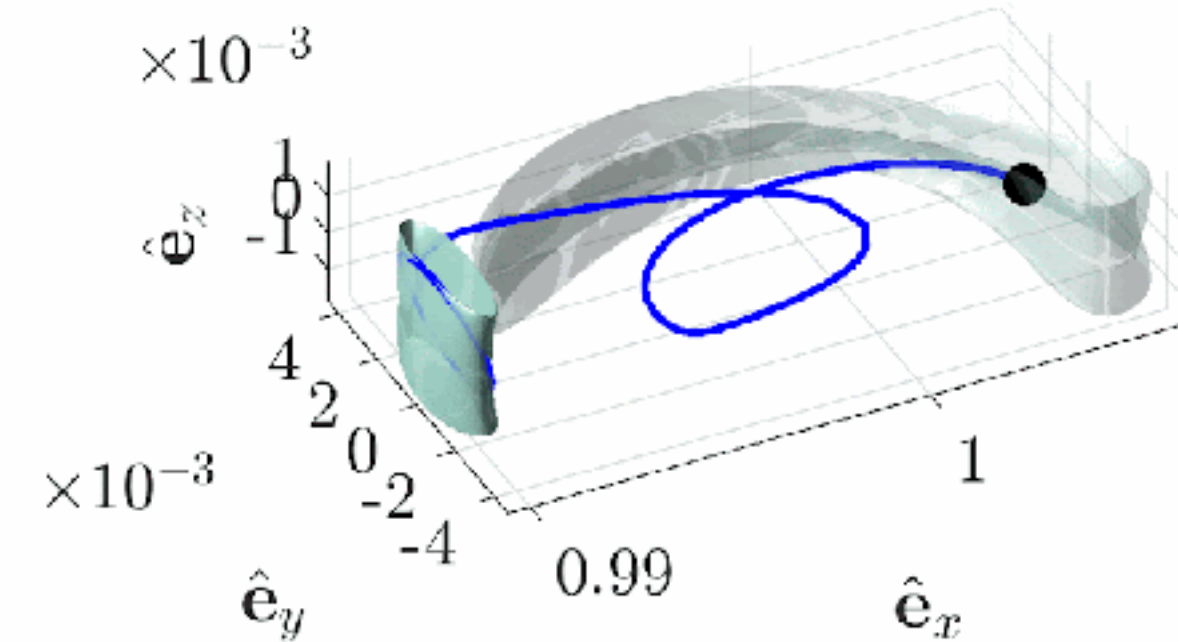
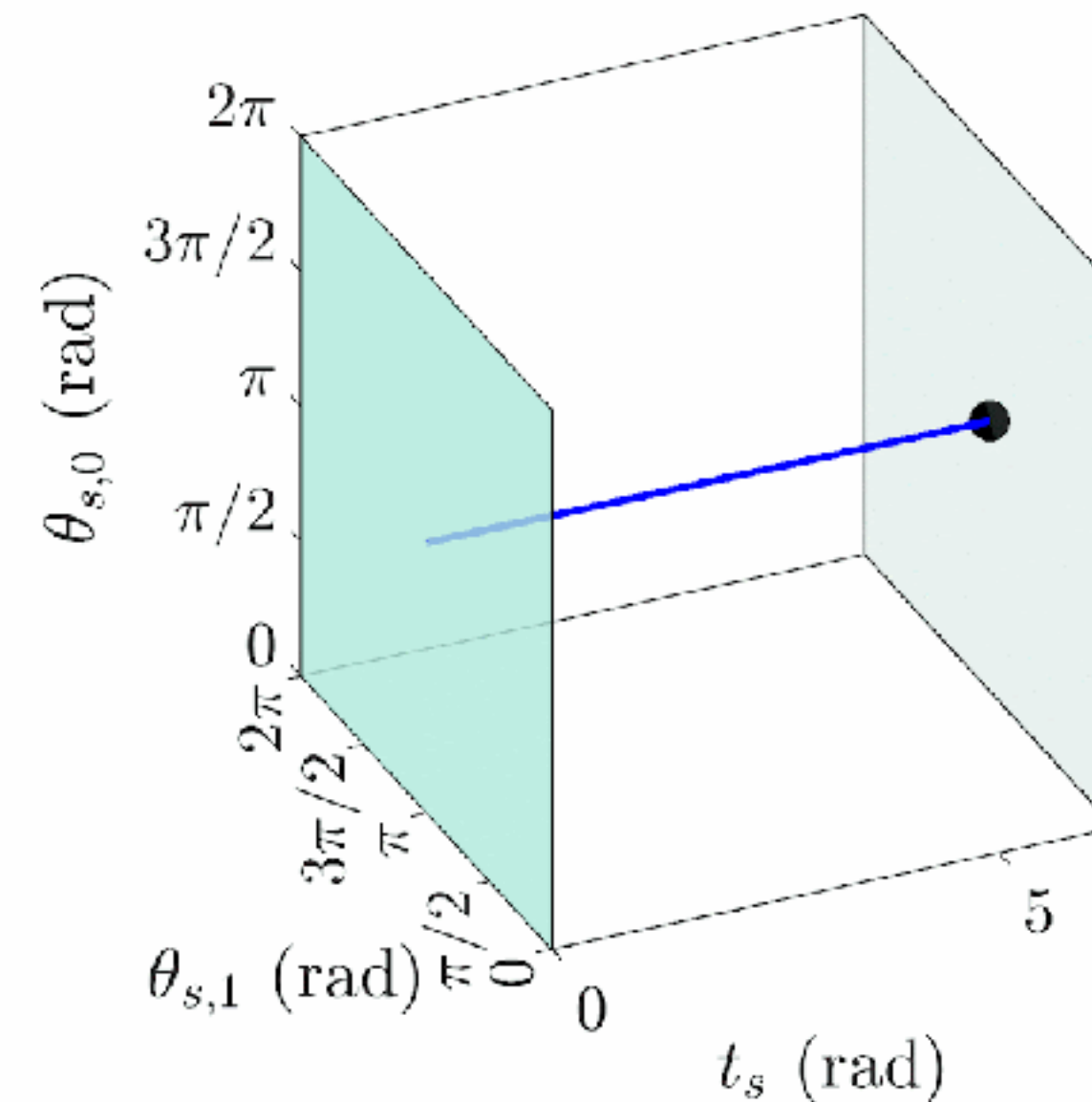
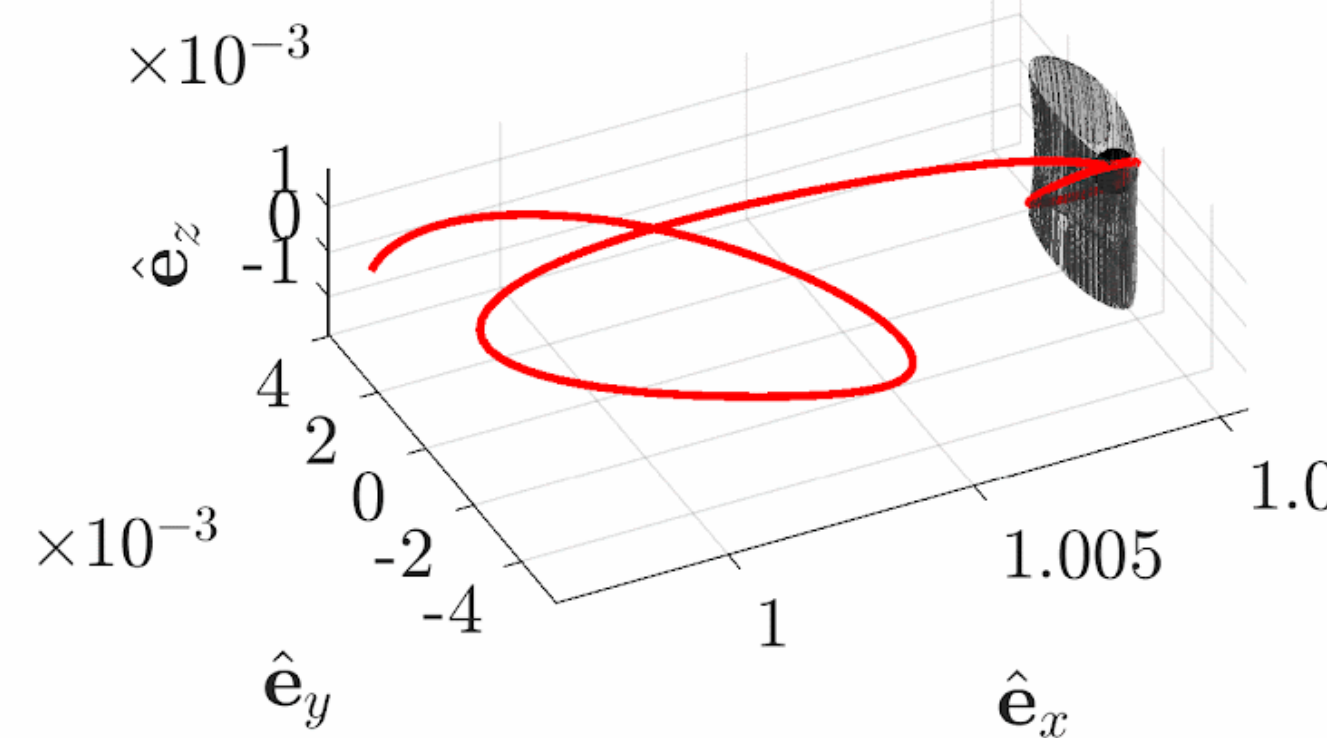
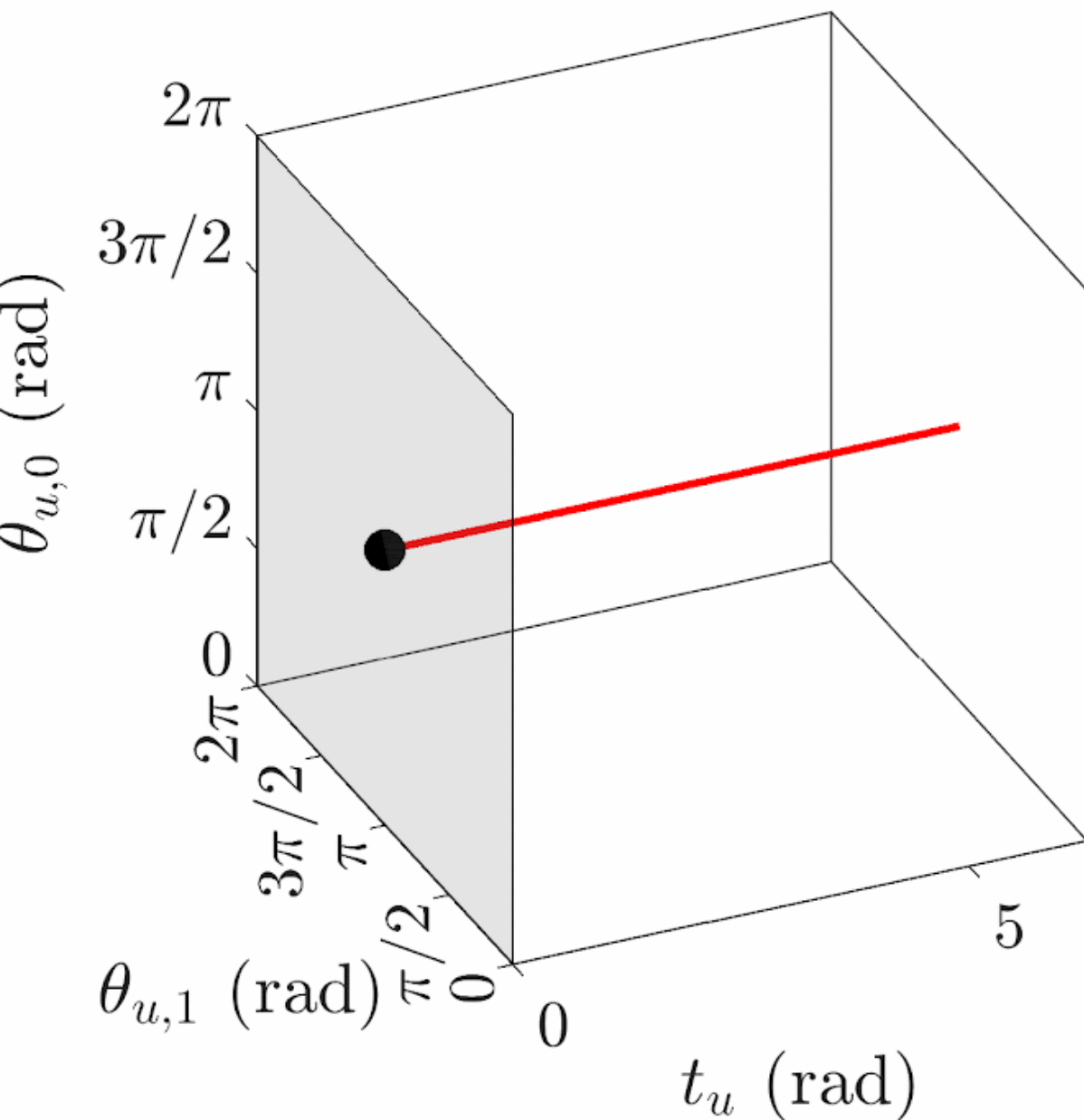
- Along the 2:1 gap we compute the associated periodic orbit family
 - The double periodic orbit families are shown below
 - A QPO from this branch are shown at right



Unstable and Stable QPO Manifolds



- As 2-D QPOs are hyperbolic, they have stable and unstable hyperbolic manifolds, forming a 3-D structure in phase space
 - Individual trajectories on these manifolds are called “whiskers”
 - These manifolds provide a rich set of trajectories that can be utilized for transfers



Family Dimensions



Torus dimension	Label	Parameters	Family dimension in Phase Space	(Un)stable manifold dimension increase
0	Equilibrium point	0	0	+1, +2
1	Periodic orbit	1	2	+1, +2
2	Quasi-periodic orbit	2	4	+1
3	Quasi-periodic orbit	3	6	N/A

QPO families fill large regions of phase space, when they exist

Take-away Message



- Quasi-periodic orbits are the “norm” and not the exception in the CR3BP
- Their numbers dominate over all other computable solutions
 - For every equilibrium point there are 1 $\rightarrow$ 2 infinities of periodic orbits
 - Every periodic orbit has 1 $\rightarrow$ 2 infinities of quasi-periodic orbits
 - Thus for every equilibrium point there are 2 $\rightarrow$ 4 infinities of quasi-periodic orbits
- QPO manifolds are of high dimension
 - It is possible to connect the stable and unstable manifolds of QPOs, forming “zero cost” heteroclinic transfers between them
 - Their overall dimensions are such that, given two QPO at the same energy level, there *should be* a heteroclinic connection between them
- If you can compute QPO, transfers between them must be ubiquitous!



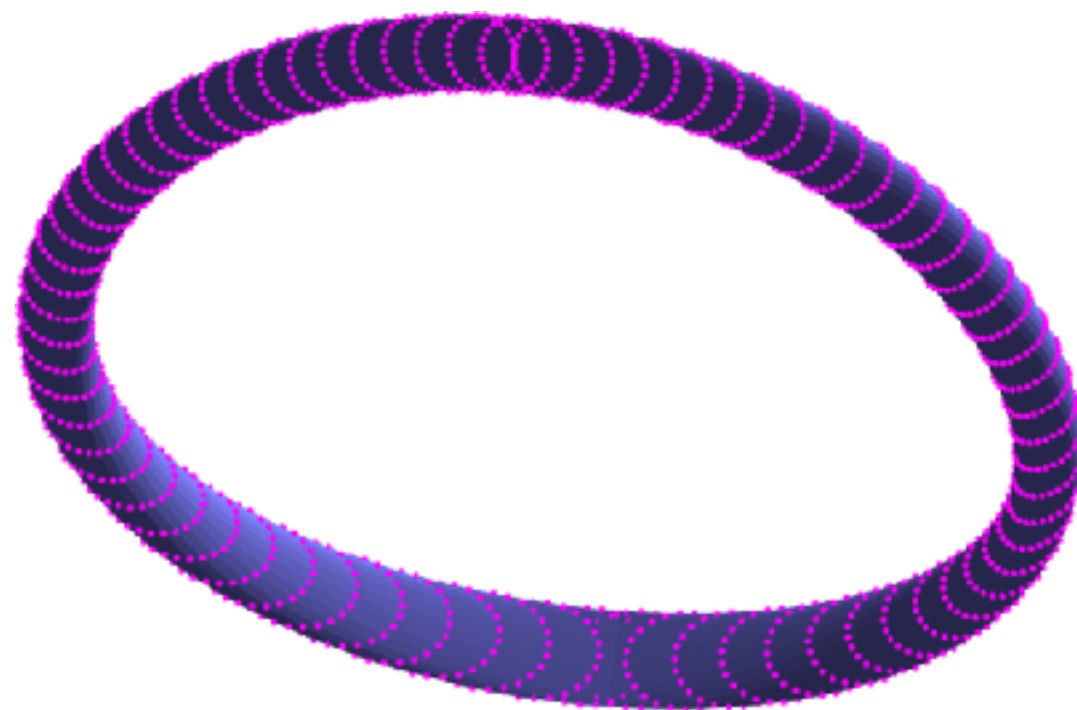
How do we compute QPOs?

Numerical Methods

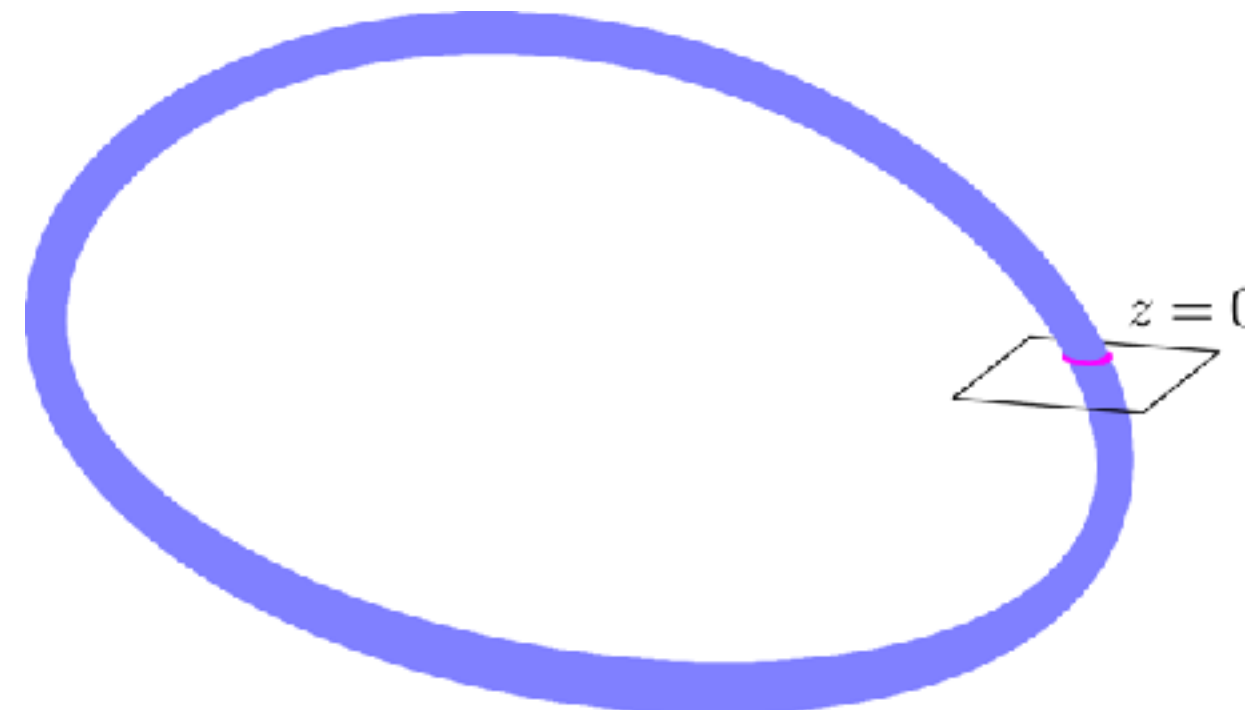


Method	Relevant Publications	Global Parametrization	Local Parametrization	Invariant Tori of Flows	Poincaré Section	Stroboscopic Map
SOV	Schilder, Osinga, and Vegt (2005) Olikara and Howell (2010)	✓		✓		
KKG	Kolemen, Kasdin, and Gurfil (2012)		✓		✓	
GMOS	Gómez & Mondelo (2001) Olikara & Scheeres (2012)		✓			✓

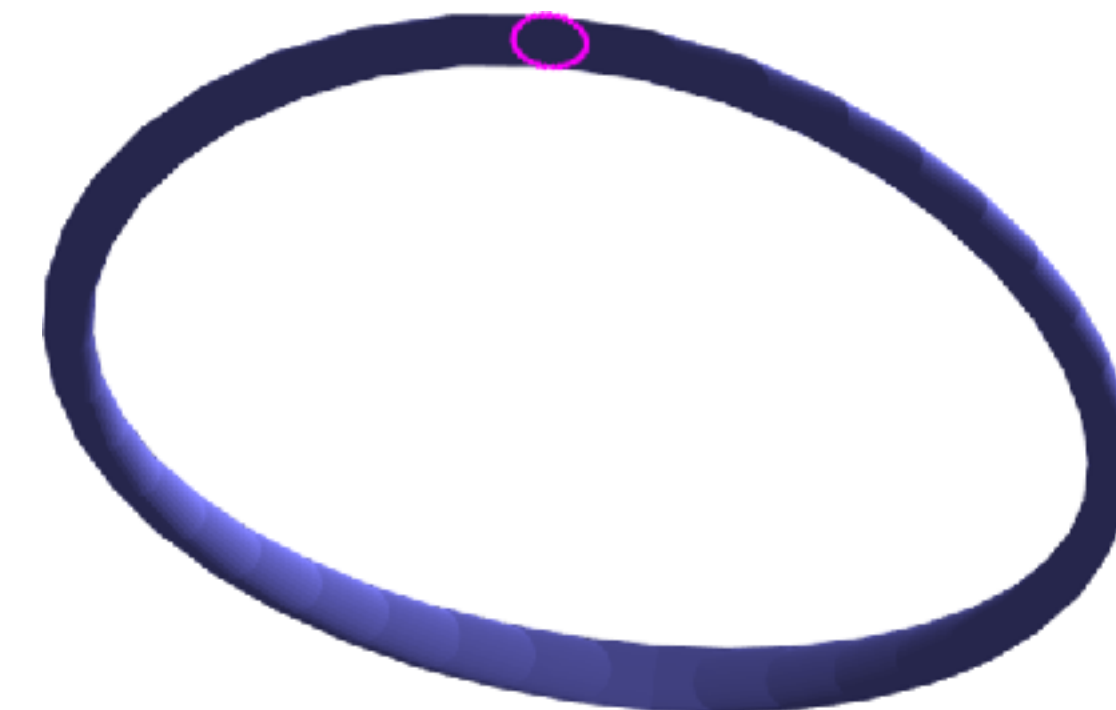
SOV



KKG



GMOS



N. Baresi, Z. Olikara and D.J. Scheeres. 2018. “Fully numerical methods for continuing families of quasi-periodic invariant tori in astrodynamics,” *Journal of the Astronautical Sciences* 65: 157-182.

2D Torus Computation



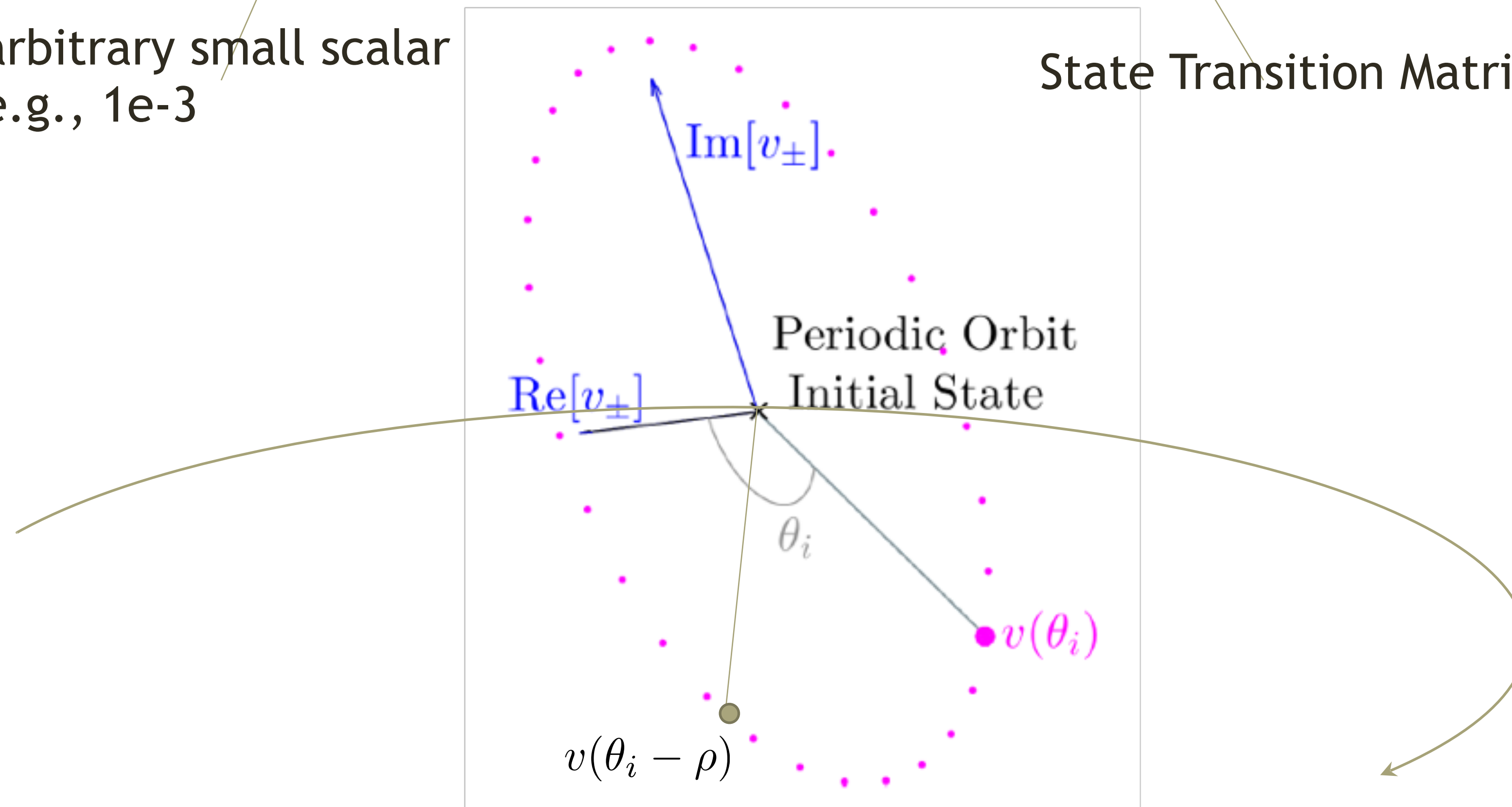
Compute QP tori via **invariant curves of a stroboscopic mapping**

Given a periodic orbit with period T and a center component $\Phi \mathbf{v}_{\pm} = e^{\pm i \rho} \mathbf{v}_{\pm}$

$$\mathbf{v}(\theta) = K \{ \cos \theta \operatorname{Re}[\mathbf{v}_{\pm}] - \sin \theta \operatorname{Im}[\mathbf{v}_{\pm}] \} \implies \Phi \mathbf{v}(\theta) = \mathbf{v}(\theta - \rho)$$

arbitrary small scalar
e.g., $1e-3$

State Transition Matrix



2D Torus Computation



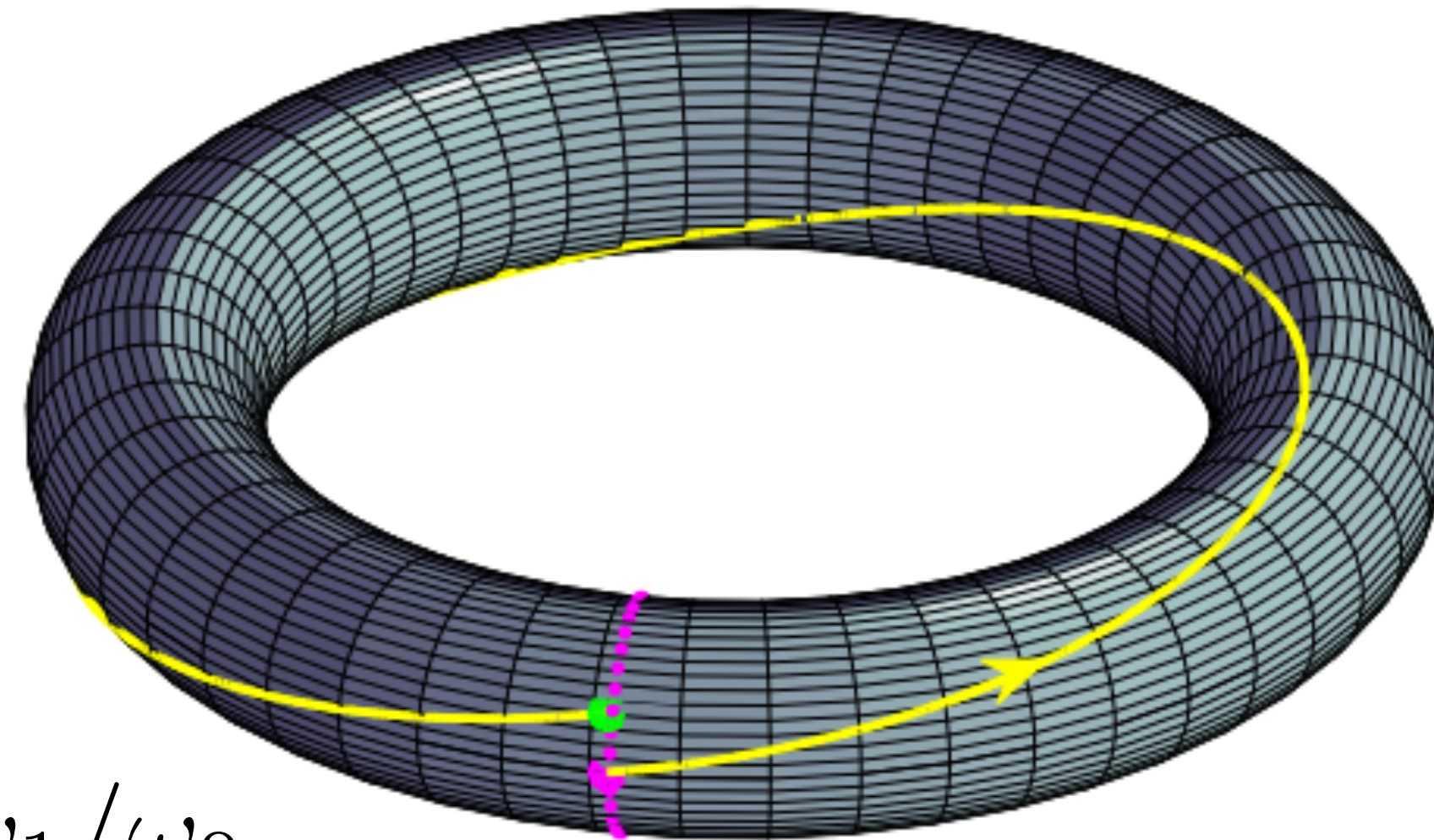
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arbitrary small scalar
e.g., $1e-3$

State Transition Matrix



$$T = 2\pi / \omega_0$$

$$\rho = \omega_1 T = 2\pi \omega_1 / \omega_0$$

ρ is known as the “rotation number” of the torus

Need to find states $\mathbf{v}$, $T(\omega_0)$, and $\rho(\omega_1)$ of a QP torus based on the information available from the Monodromy Matrix

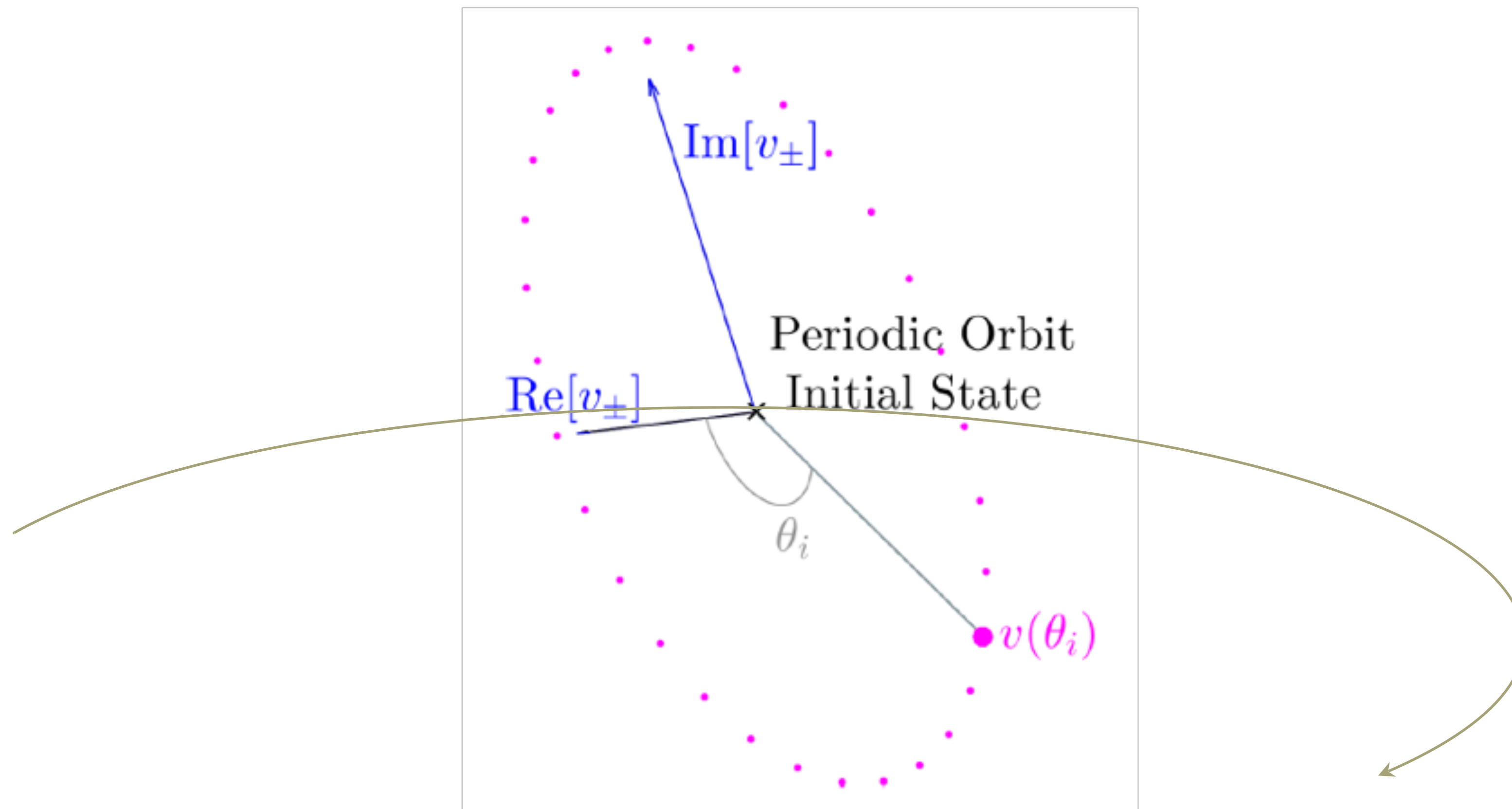
2D Torus Computation



Main Idea: Update initial info provided by PO via **Newton's method**

Given a periodic orbit with period T and a center component $\Phi v_{\pm} = e^{\pm i \rho} v_{\pm}$

$$v(\theta) = K \{ \cos \theta \operatorname{Re}[v_{\pm}] - \sin \theta \operatorname{Im}[v_{\pm}] \} \implies \Phi v(\theta) = v(\theta - \rho)$$

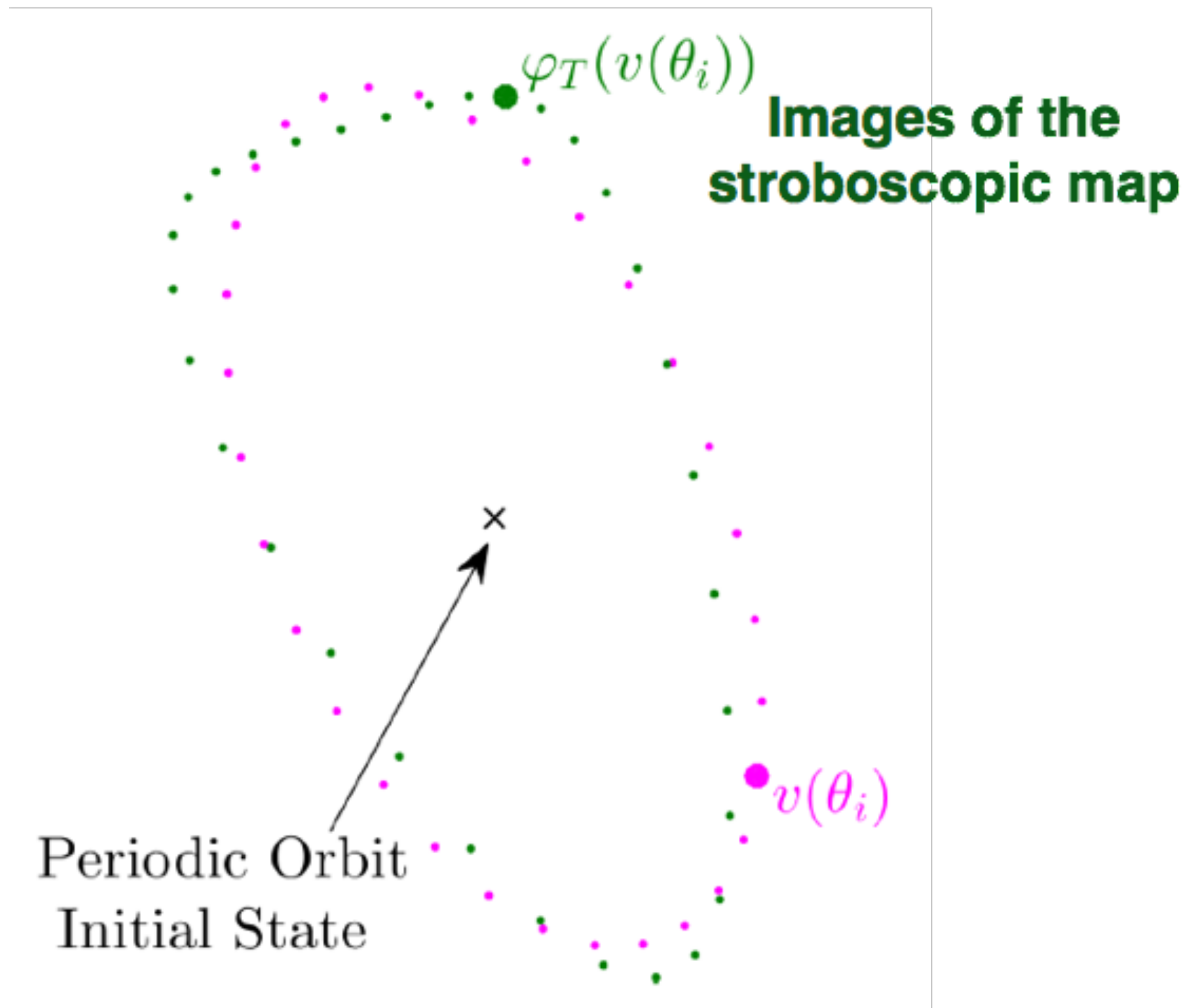


2D Torus Computation



Propagate each of the N “solution points” for time T

$$\dot{\mathbf{x}} = \mathbf{f}(t, \mathbf{x}) \implies \varphi_T(\mathbf{v}(\theta_i))$$

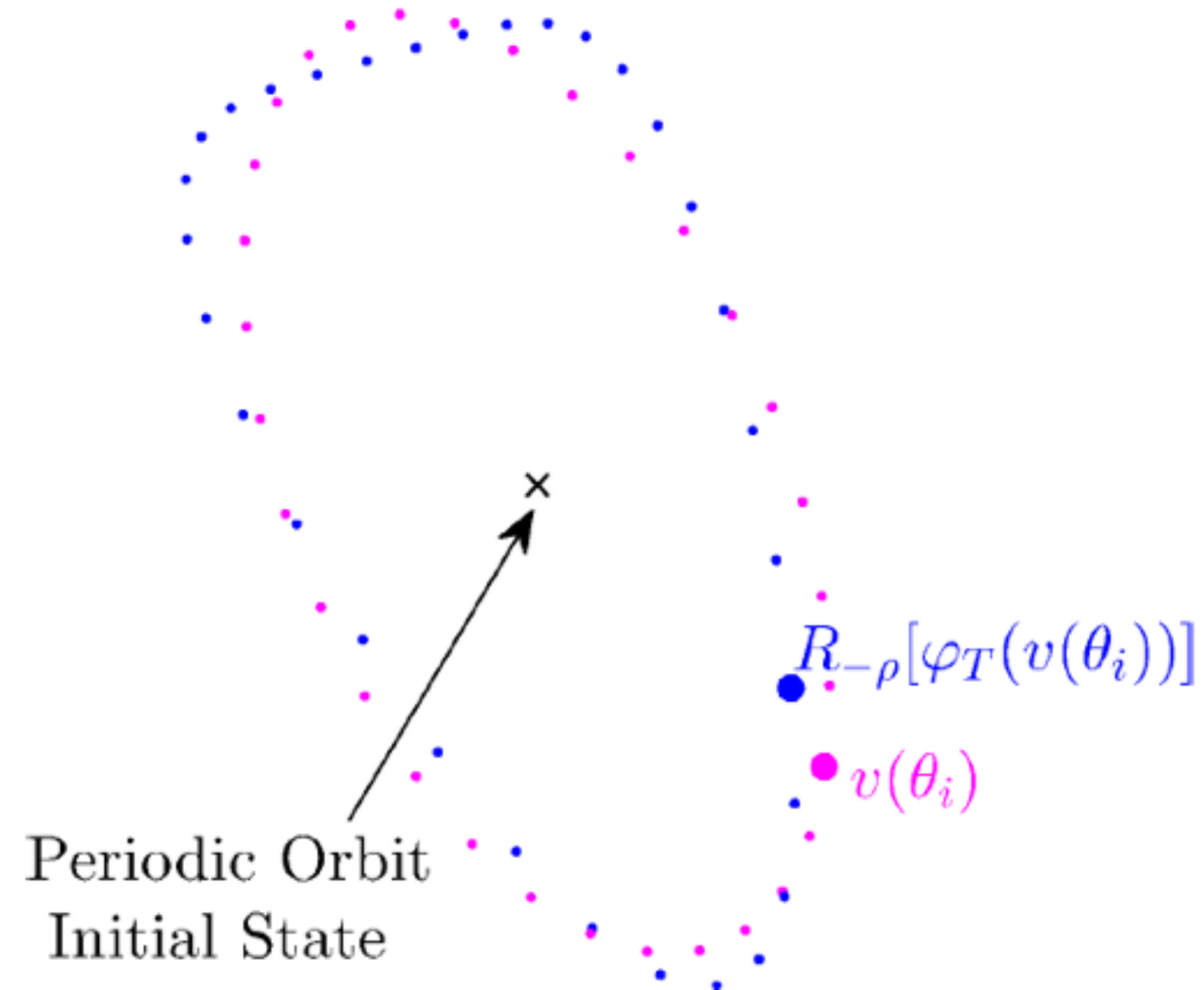


2D Torus Computation



Rotate the N “solution points” via Discrete Fourier Transform

$$F(\mathbf{z}) := R_{-\rho}[\varphi_T(\mathbf{v}(\theta_i))] - \mathbf{v}(\theta_i) = \mathbf{0}$$

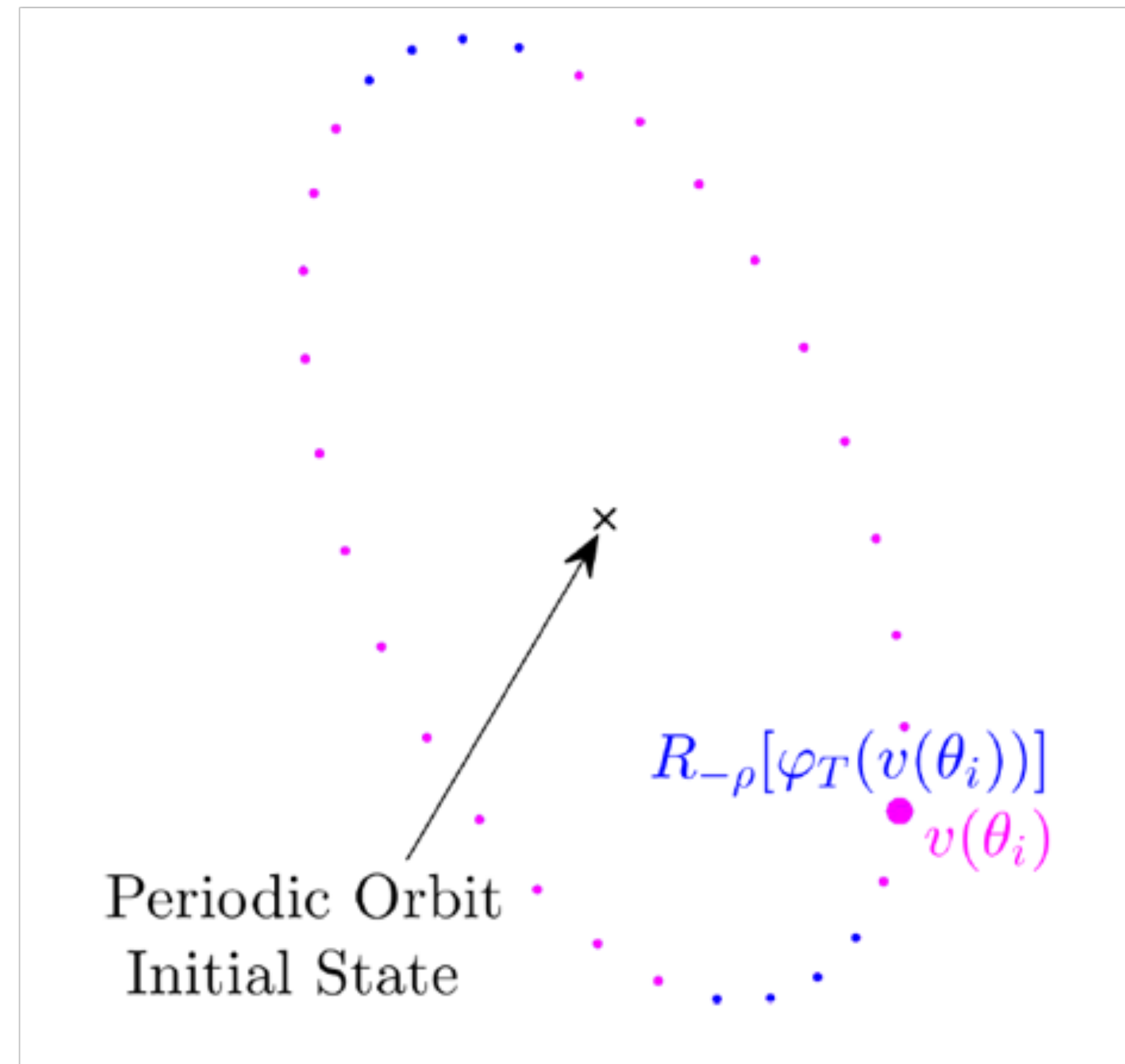


2D Torus Computation



Update initial guess via Newton's method and **iterate** until convergence

$$\mathbf{z} = [\mathbf{v}(\theta_i), \quad T, \quad \rho]^T \in \mathbb{R}^{(6N+2)}, \quad \mathbf{z}^{(k+1)} = \mathbf{z}^{(k)} - D\mathbf{F}(\mathbf{z}^{(k)})^\dagger \mathbf{F}(\mathbf{z}^{(k)})$$

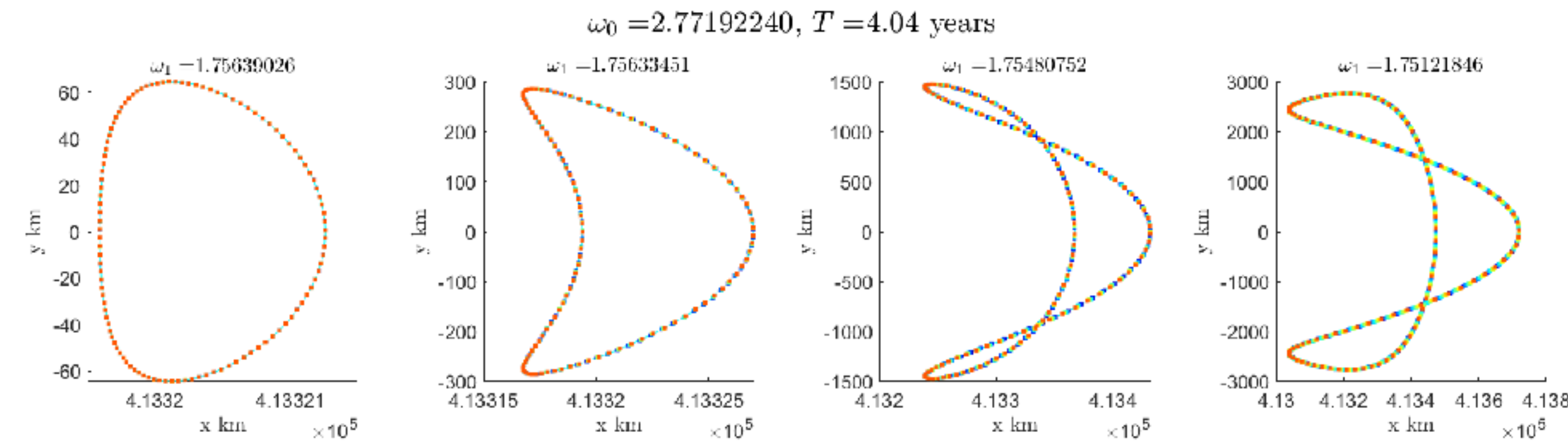


2D Torus Computation



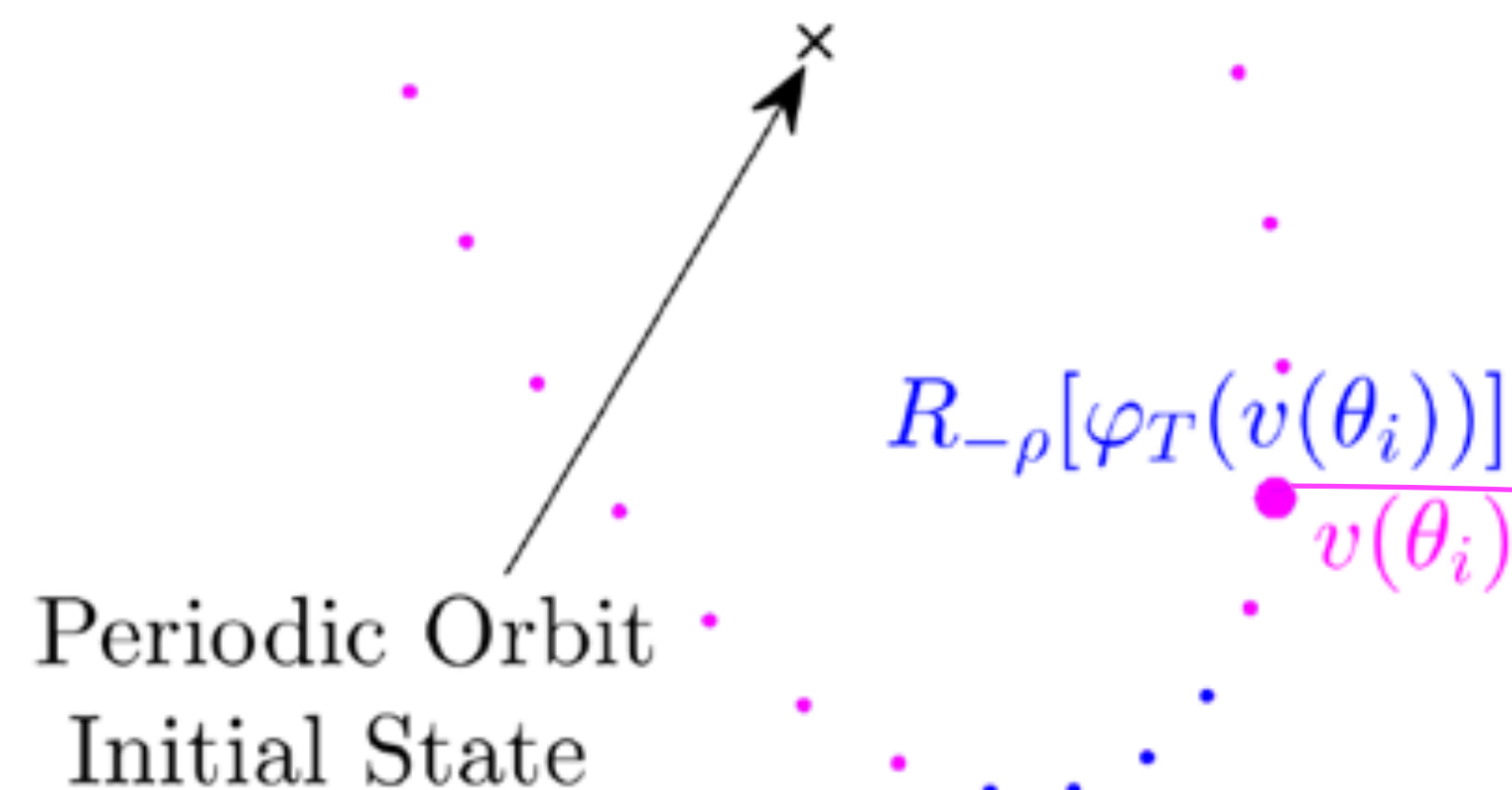
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Quasi-periodic trajectory

Converged surface for a
2-D QPO is a 1-D Circle



Stability Analysis



Linear stability can be investigated as a **byproduct** of Newton's method:

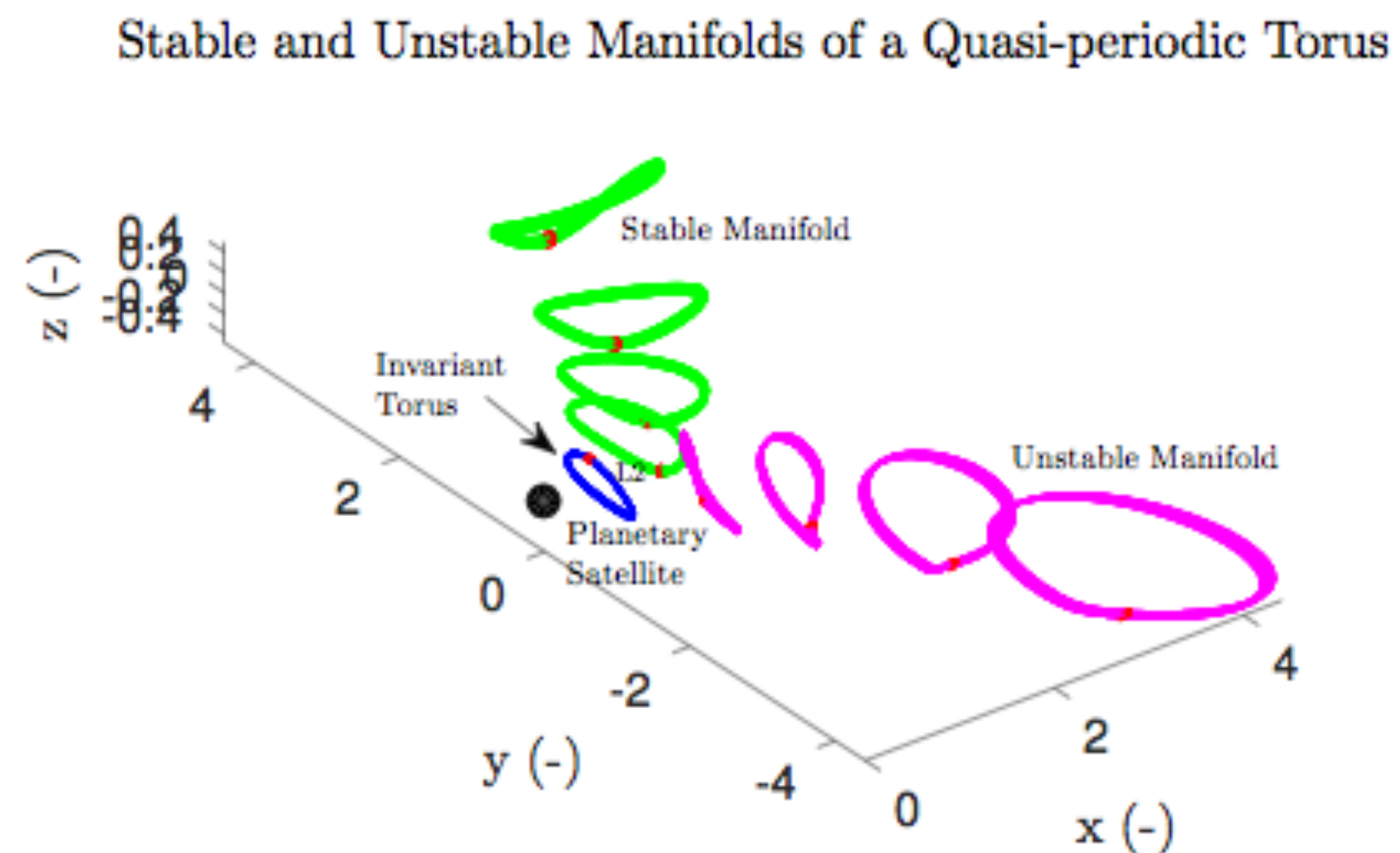
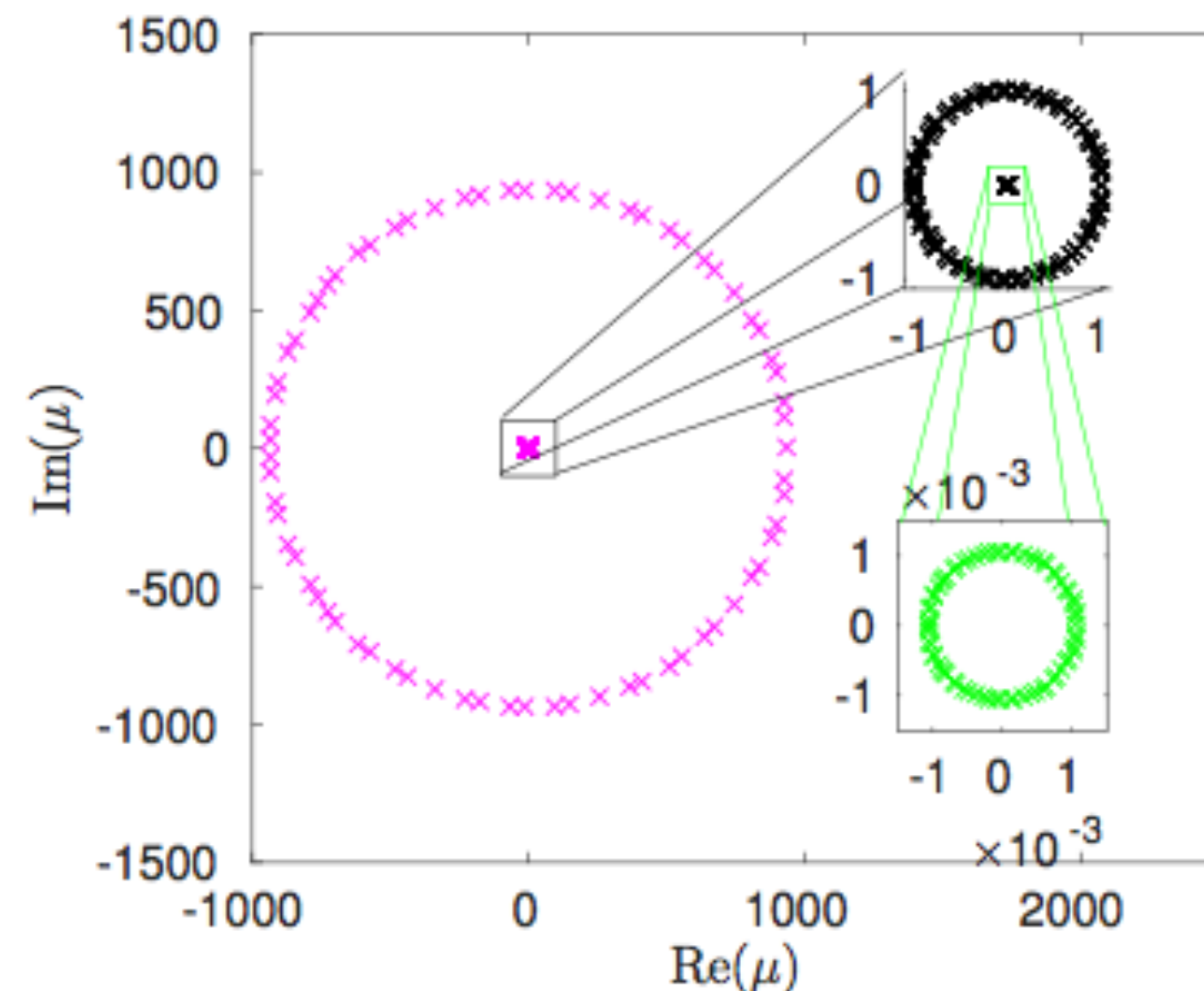
Recall

$$R_{-\rho}[\varphi_T(\mathbf{v}(\theta_i))] - \mathbf{v}(\theta_i) = \mathbf{0} \implies G_{-\rho}(\mathbf{v}(\theta_i)) = \mathbf{v}(\theta_i)$$

$$G_{-\rho} : R_{-\rho} \circ \varphi_T$$

Stability of QP Invariant Tori:

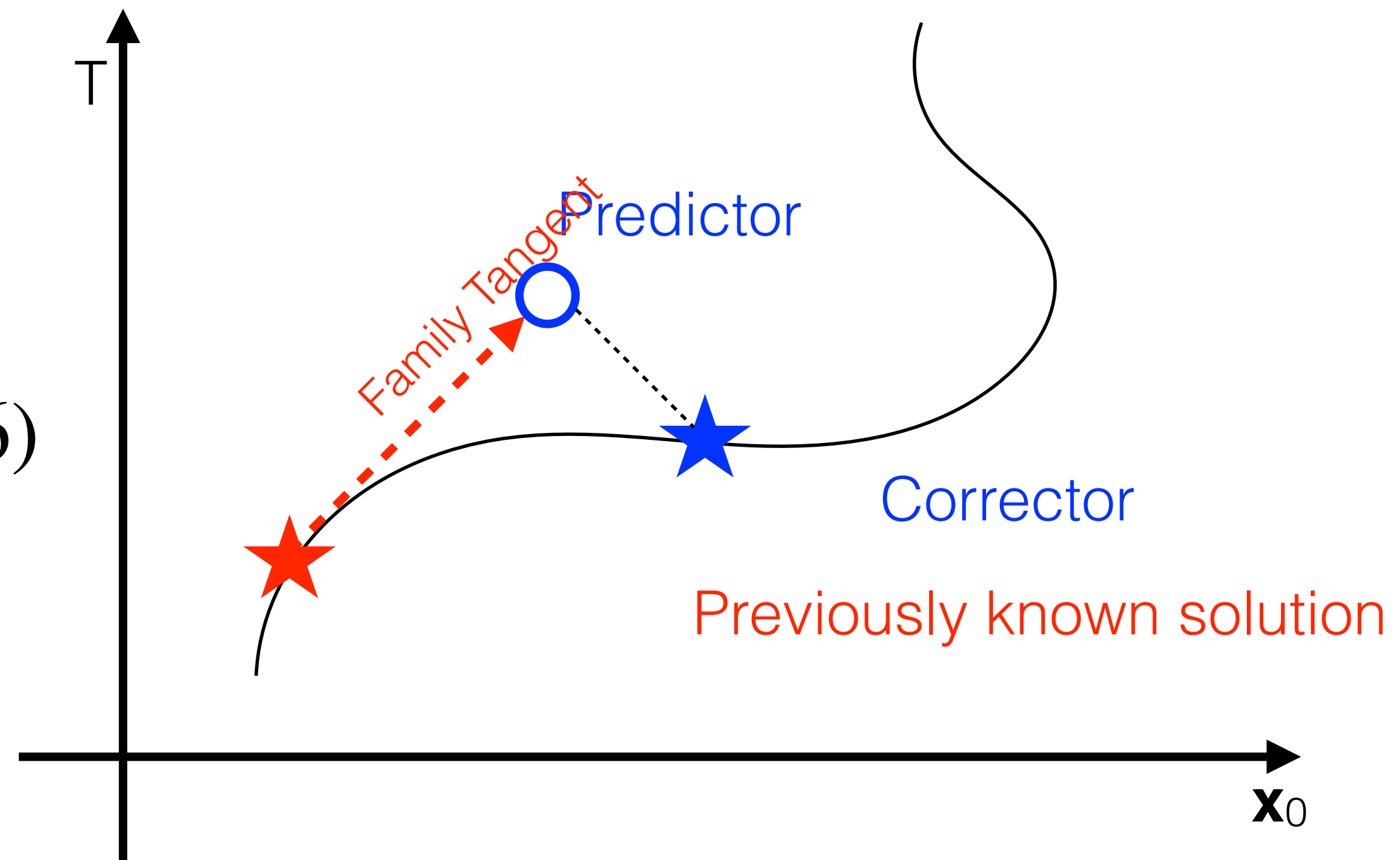
Study the eigenvalues of the Jacobian $DG_{-\rho}(\mathbf{v}(\theta_i)) = R_{-\rho} \circ \Phi$ to infer the stability of the QP invariant torus (Jorba, 2001)





Continuation of Families

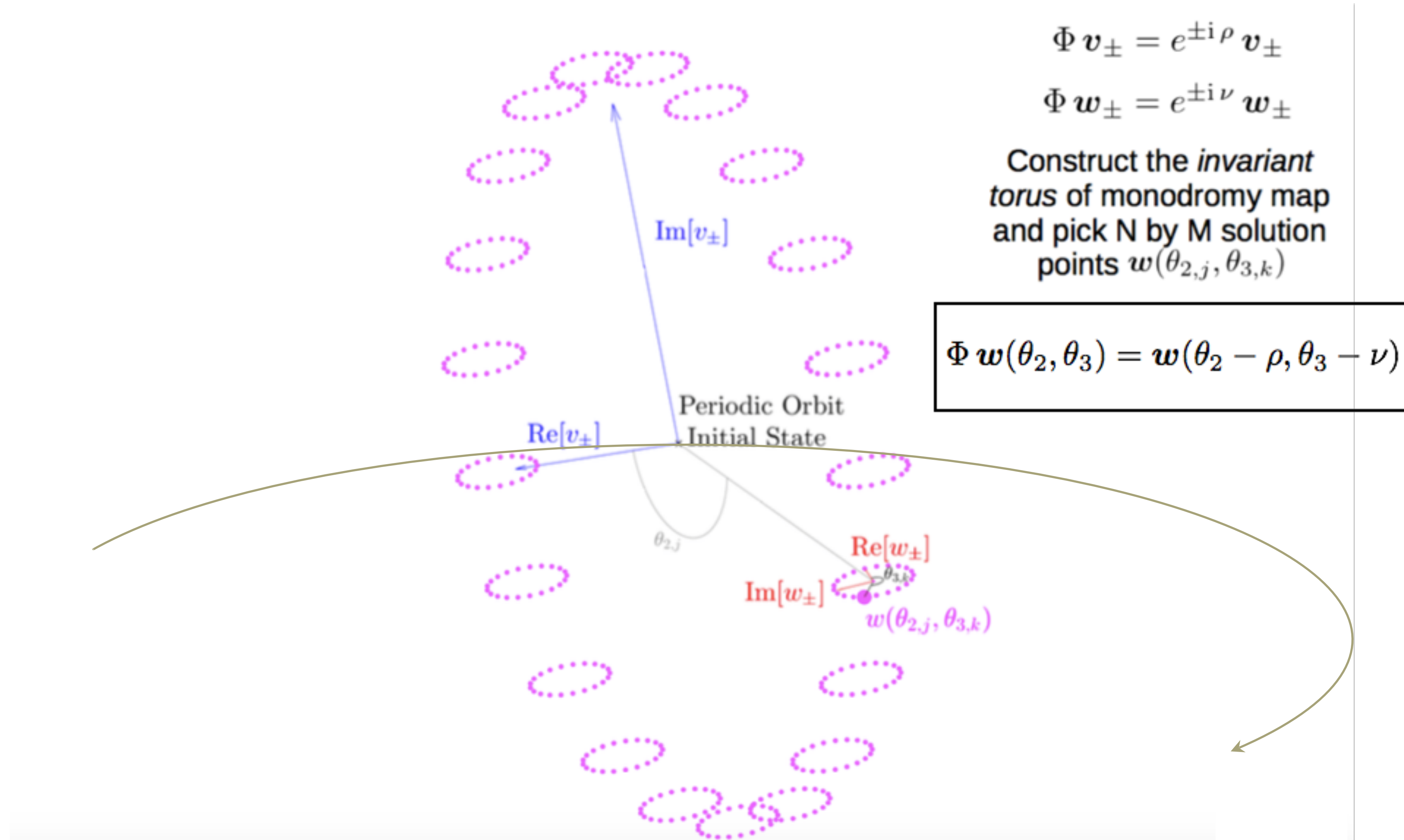
- To move from one QPO to another we use pseudo-arclength continuation
- Implemented using *multiple-shooting* and *co-location* methods (Olikara 2016)
- Significant benefits compared to other published methods



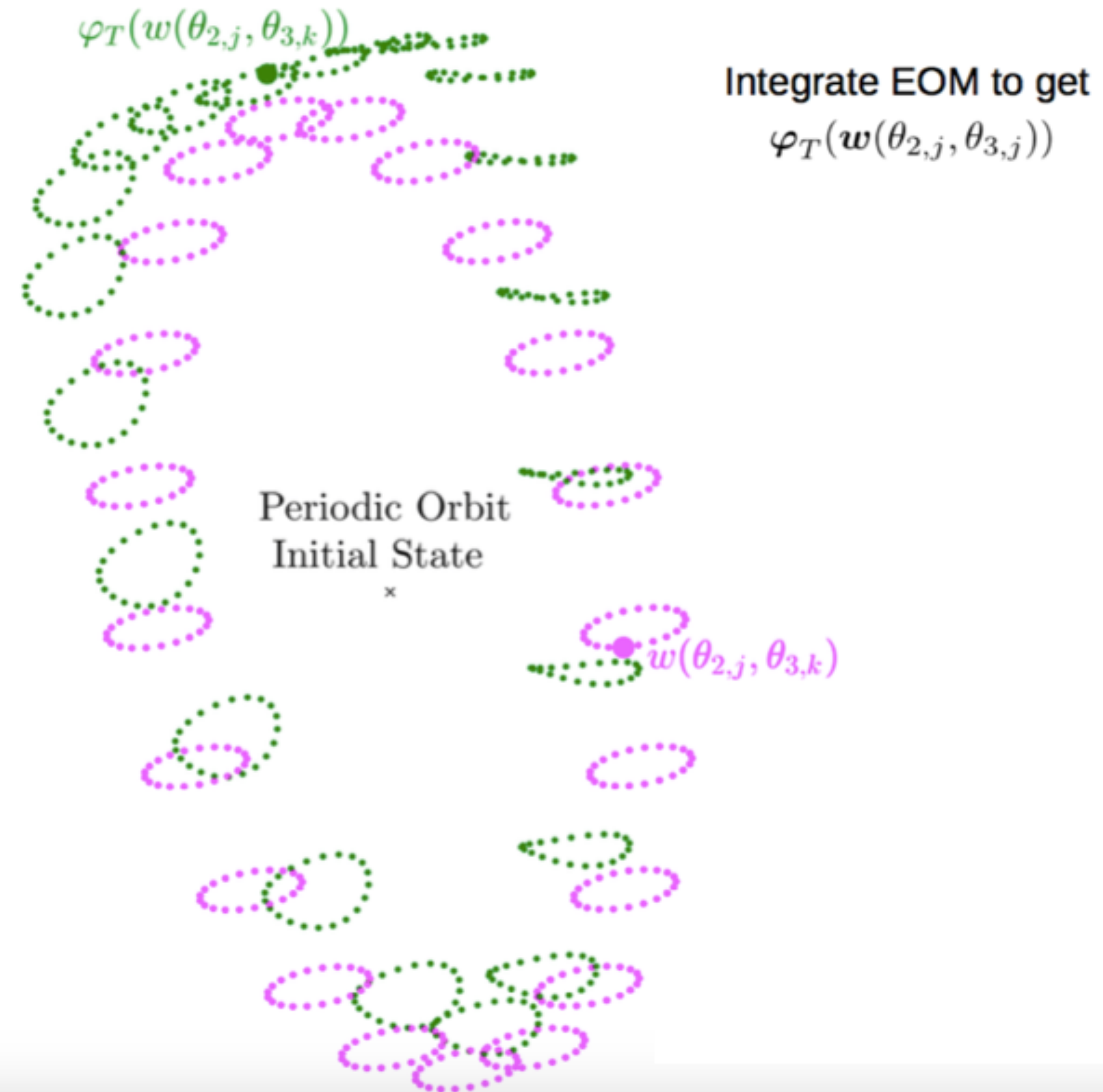
- Naturally provide information on stability
- Rapid computation for a given accuracy
- Generalizable to N -dimensional tori (with a computational cost)

See: N. Baresi, Z. Olikara and D.J. Scheeres. 2018. “Fully numerical methods for continuing families of quasi-periodic invariant tori in astrodynamics,” *Journal of the Astronautical Sciences* 65: 157-182.

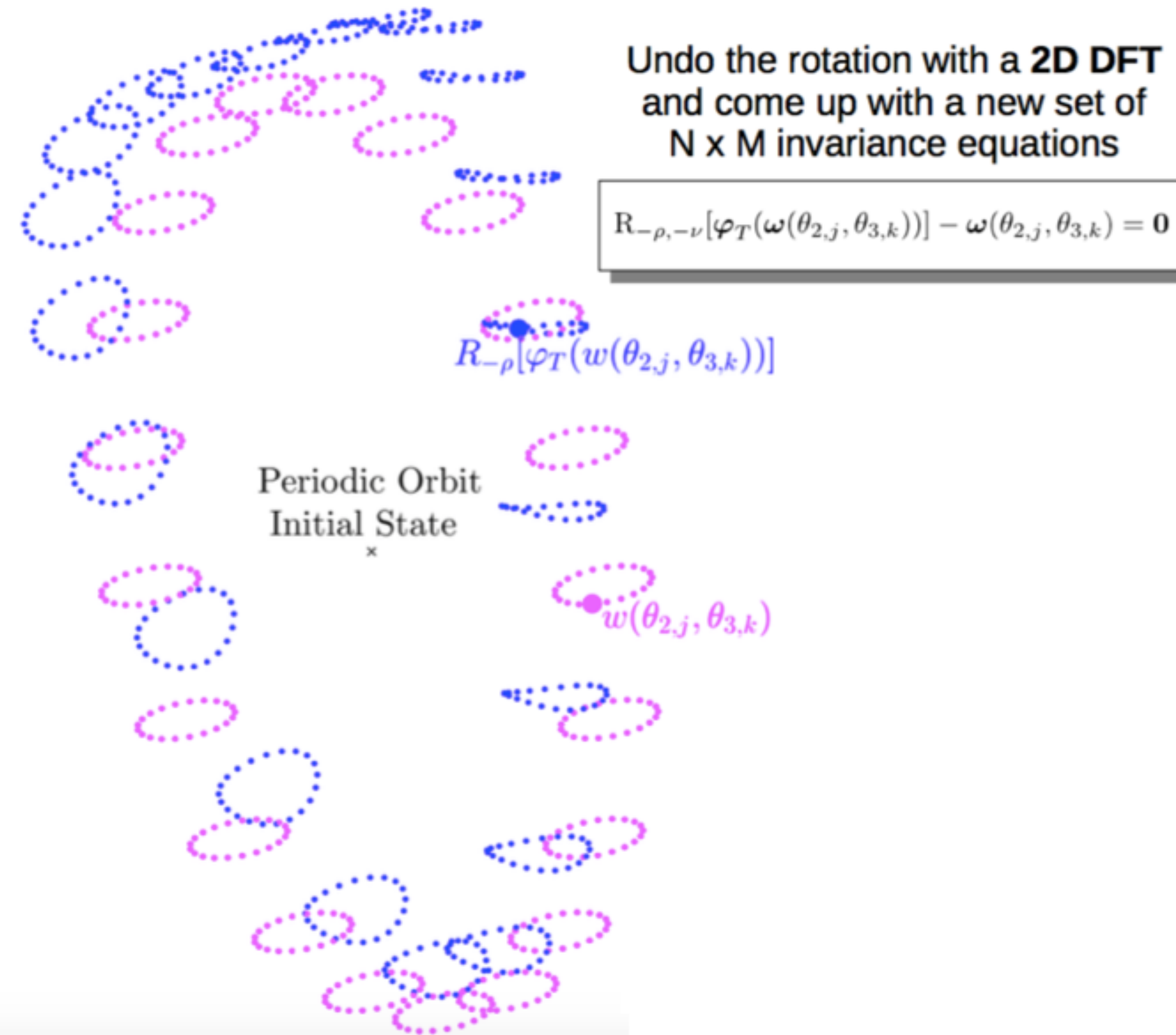
3D Torus Computation



3D Torus Computation



3D Torus Computation



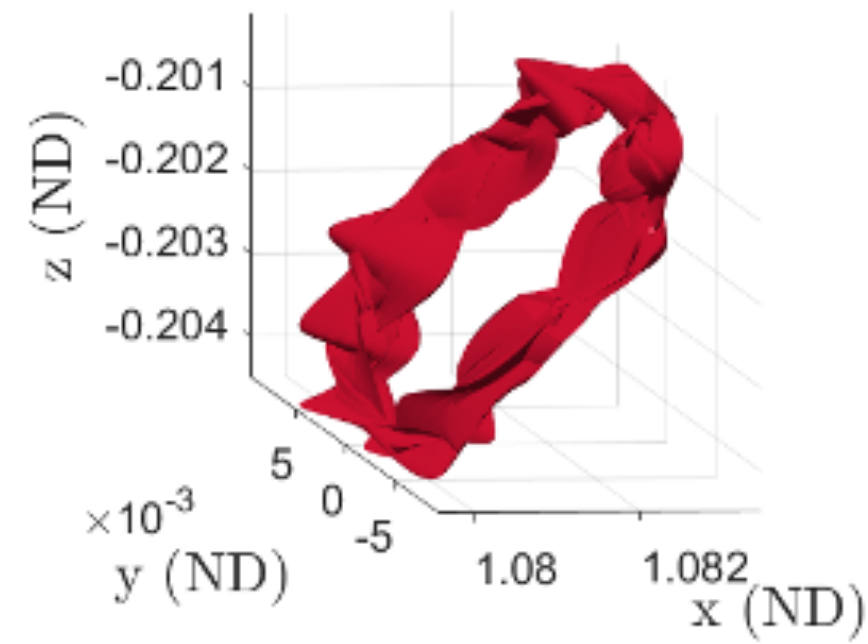
3D Torus Computation



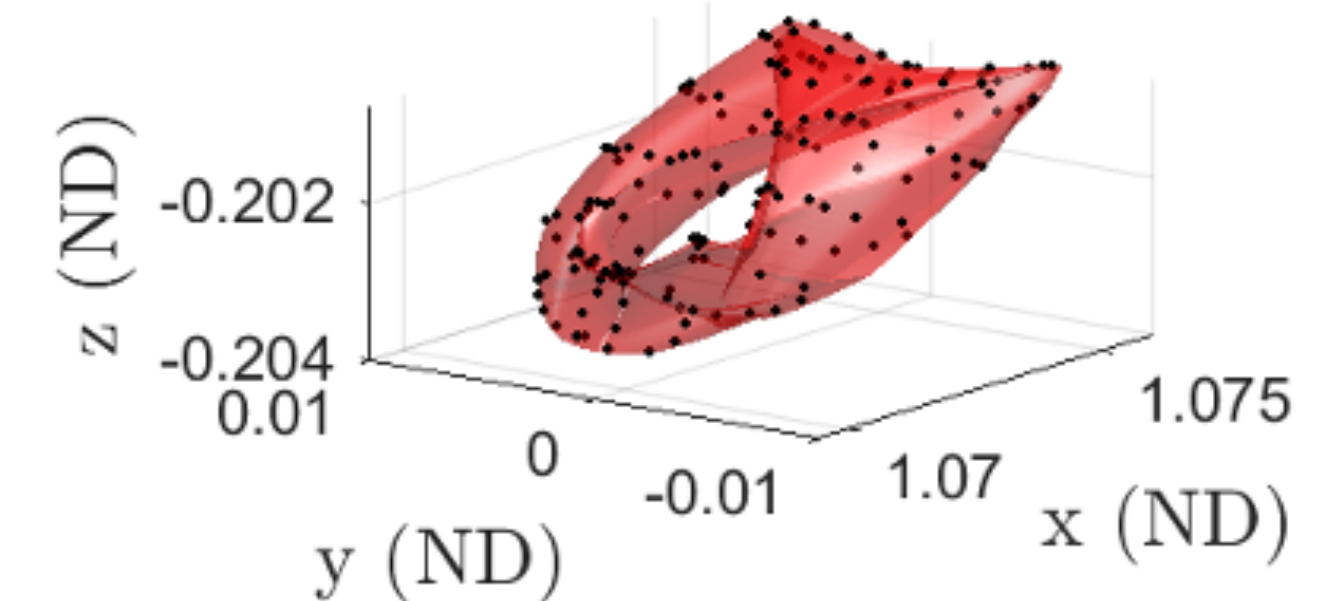
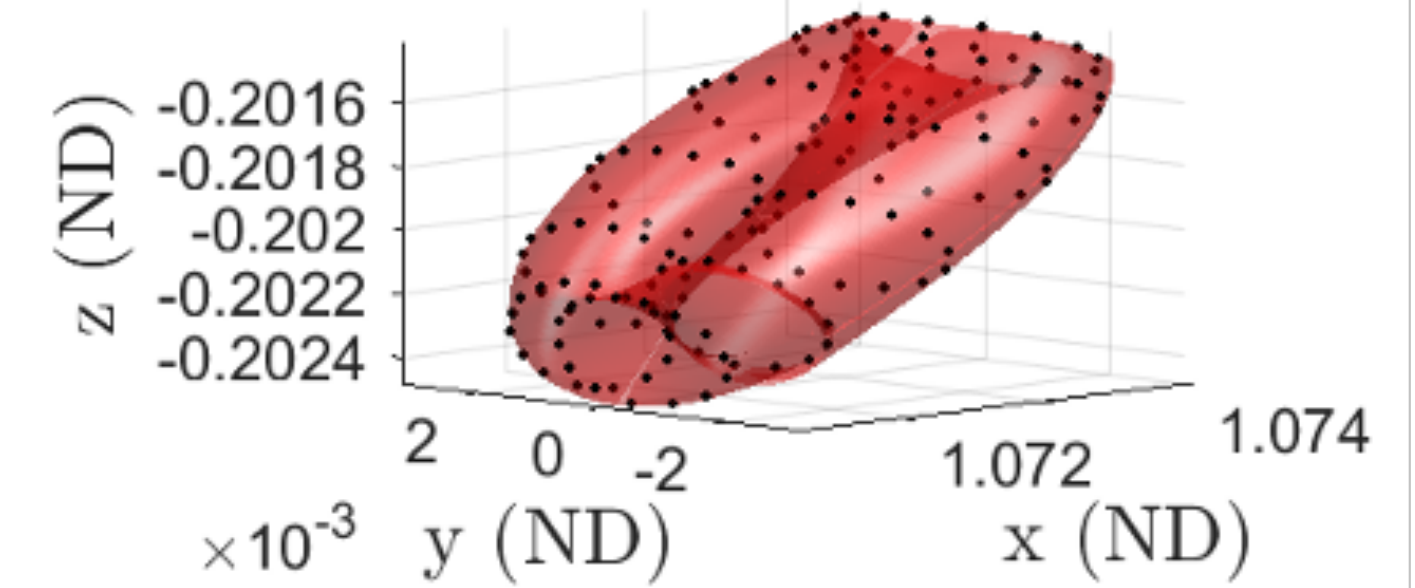
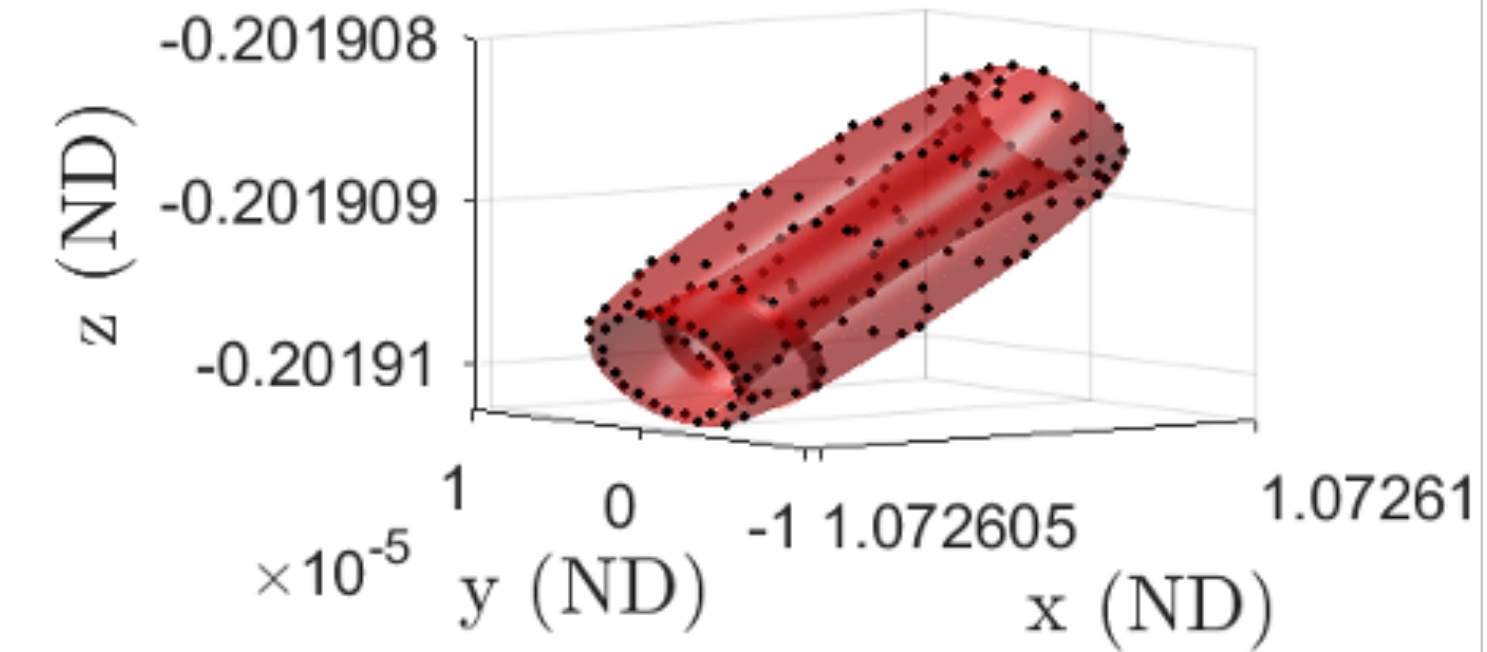
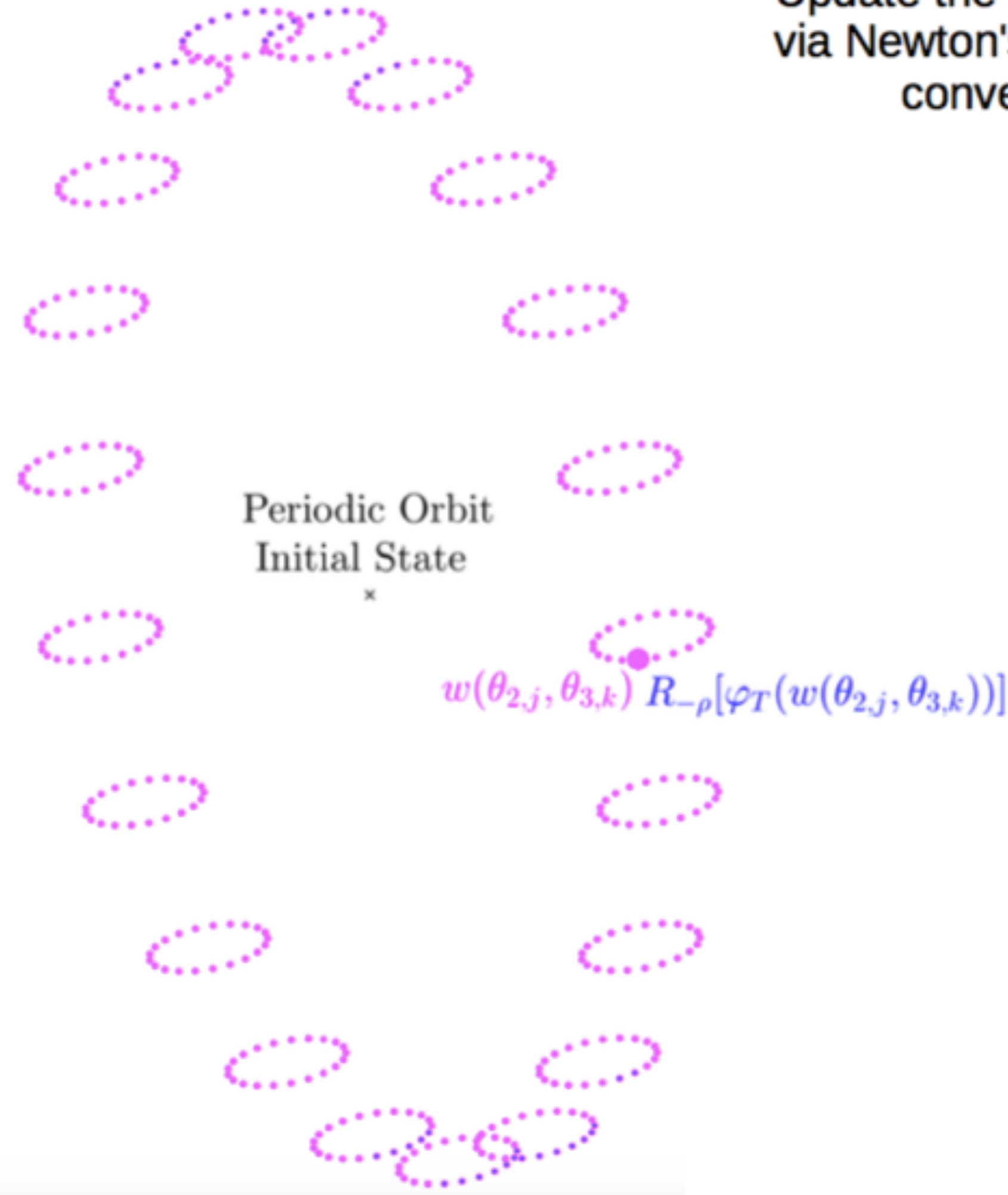
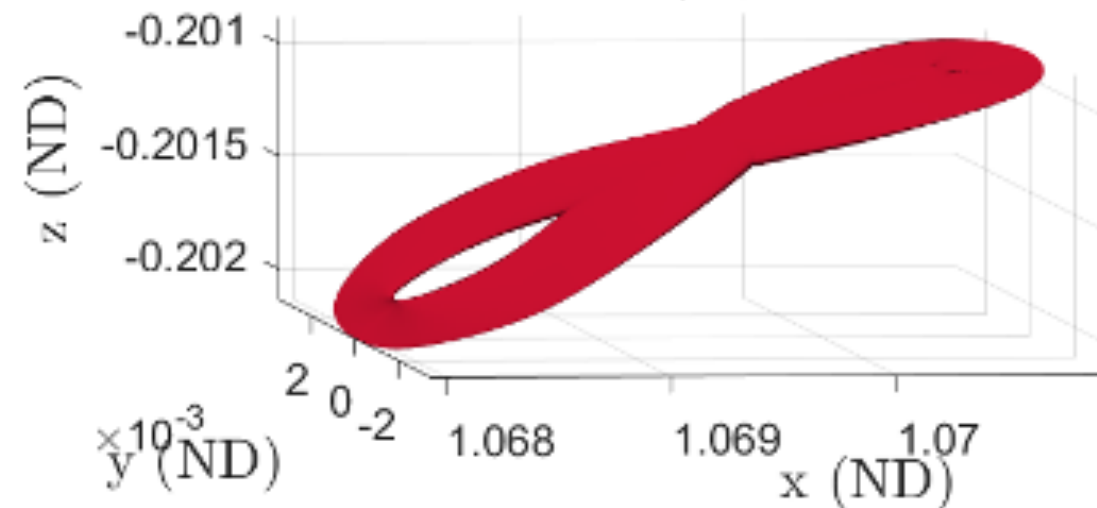
Converged surface for a
3-D QPO is a 2-D QPO

Update the solution points
via Newton's method until
convergence

$\omega_0=2.66095042$, $\omega_1=1.69221669$, $\omega_2=2.36231833$
 $J=3.01517231$
 $A_1=7.8720\text{e-}03$, $A_2=3.6182\text{e-}03$

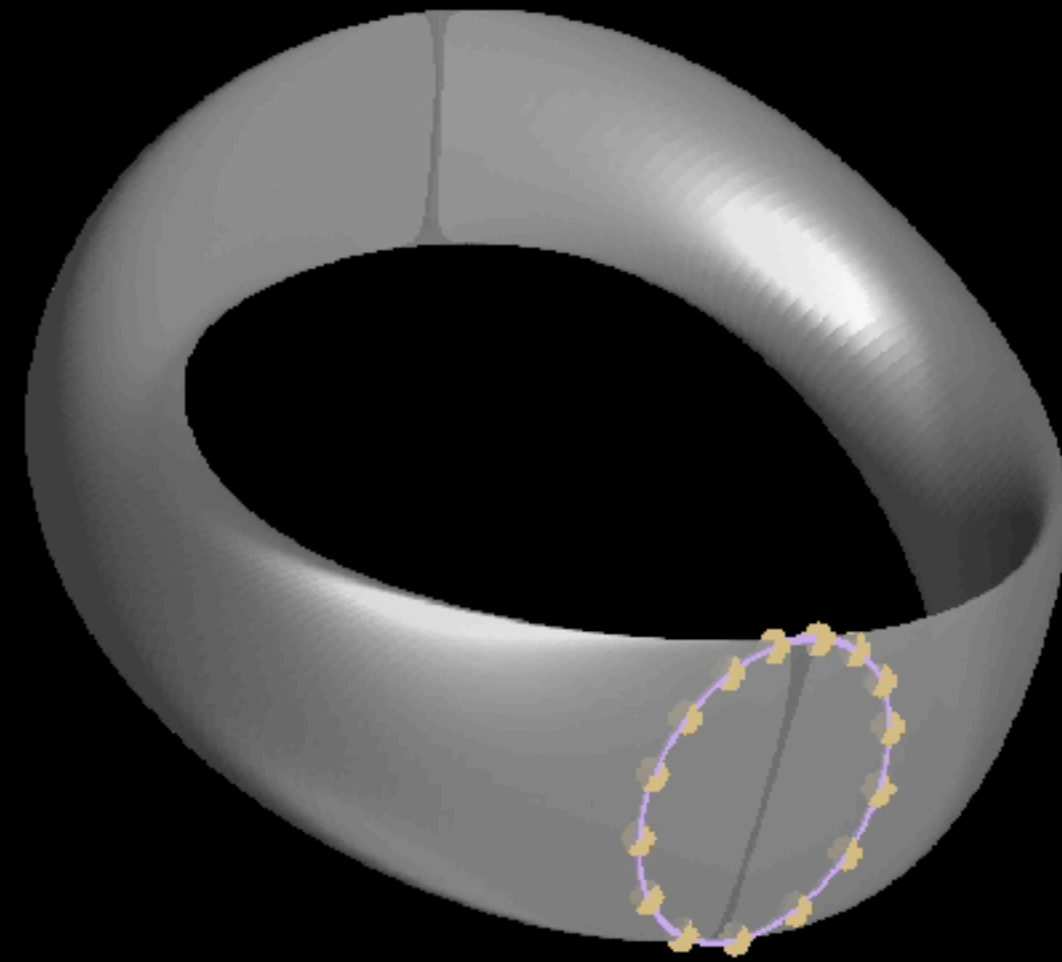


$\omega_0=2.88586765$, $\omega_1=1.56171044$, $\omega_2=2.11879967$
 $J=3.01700225$
 $A_1=1.1759\text{e-}03$, $A_2=3.5496\text{e-}03$



Computing Quasi-periodic Orbits and their Stability with the Flow Map Parameterization Method

Quasi-periodic Orbit Computation



GMOS: The Large Matrix Problem

Corrections for states $6N$ = $6N \times 6N$ State Variations $6N$ Error in states

⁻¹

Flow-map Parameterization Methods (Haro & Mondelo, Kumar et al, Fernández-Mora et al)

- Take advantage of the structure around orbits to create upper triangular system of equations
- Developed for certain class of orbits

$6N$ = $6N \times 6N$ ⁻¹



Adapted Frame

The Parameterization Method

We solve for the corrections in a frame that leverages the (approximate) (un)stable, elliptic, and tangent directions of the orbit

Adapted Frame

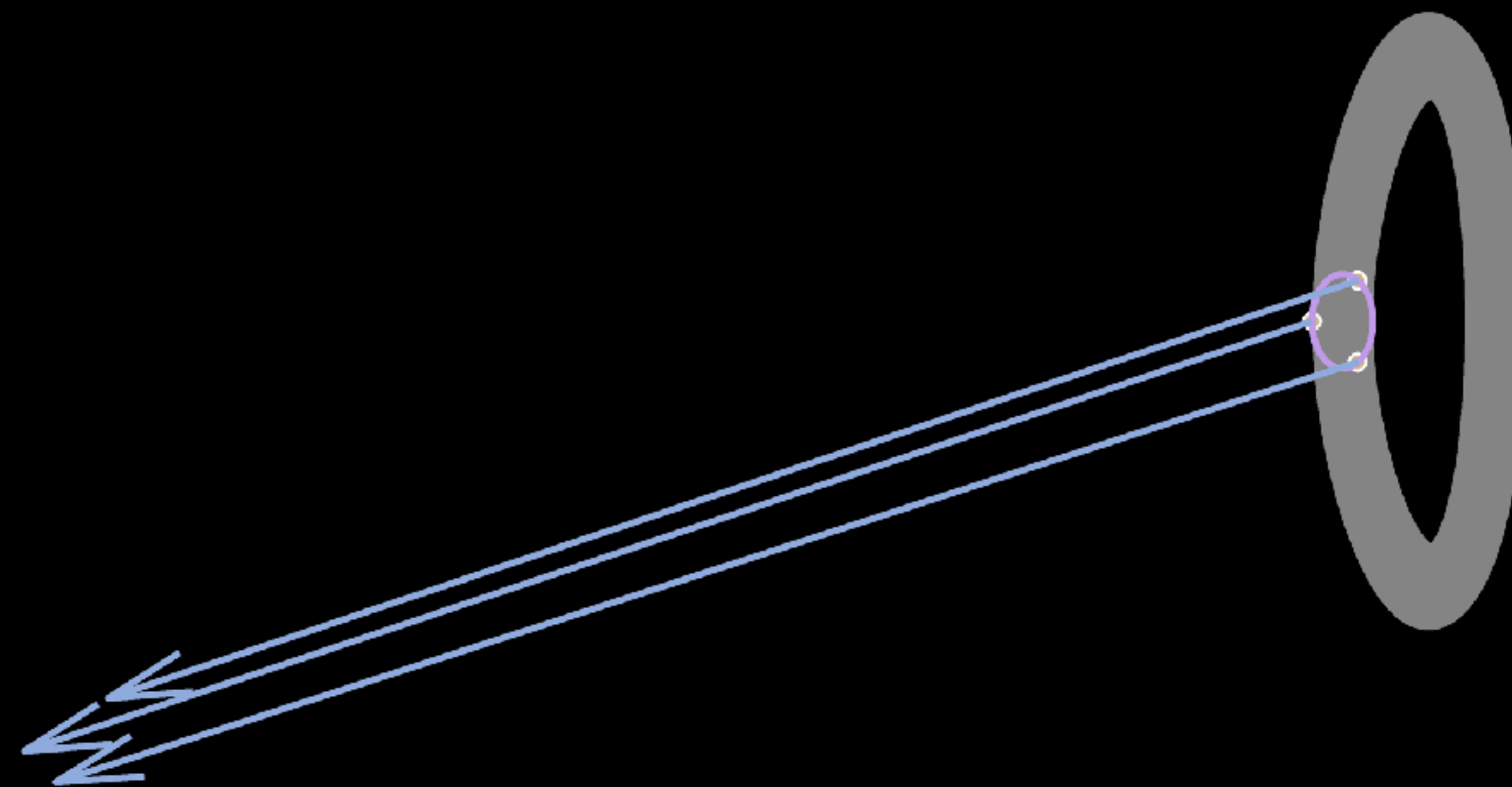
$$P(\theta) = \begin{bmatrix} D_\theta u(\theta) & w_s(\theta) & w_c(\theta) & w_f(\theta) & w_u(\theta) & \bar{w}_c(\theta) \end{bmatrix}$$

Tangent direction
Stable direction
Center direction
Family direction
Unstable direction

$$D_x \varphi_T(u(\theta)) P(\theta) = P(\theta + \rho) \begin{pmatrix} \Lambda & S \\ 0 & \Lambda^{-1} \end{pmatrix}$$

$$\Lambda = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \lambda_s & 0 \\ 0 & 0 & \lambda_c \end{pmatrix} \quad S = \begin{pmatrix} S_{11} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

Stable Directions



A planar Lyapunov-like quasi-periodic orbit



Main Speed-Up: Creates an Upper Triangular Update Matrix

Upper Triangulation

Expression for
orbit correction

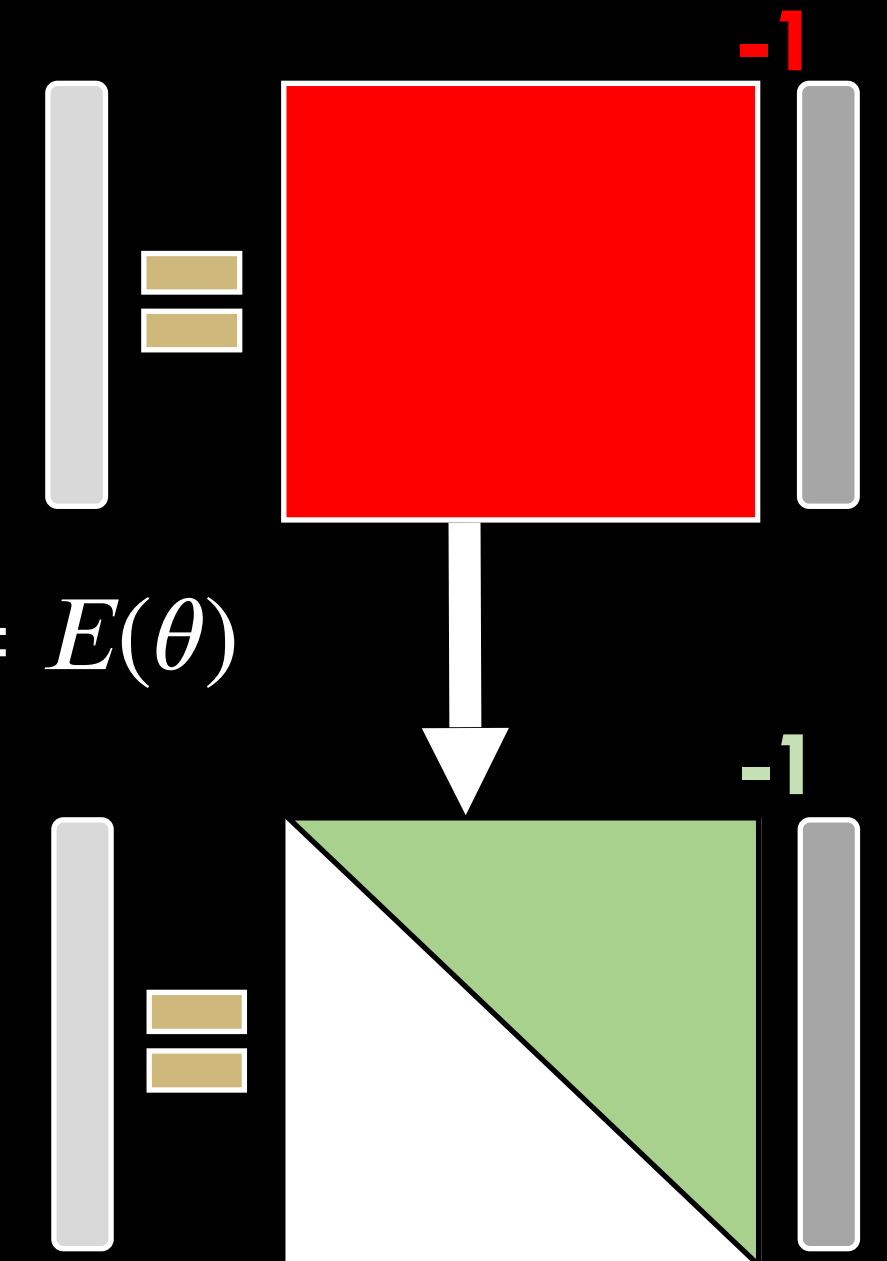
$$D_x \varphi_T(u(\theta)) \Delta u(\theta) - \Delta u(\theta + \rho) = E(\theta)$$

Correction in
adapted frame

$$\Delta u(\theta) = P(\theta) \Delta \xi(\theta) \Rightarrow D_x \varphi_T(u(\theta)) P(\theta) \Delta \xi(\theta) - P(\theta + \rho) \Delta \xi(\theta + \rho) = E(\theta)$$

Pre-multiply by
 $P(\theta + \rho)^{-1}$

$$\begin{pmatrix} \Lambda & S \\ 0 & \Lambda \end{pmatrix} \Delta \xi(\theta) - \Delta \xi(\theta + \rho) = \eta(\theta)$$



Flow Map Parameterization Algorithm

Procedure

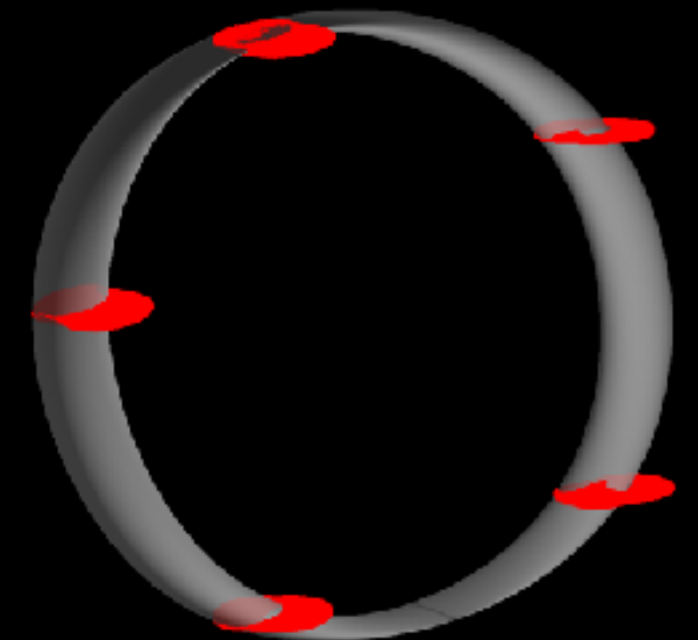
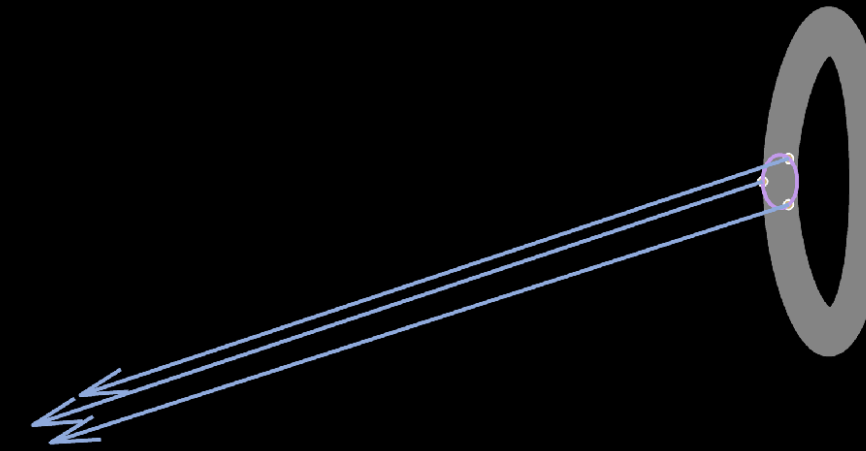
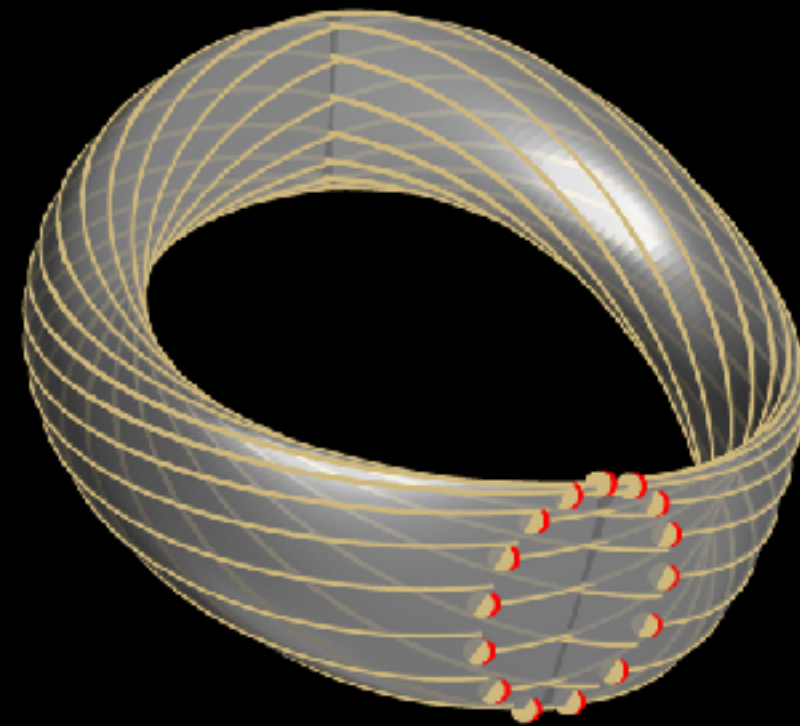
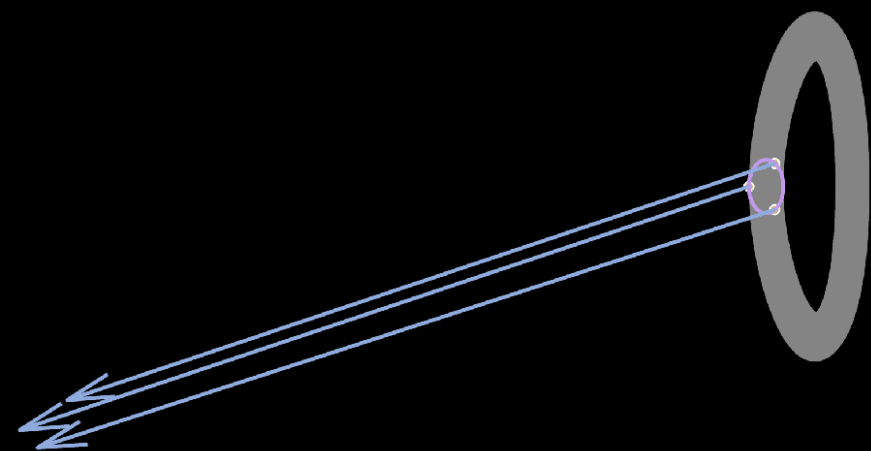
Initial guess for
torus and stability

Compute Adapted
Frame

Correct Torus

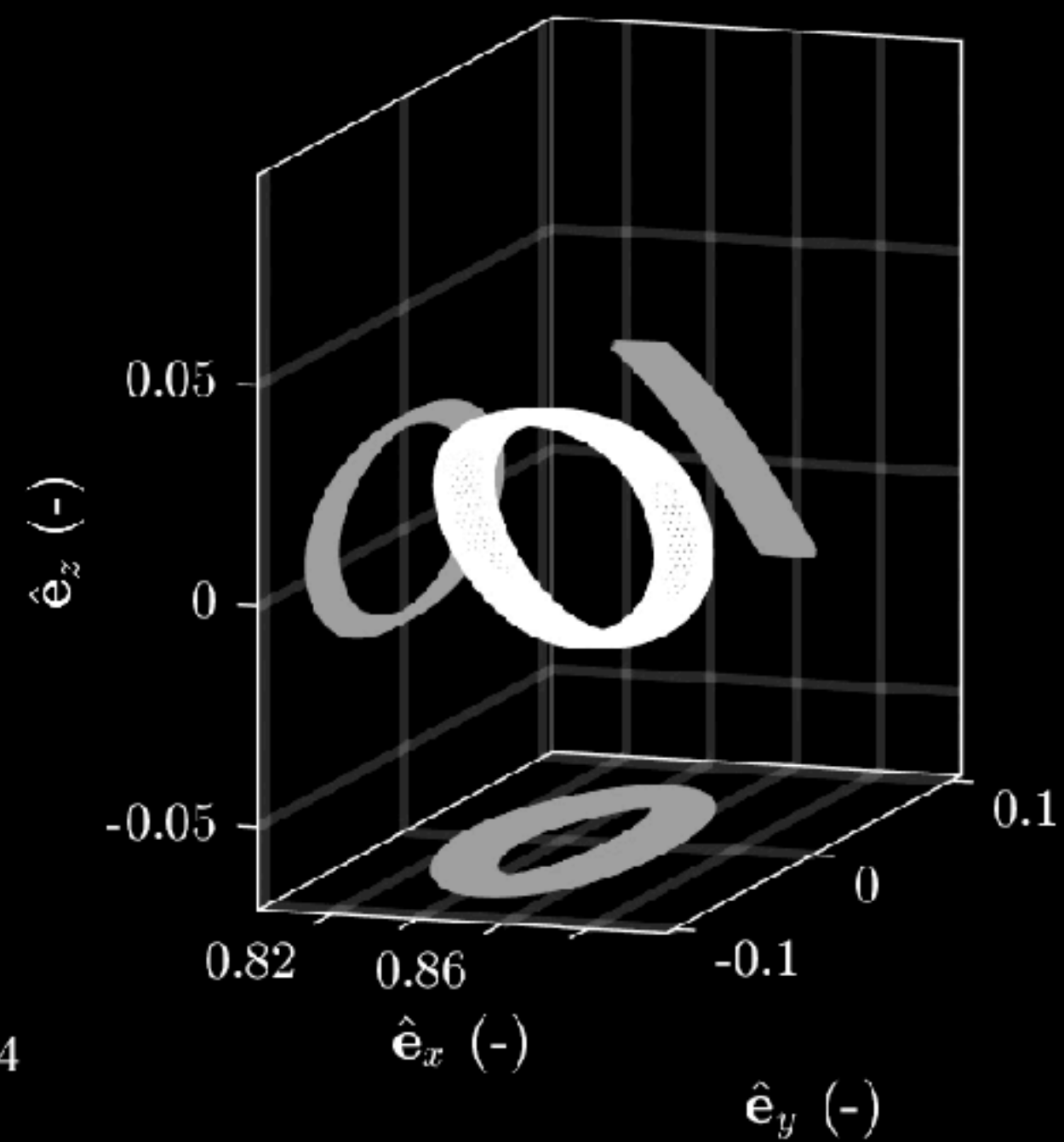
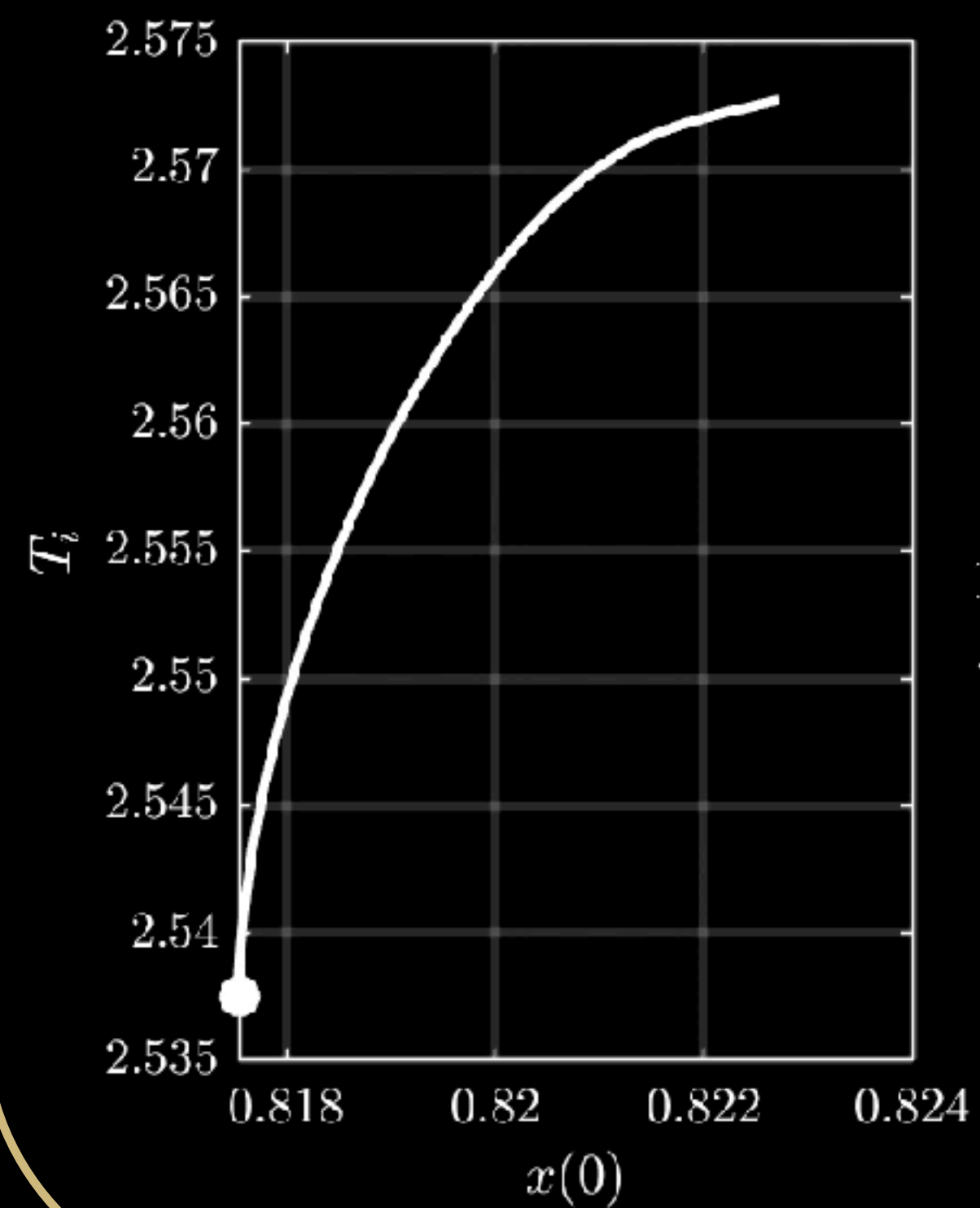
Compute Adapted
Frame

Correct bundles

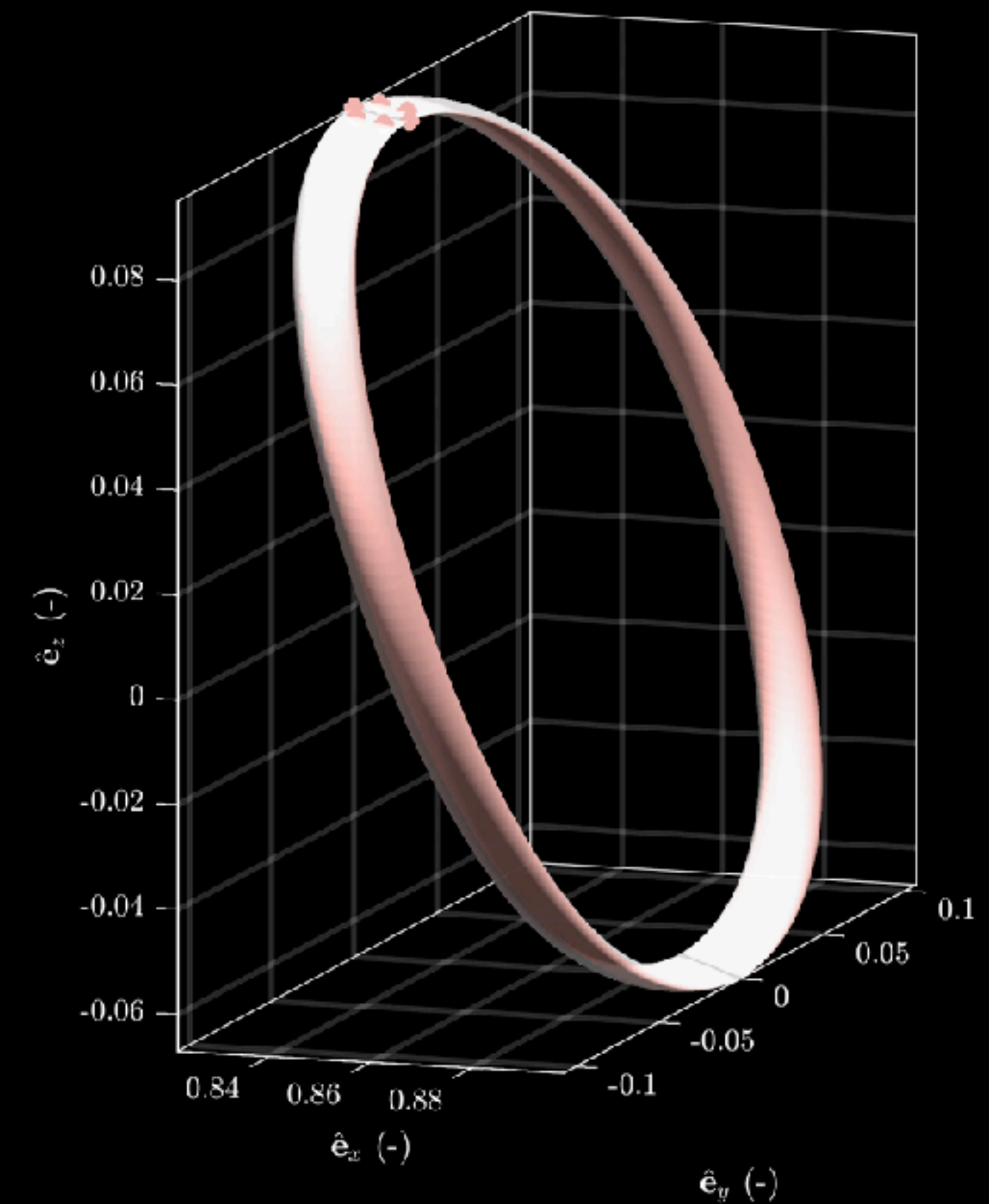


Example: The L1-H1 Family

L1-H1 Family Continuation

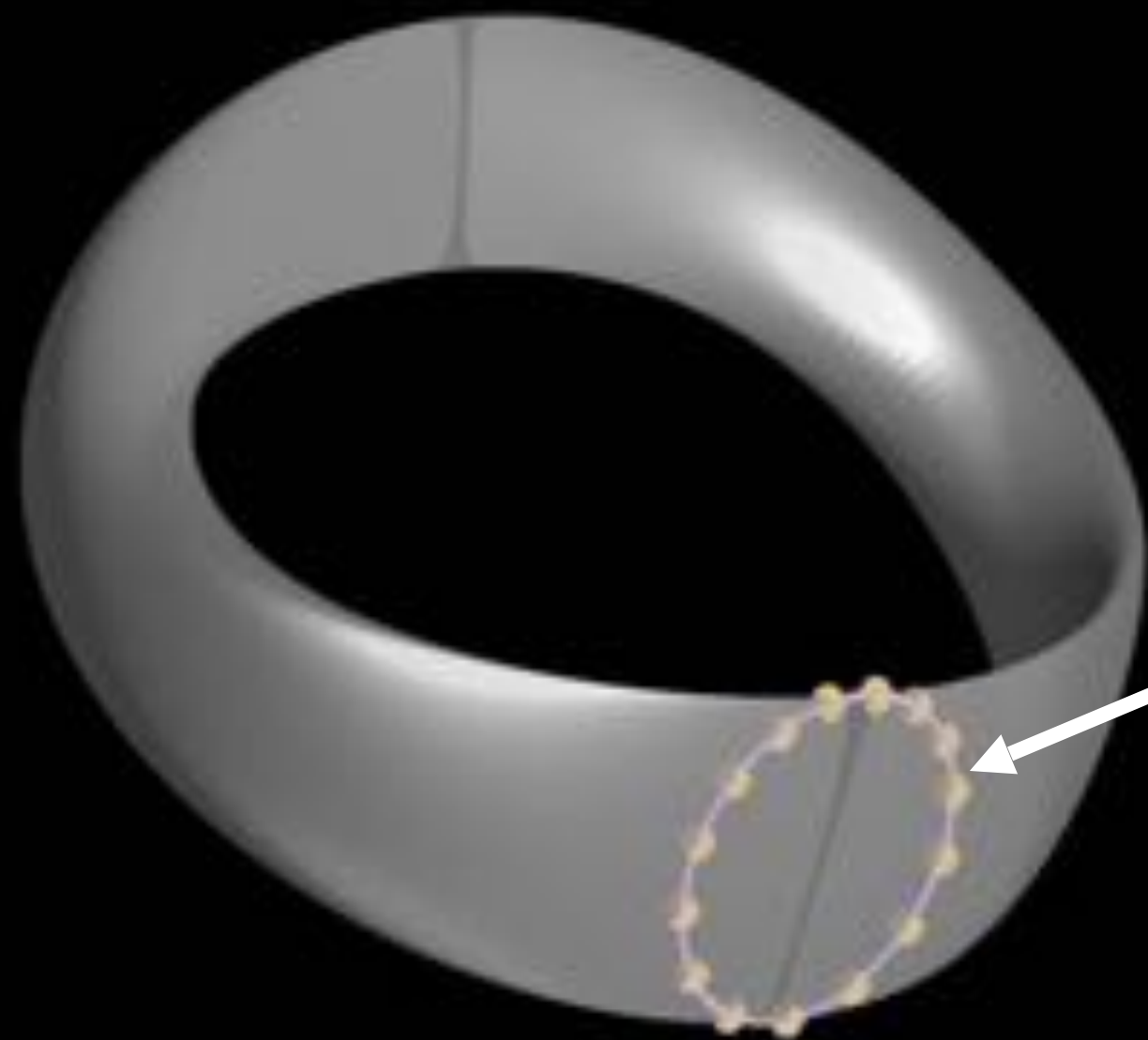


L1-H1 Unstable Manifold



Grid Point Analysis

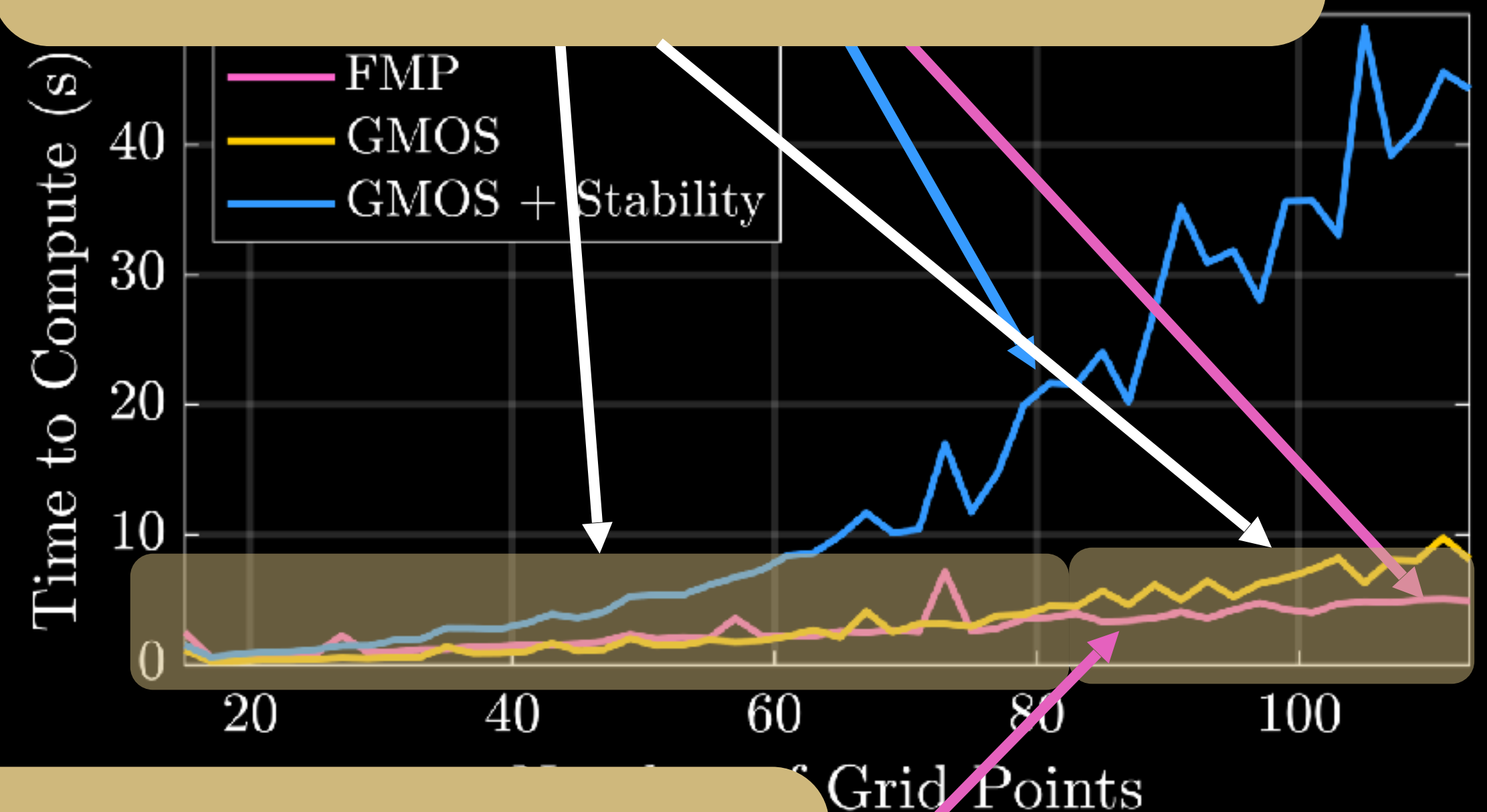
Grid Point Schematic



of points used to approximate the invariant curve

Takeaway #3

$\text{Time}(\text{FMP}) < \text{Time}(\text{GMOS})$ for large N



Important!

$\text{Time}(\text{FMP})$ can be decreased with more parallelization!

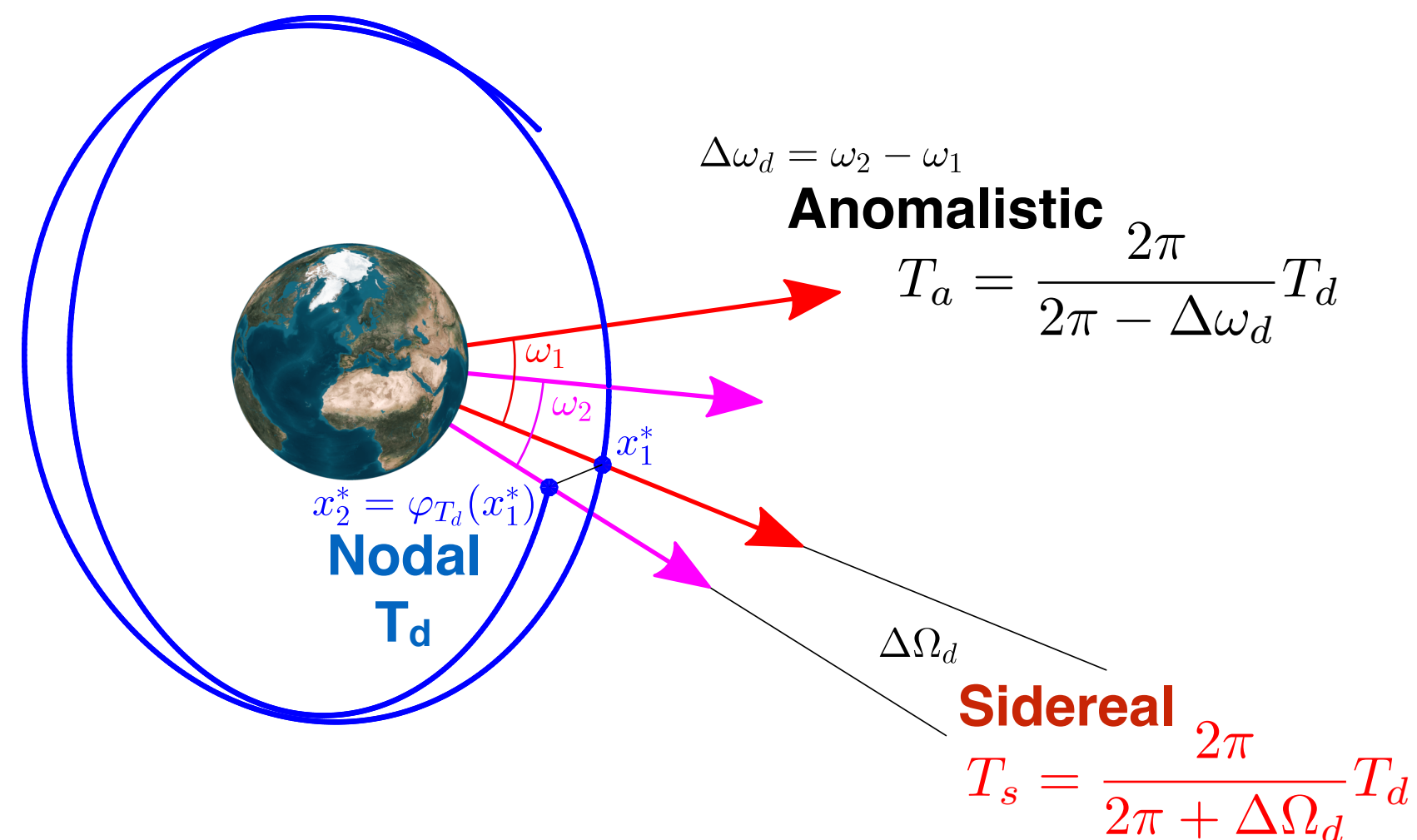




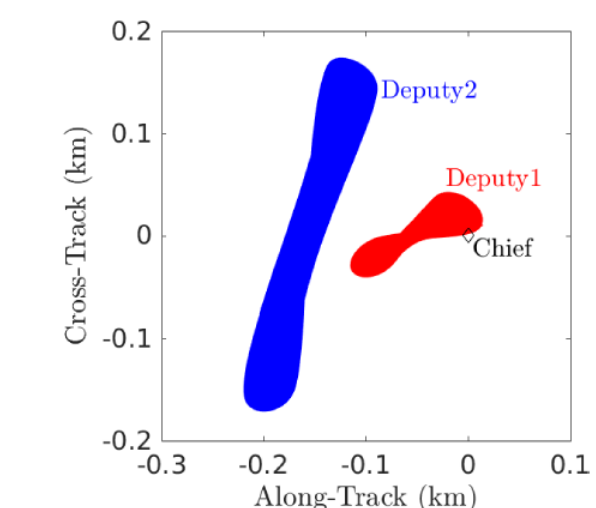
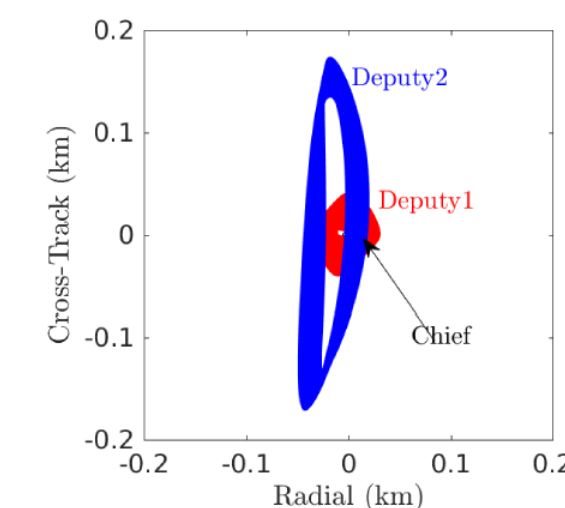
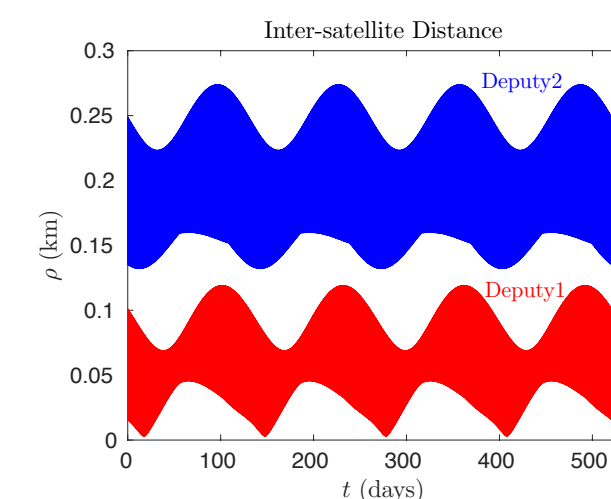
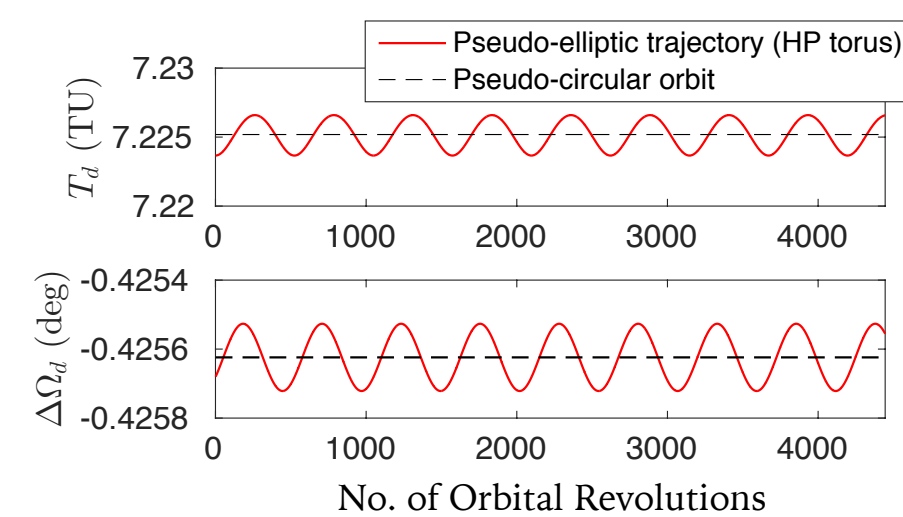
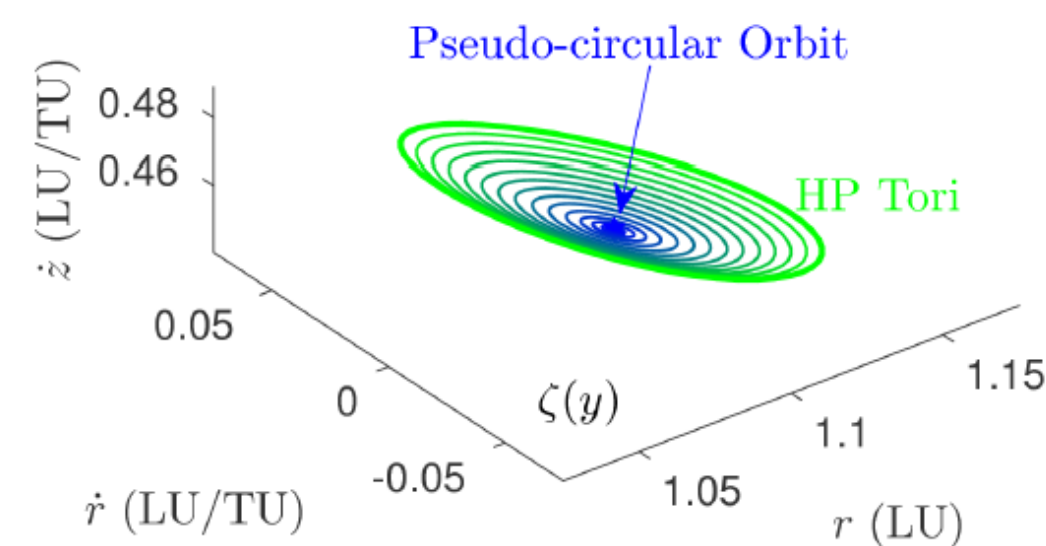
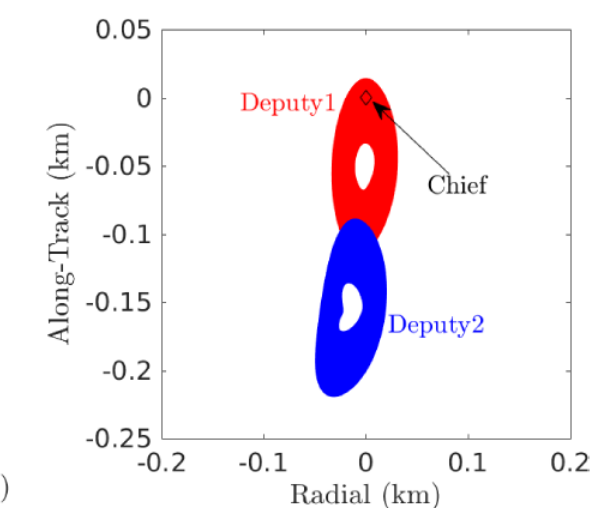
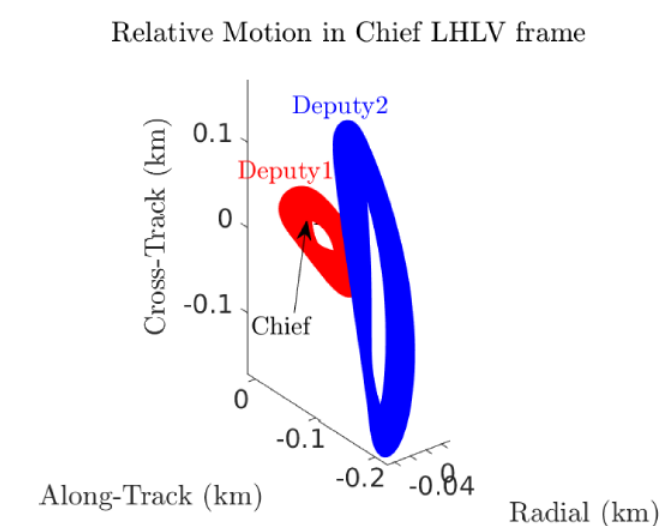
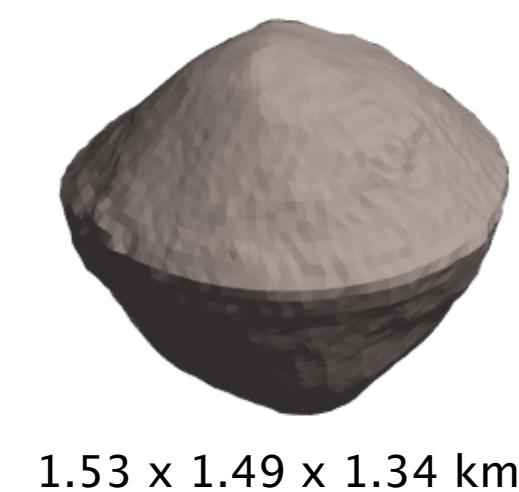
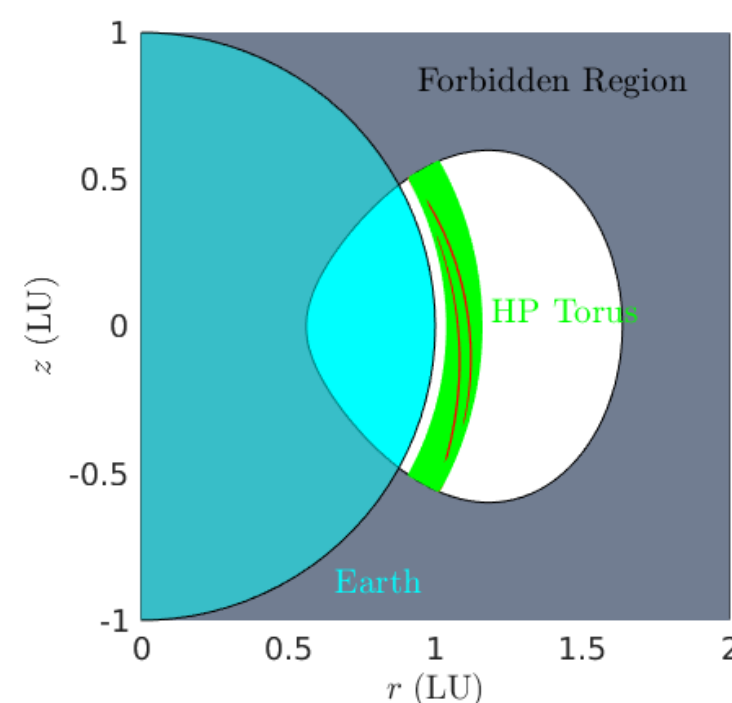
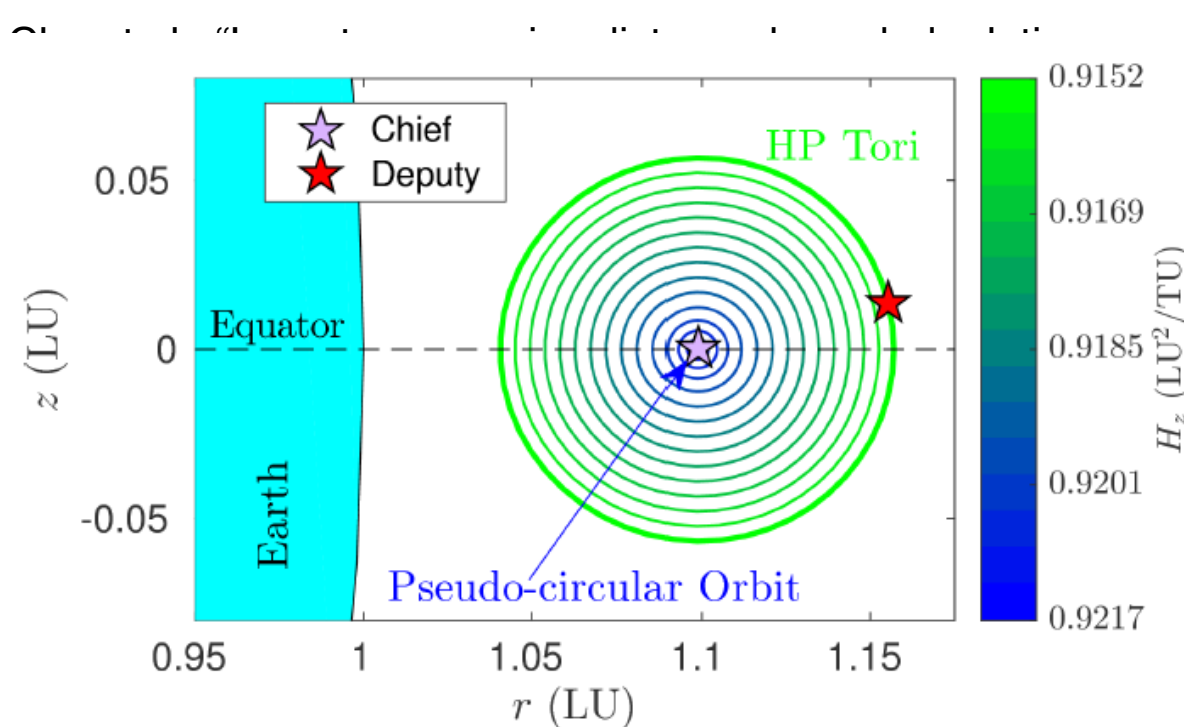
How can QPOs be used?



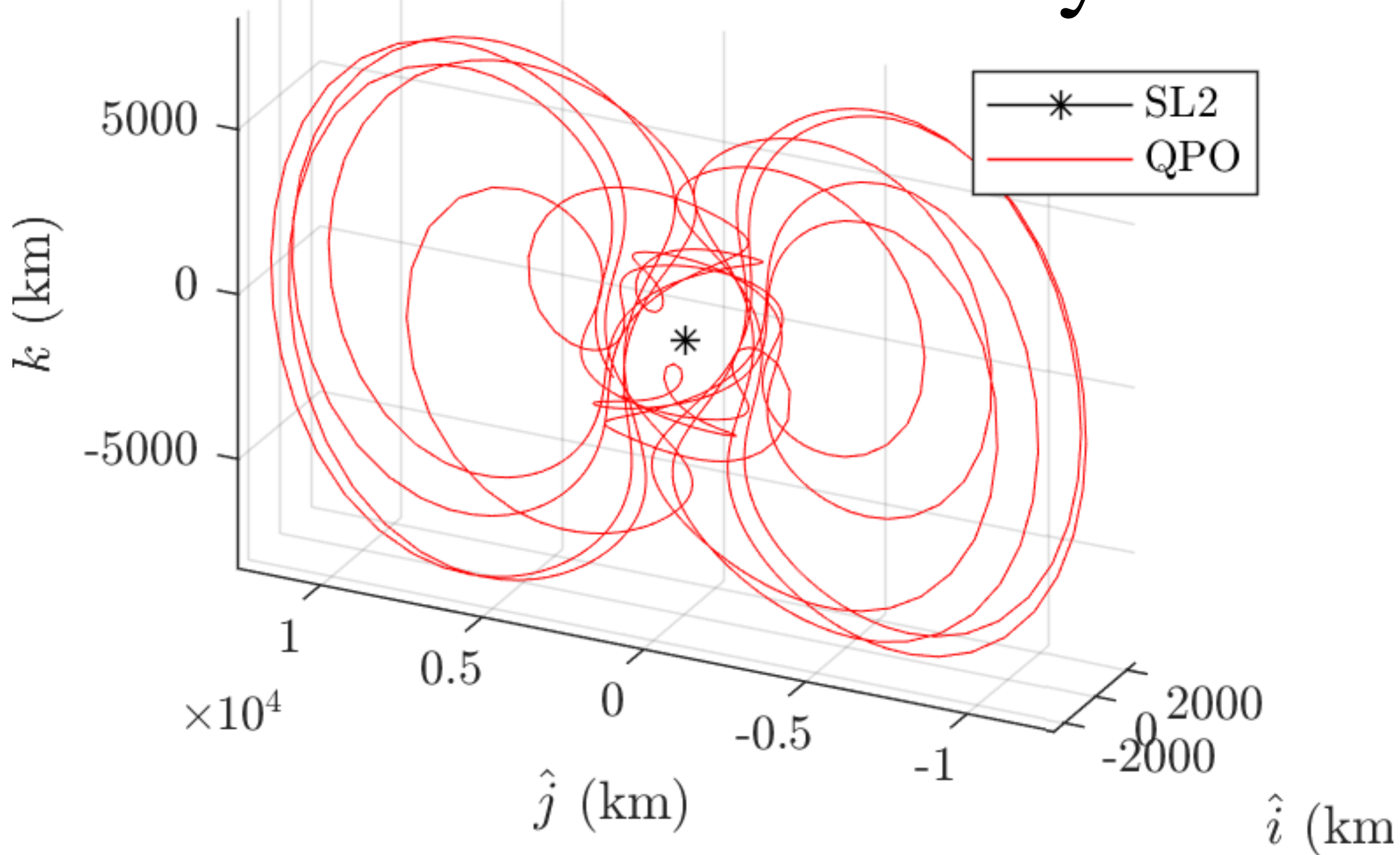
Formation Flight



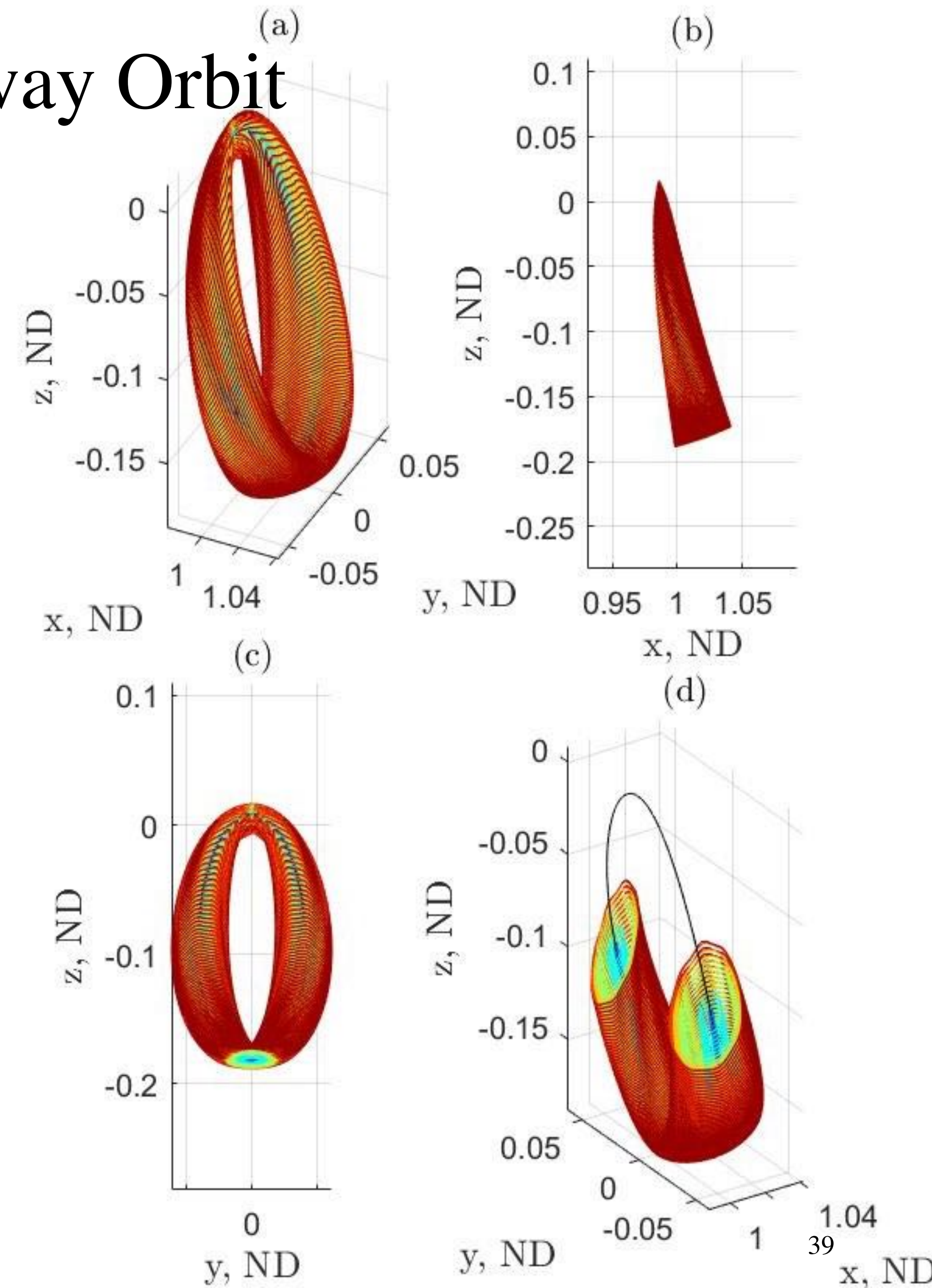
- Motion of multiple satellites on a QPO = bounded motion — many applications for formation flight.
- Motion about an oblate planet = motion on a 3-D torus... can solve for the torus families and use for orbit design.



9:2 Synchronous Gateway Orbit



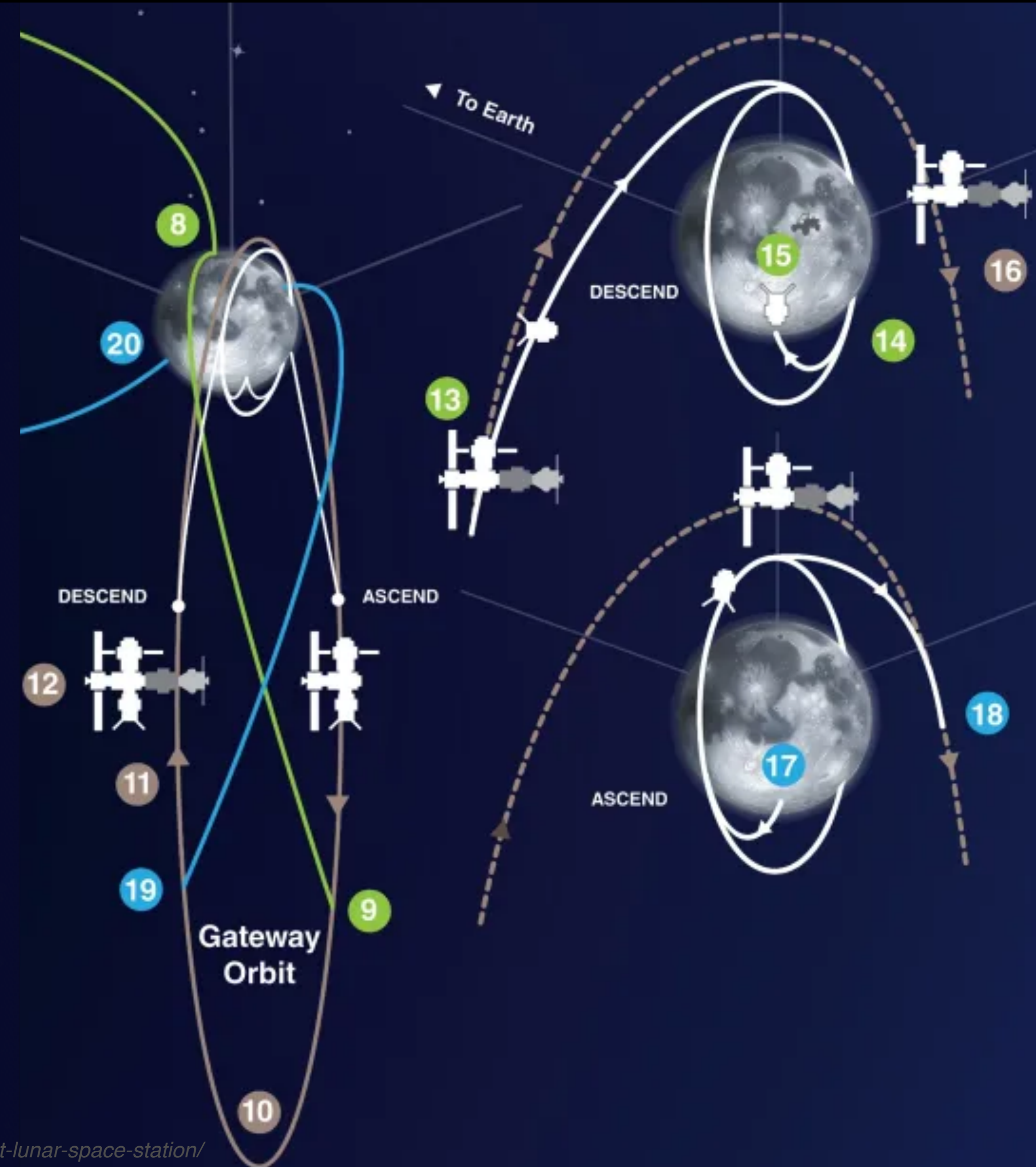
- 9:2 halo orbit staging orbit for the Gateway
- 9:2 synchronous quasi-halos have the same stroboscopic time, giving them bounded relative motion
- The Gateway orbit is surrounded by QPOs with a shared period as Gateway, keeping them in a long-term bounded relative motion



Motivation

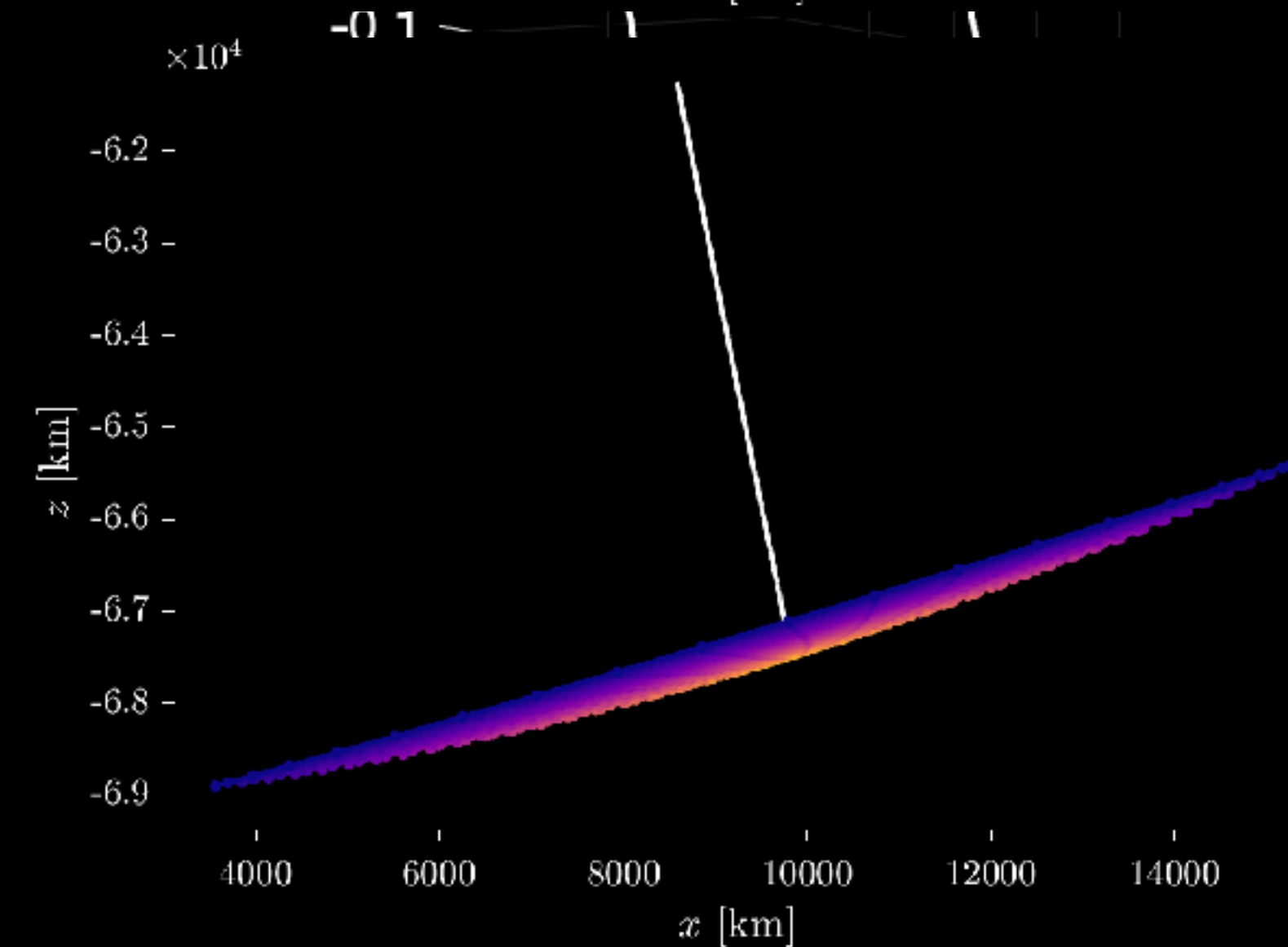
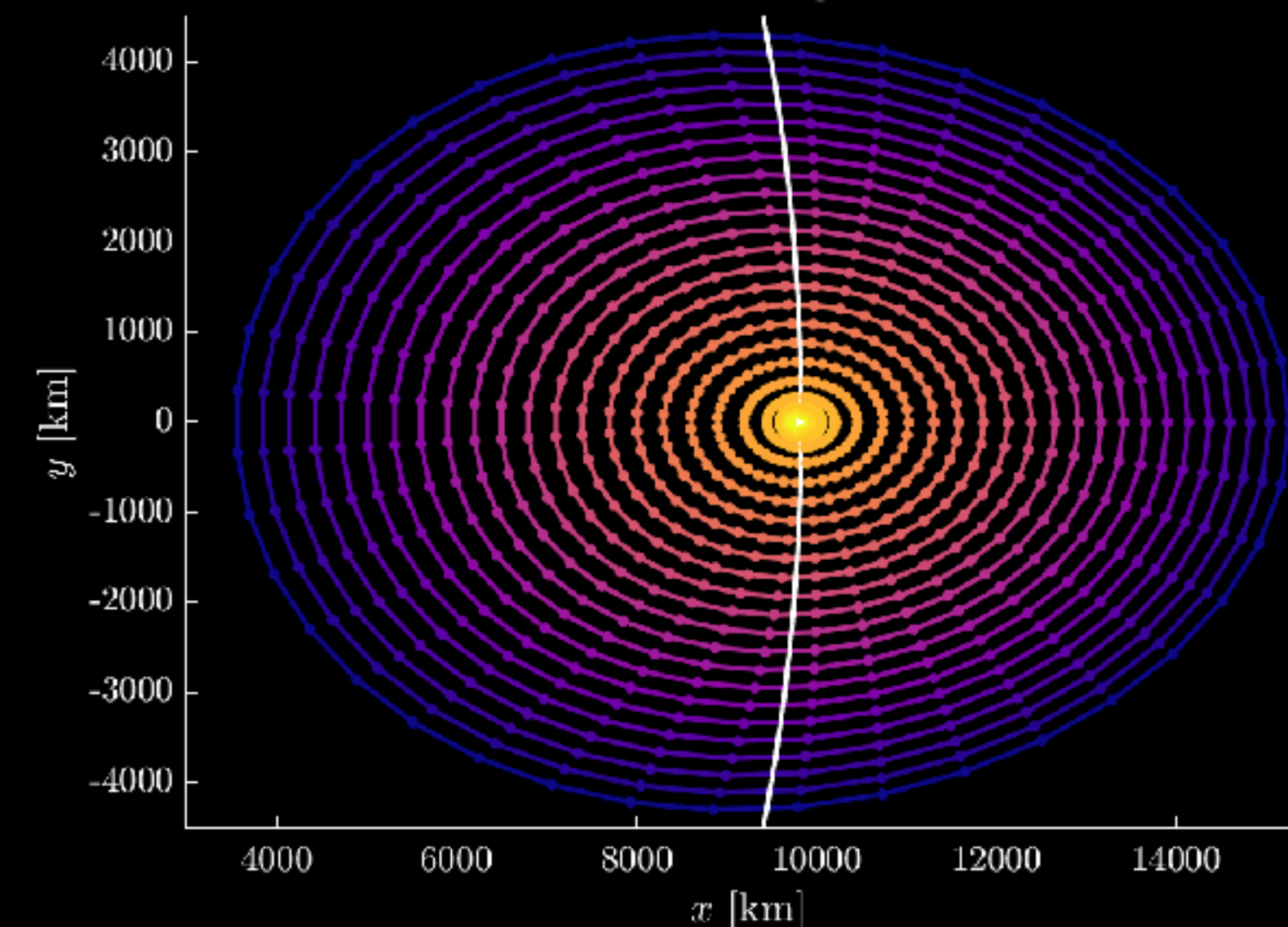
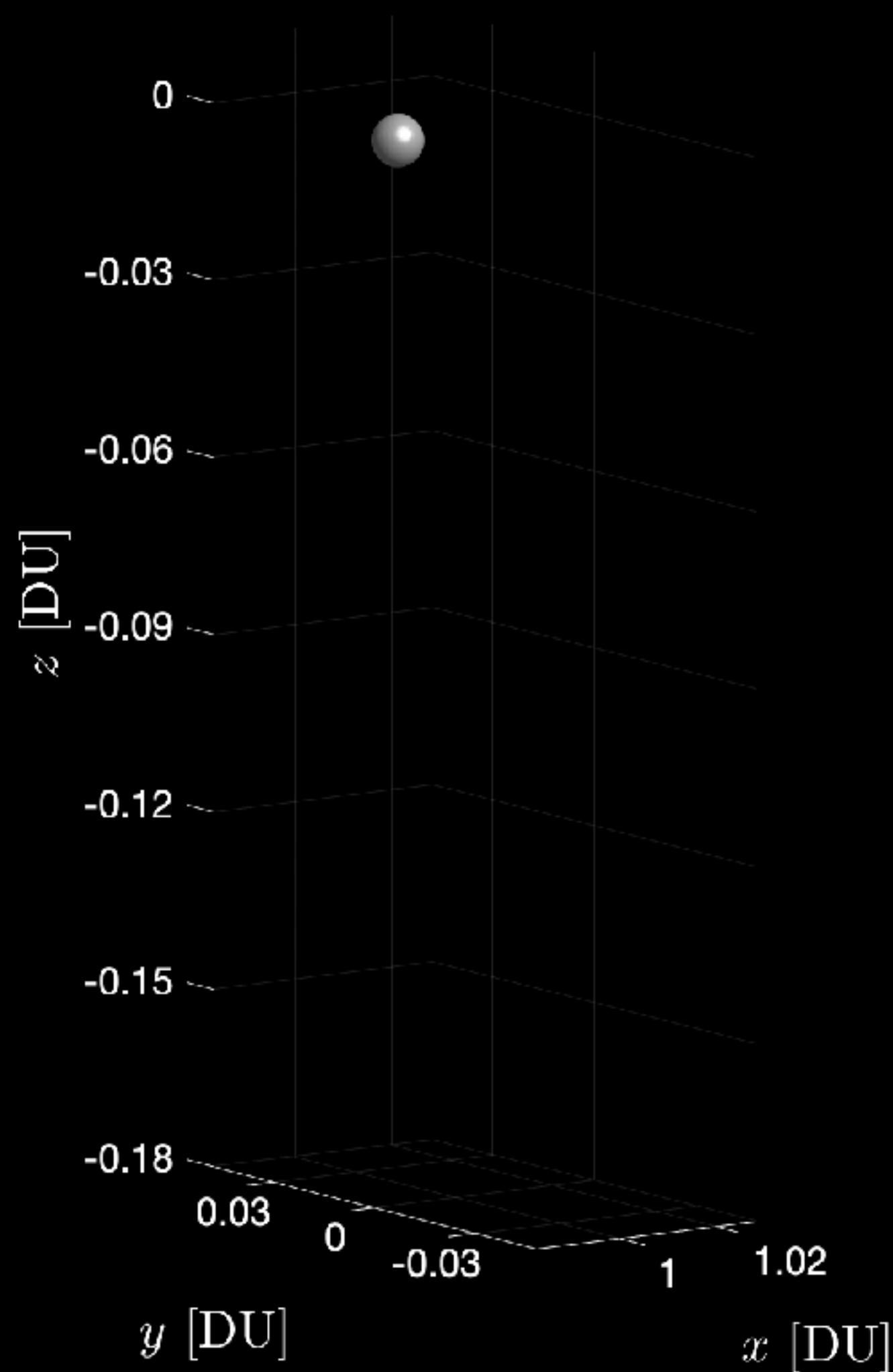
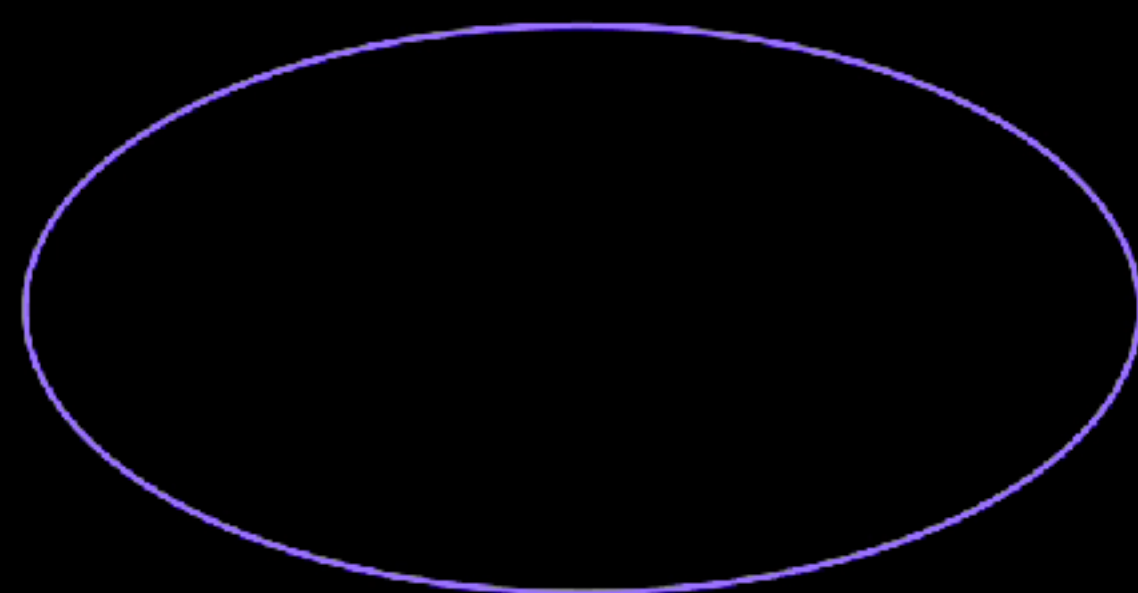


Complex mission architecture...
Need numerous options for loitering around Gateway

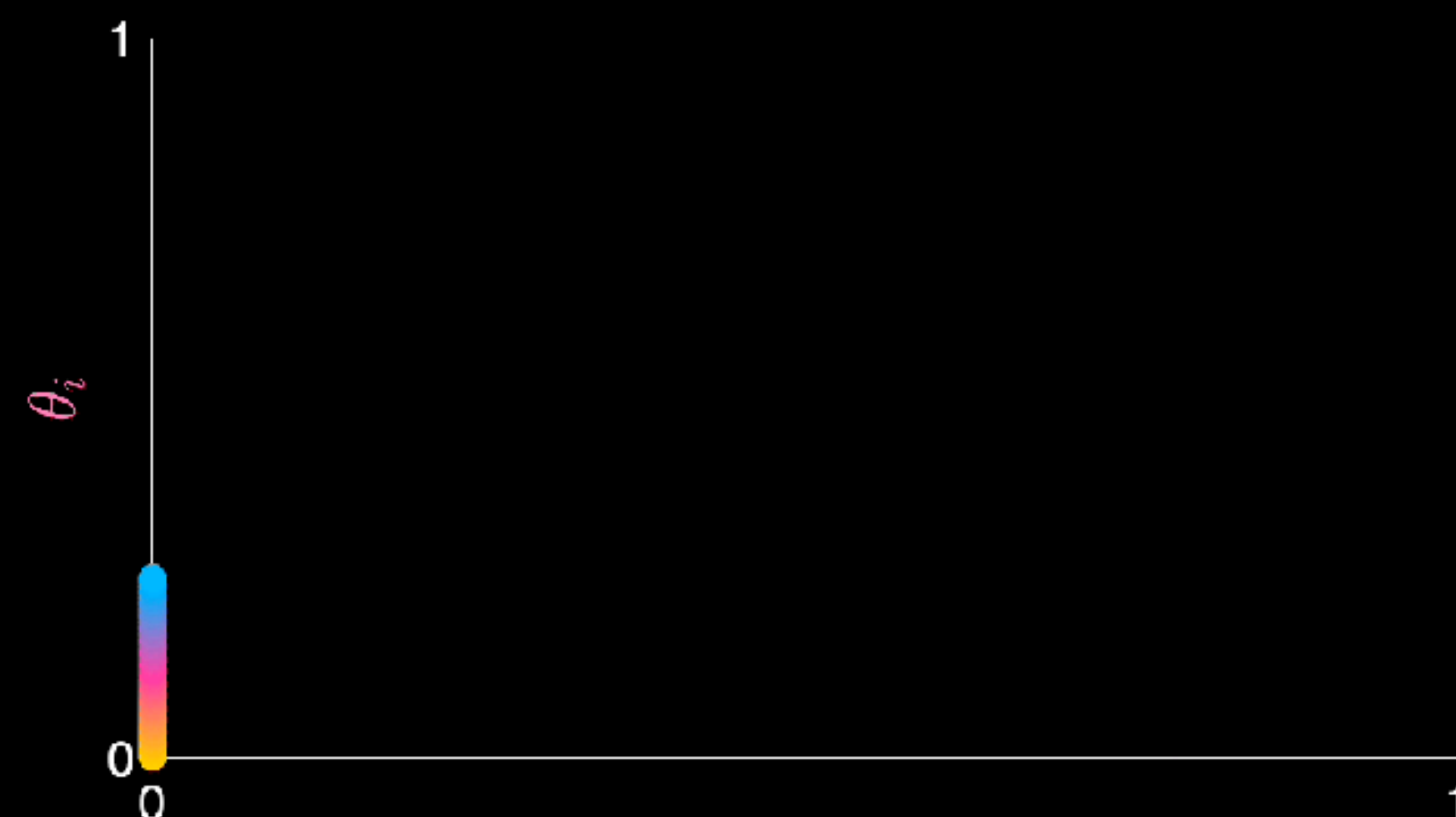
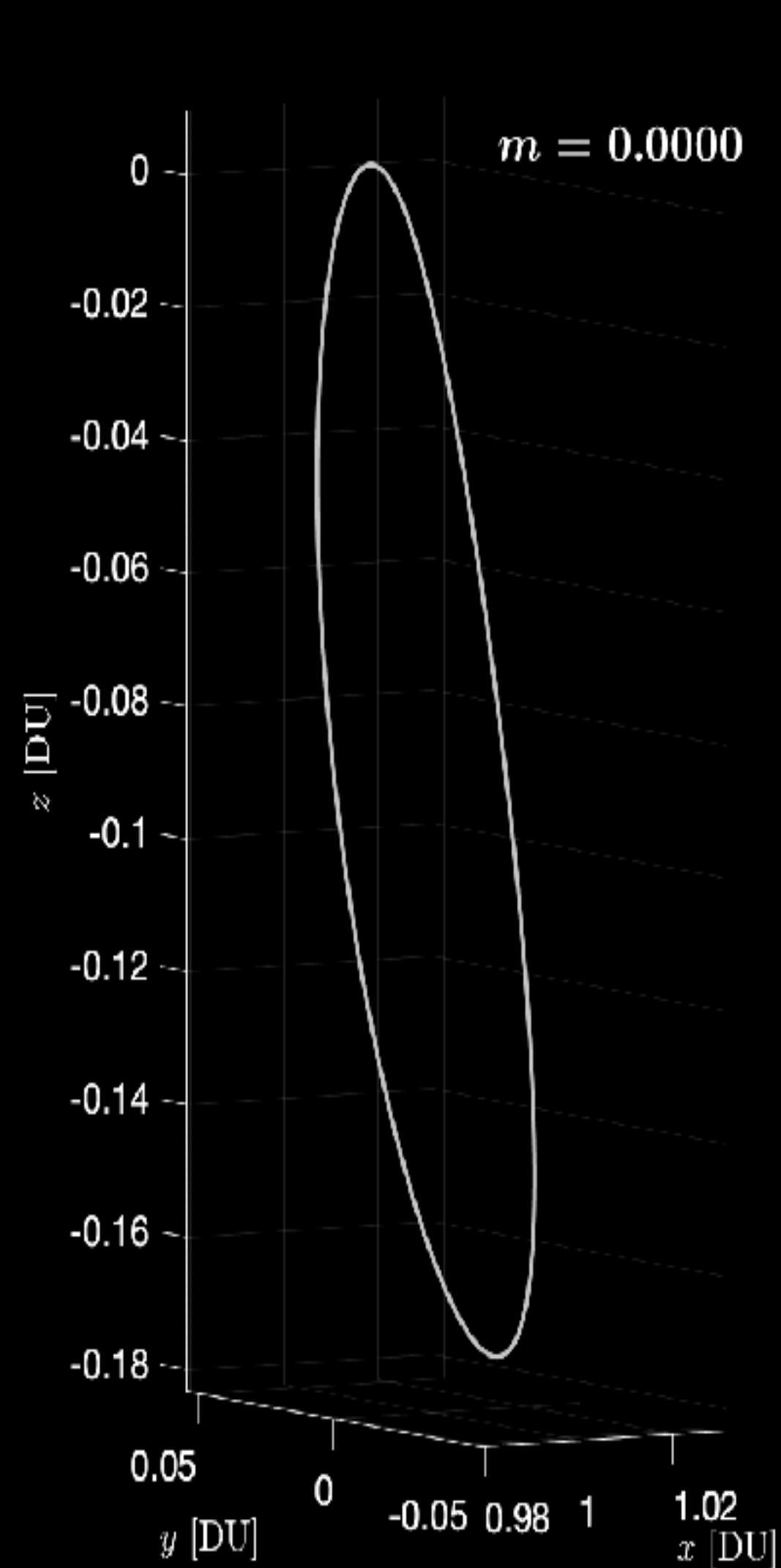


Periodic & Quasi-Periodic Orbits

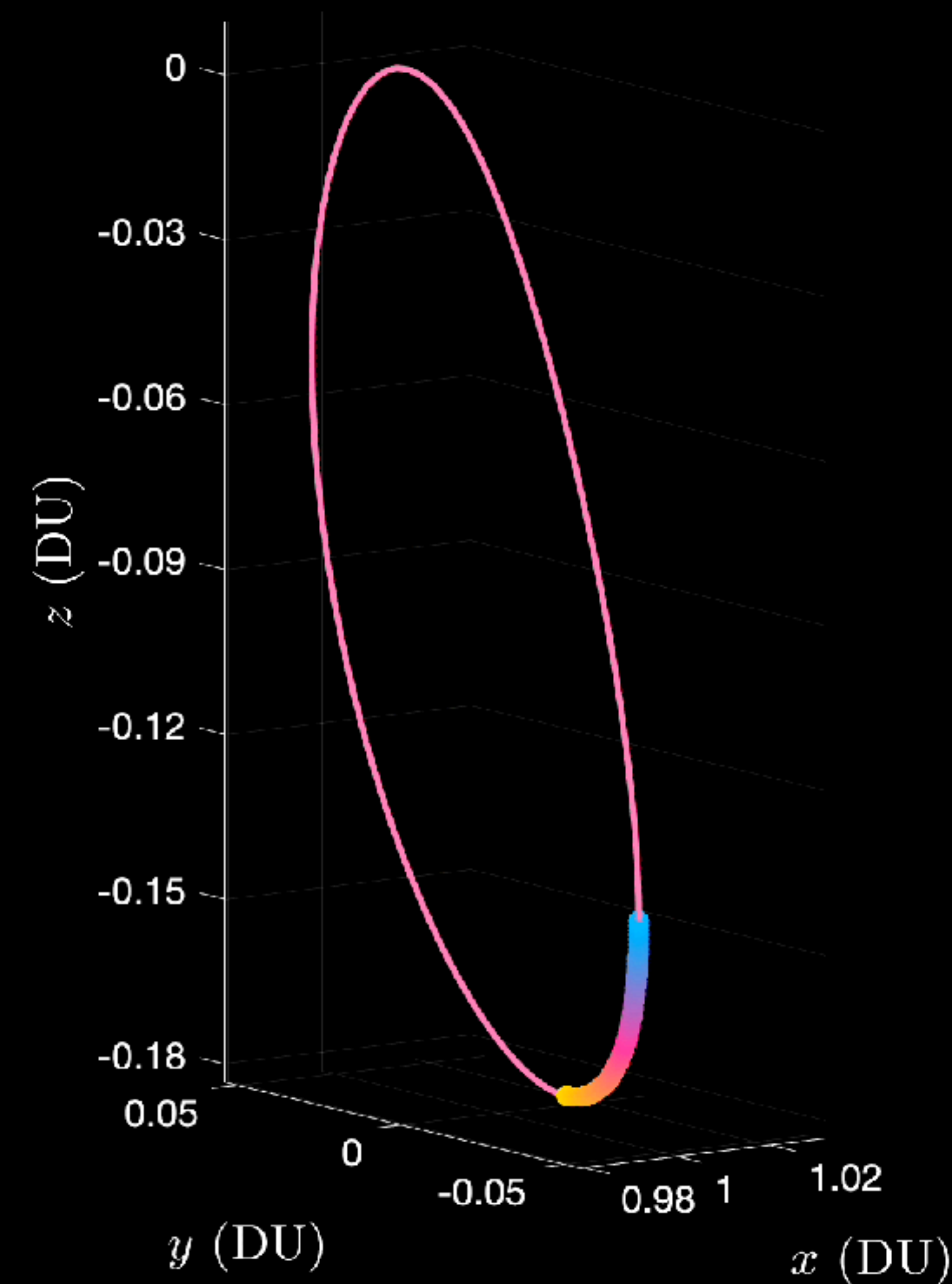
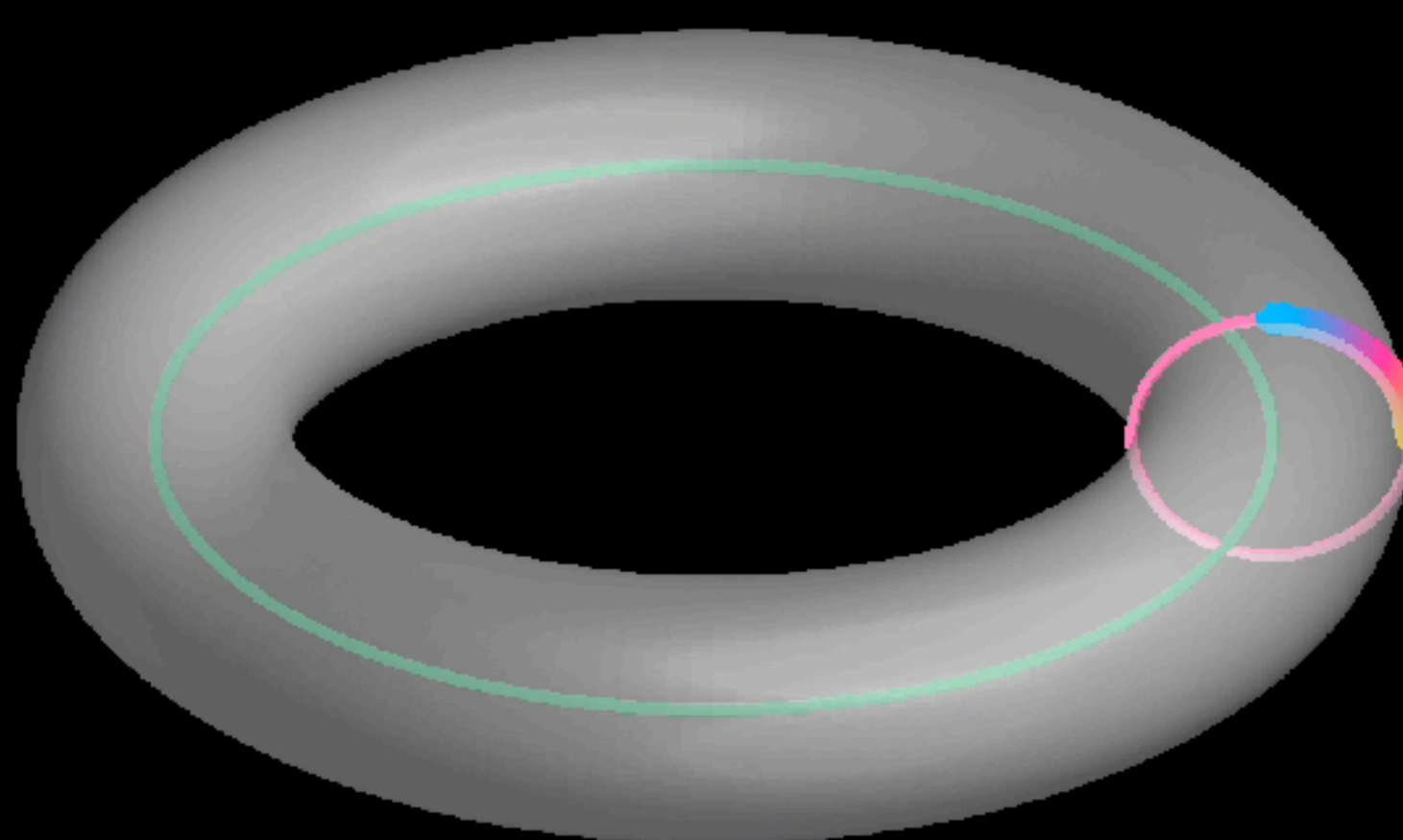
0 0.5 1 1.5 2
 θ_0/π
 9:2 NRHO $\tau = 0.00\pi$



Gateway's Planned Orbit



PO 0 to **PO F** $\tau = 0.0000\pi$



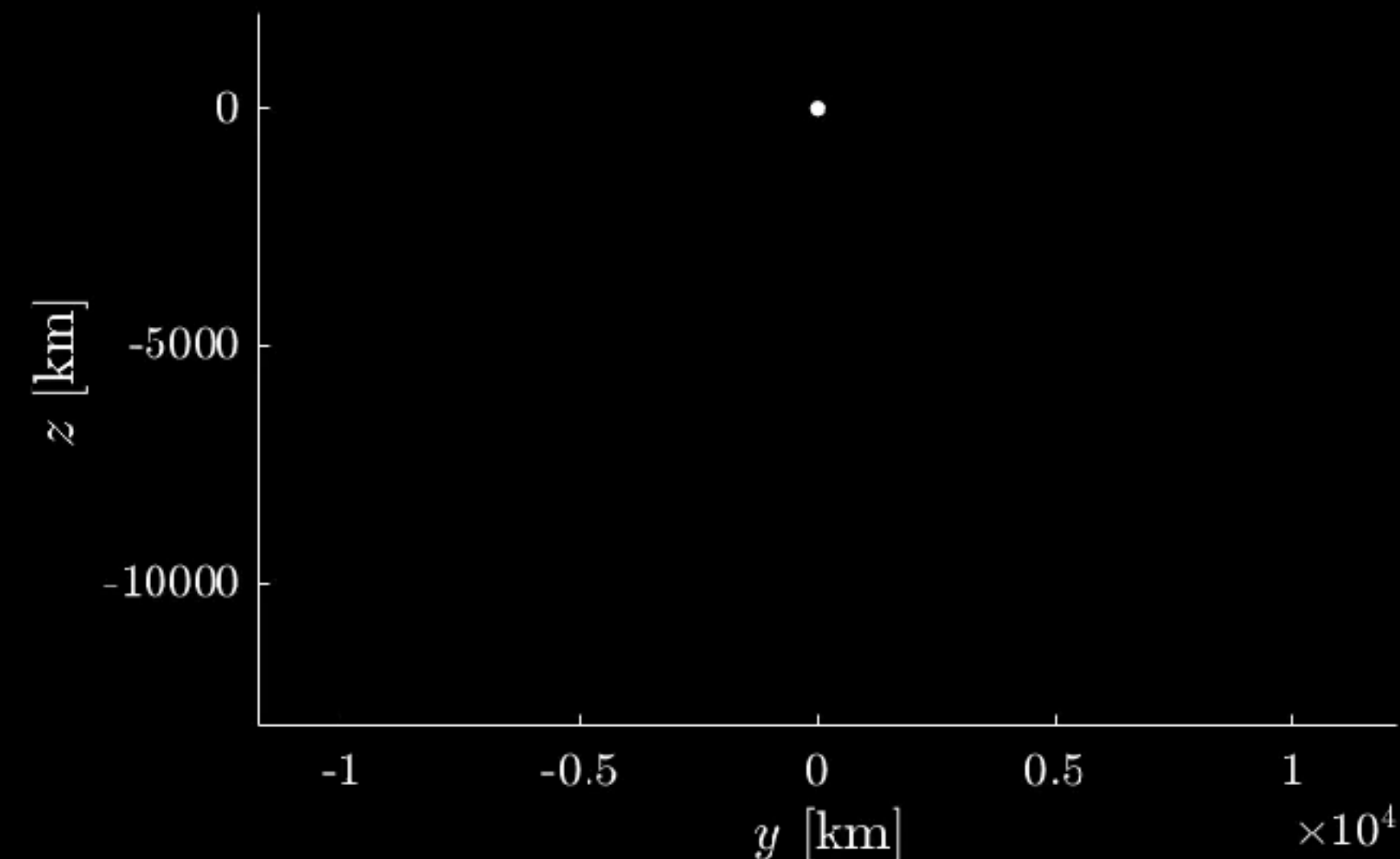
SEM HR4BP QPOs Around Gateway



Moon-Centered Earth-Moon Axes
 $t = 0.0$ days

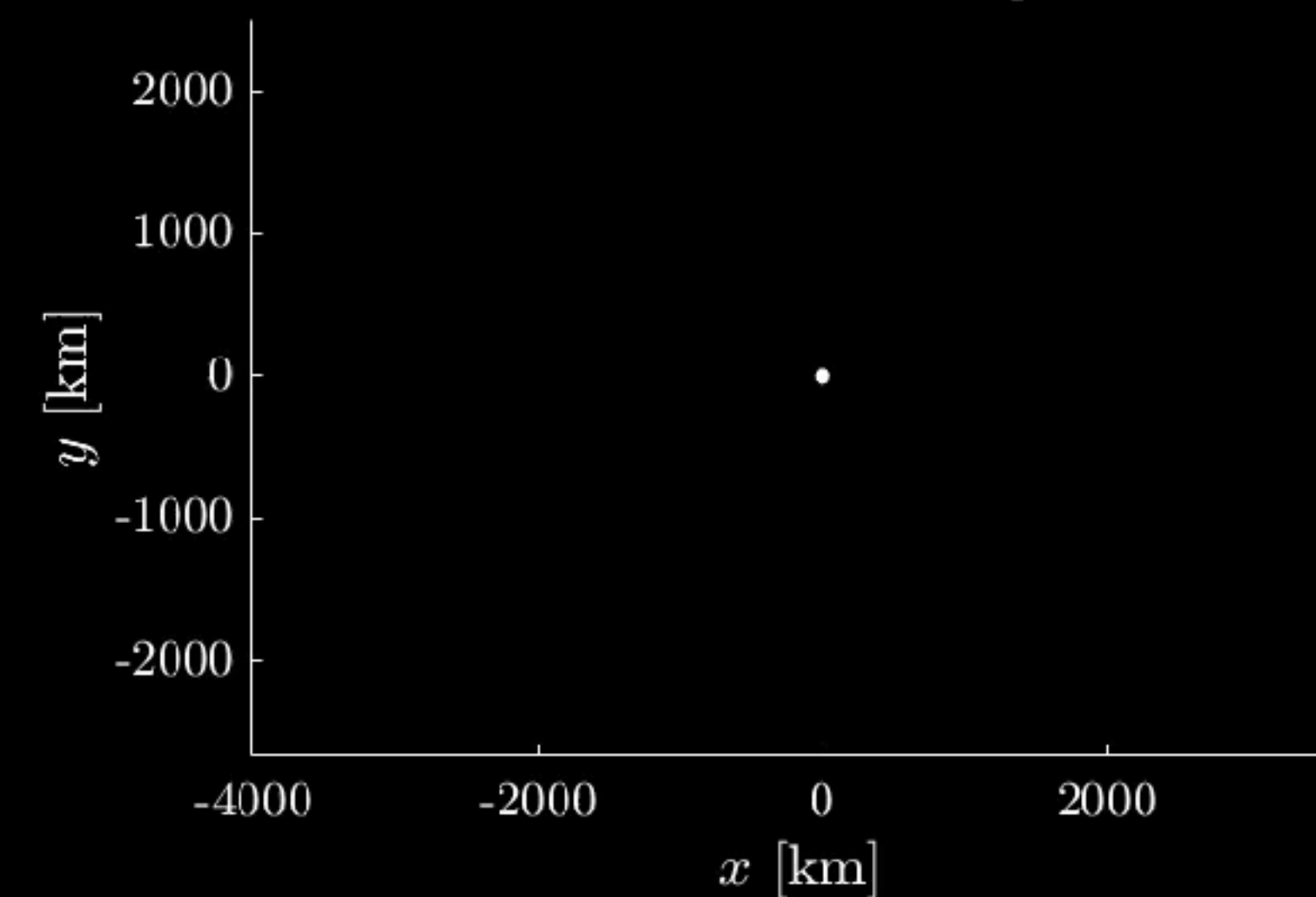
Gateway

Relative Motion Near Perilunes



Relative Motion

Relative Motion Near Apolunes

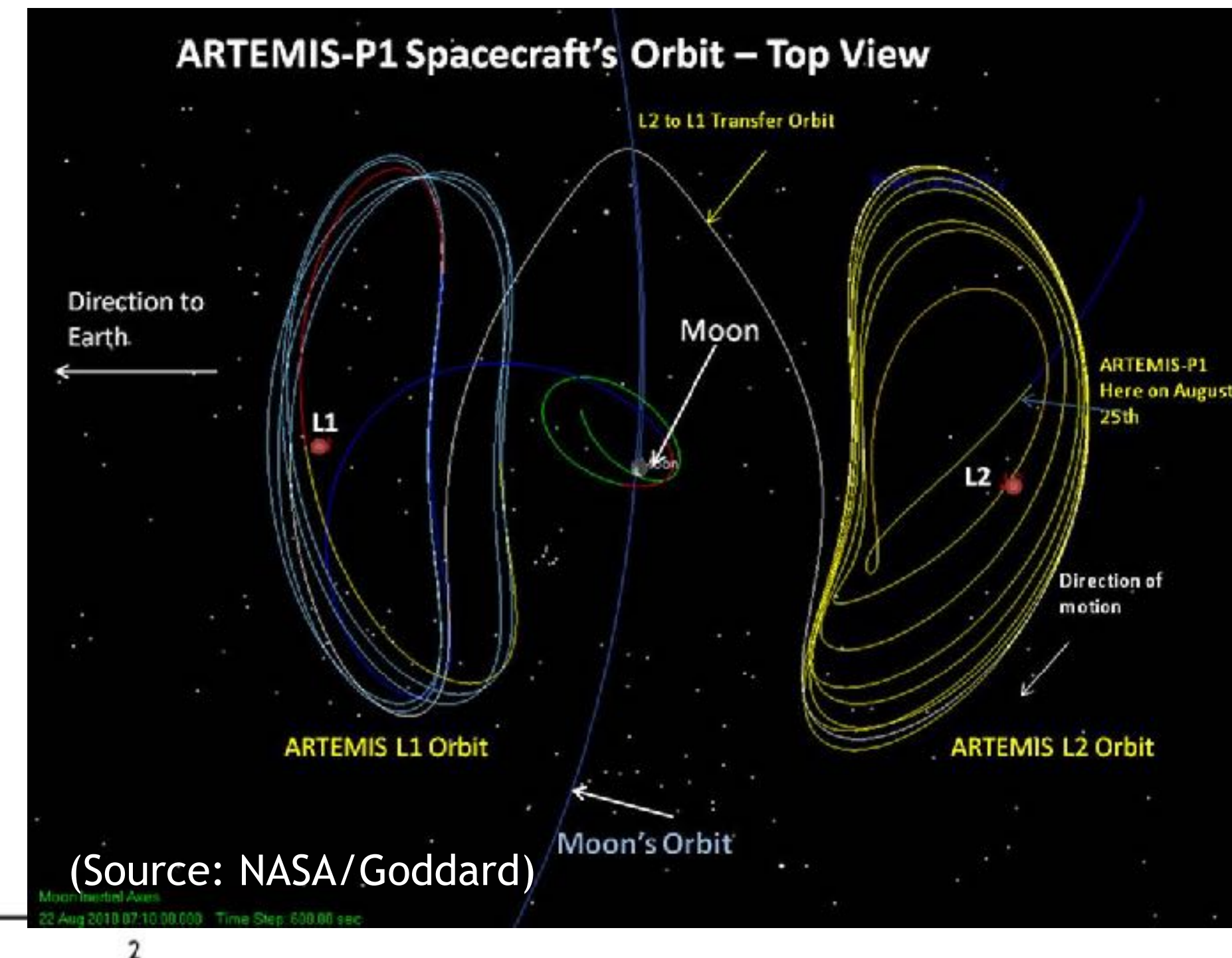
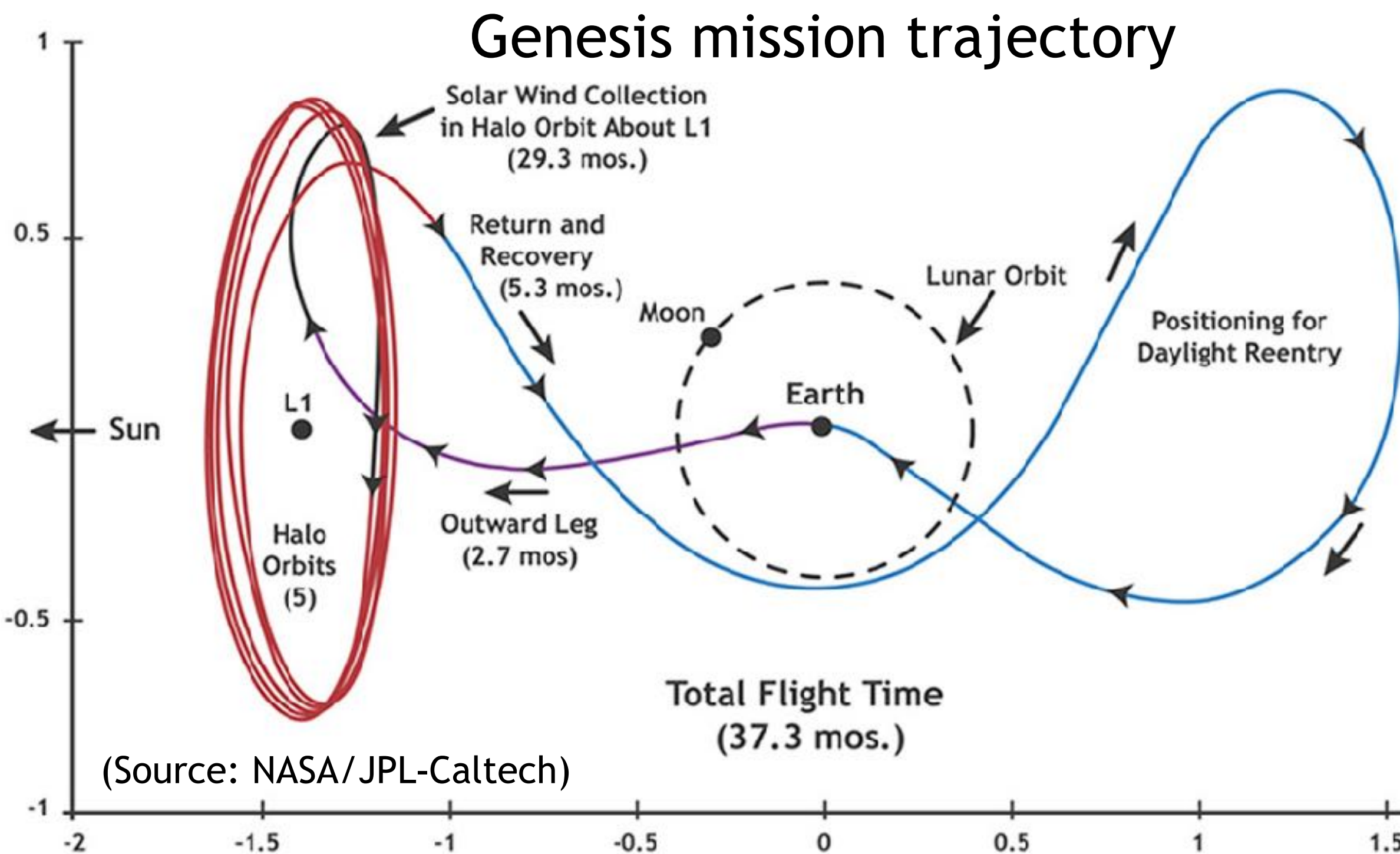


24 trajectories:
3 QPOs
8 points / QPO

Mission Design 101



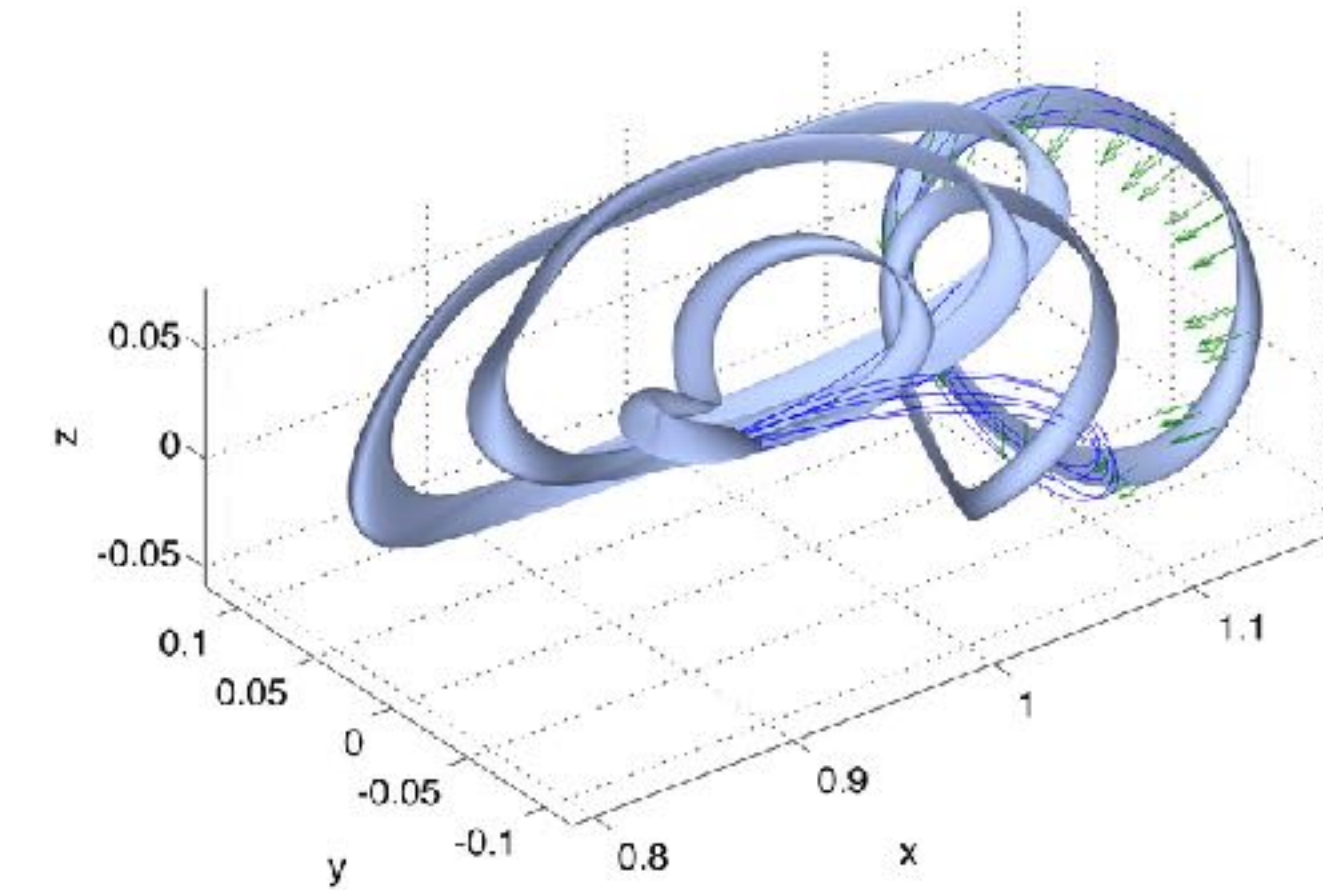
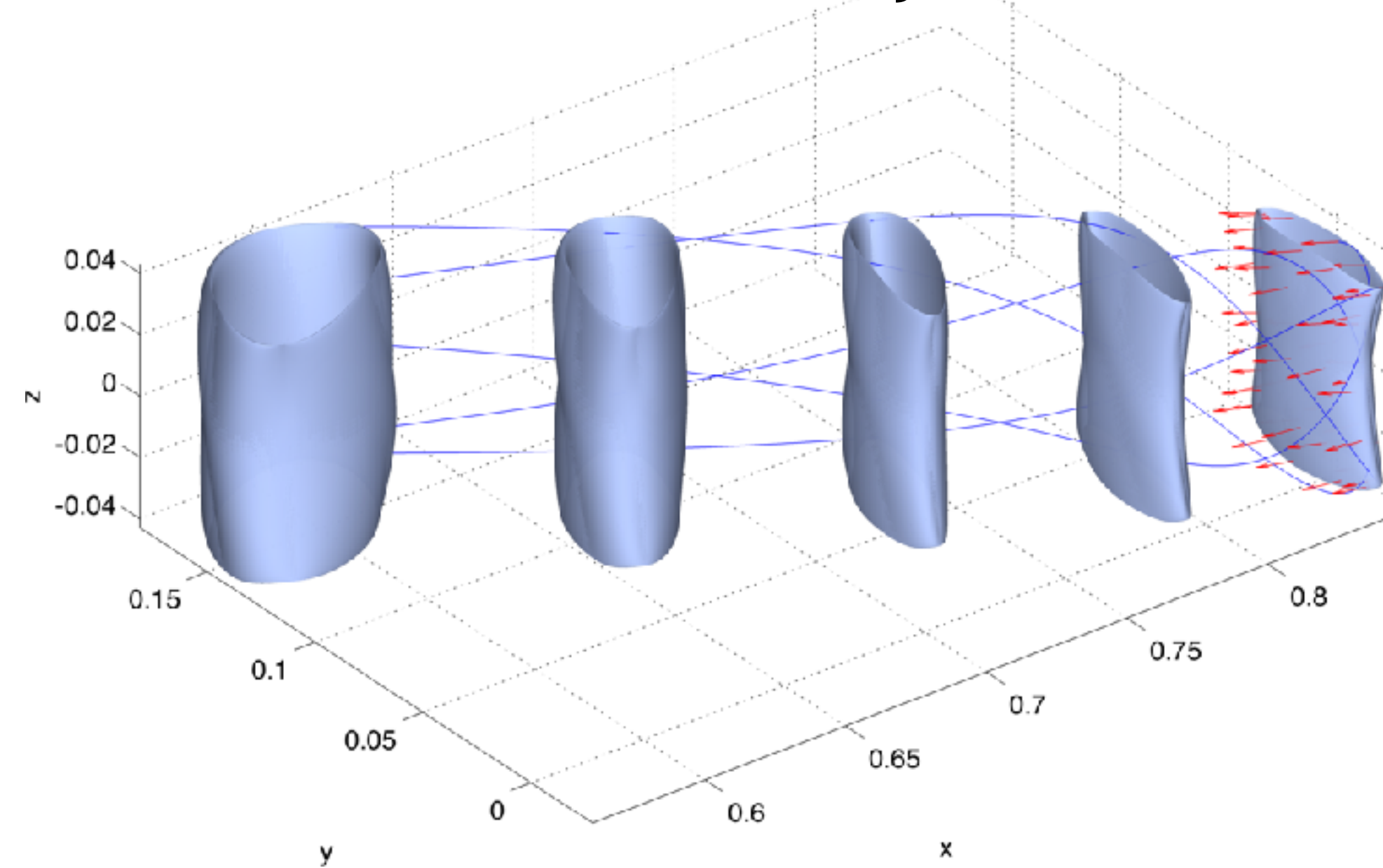
- Mission design engineers use stable and unstable manifolds to travel between solutions, requiring minimal propellant
 - Implemented in many missions, e.g., Genesis and ARTEMIS...
 - Are called “heteroclinic connections” or “whisker transfers”
- Heteroclinic connections between QPO manifolds can expand mission options



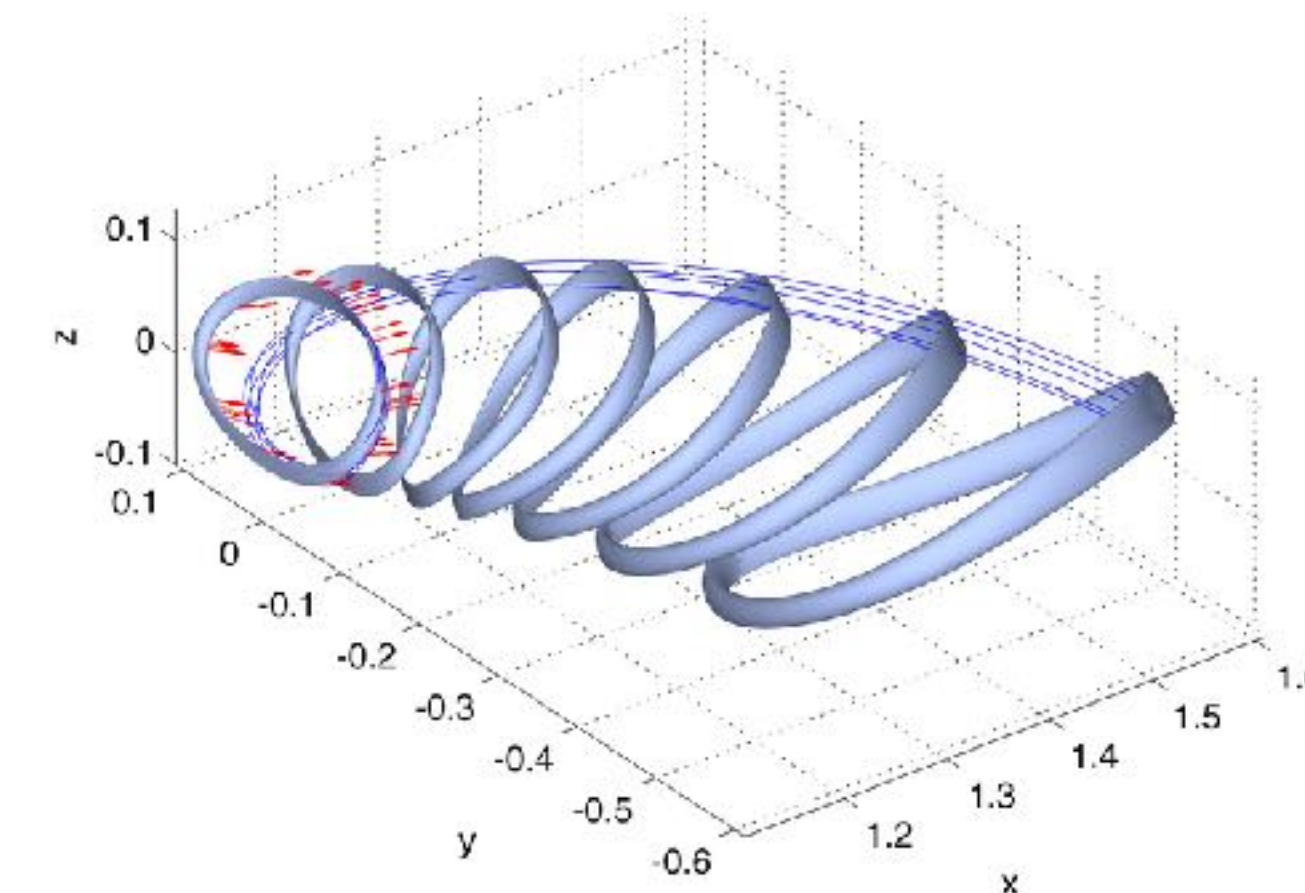
Targeting QPOs using Manifolds



Unstable manifold from a Lissajous orbit



Stable manifold from a halo orbit



Unstable manifold from a halo orbit

2-D QPO manifolds occupy 3-D in phase space
 Intersections between manifolds and single
 trajectories occur generically
 Provide a larger region over which to optimize
 maneuvers that “enter onto” a manifold

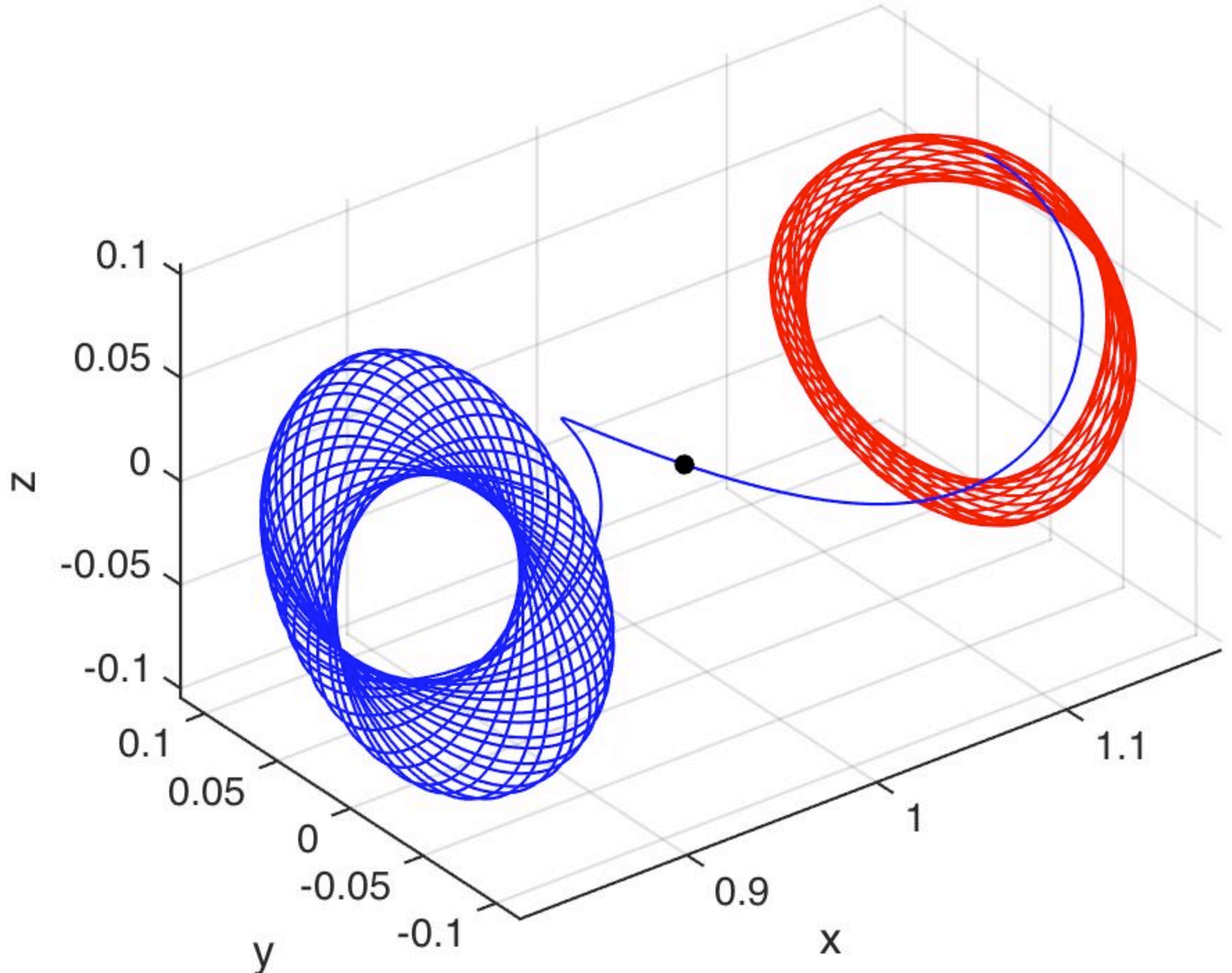
Heteroclinic Connections for QPOs



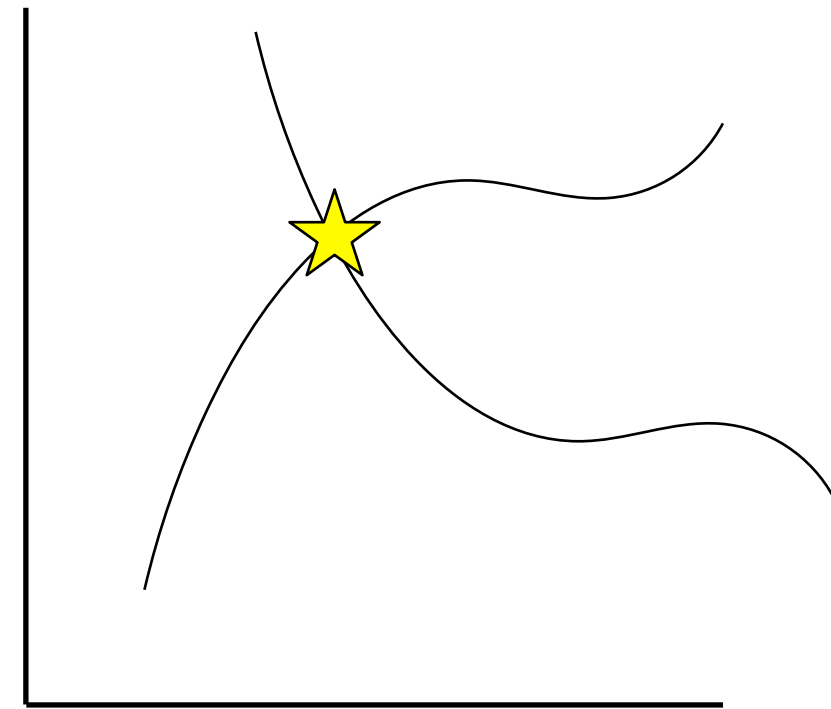
A single QPO can connect to an entire family of different QPOs at the same value of the Jacobi constant.

This works because the Co-Dimension between the hyperbolic manifolds of two QPOs at the same level of Jacobi energy equals 1.

The key for visualization is to understand the Co-Dimension between objects in higher dimensional spaces.



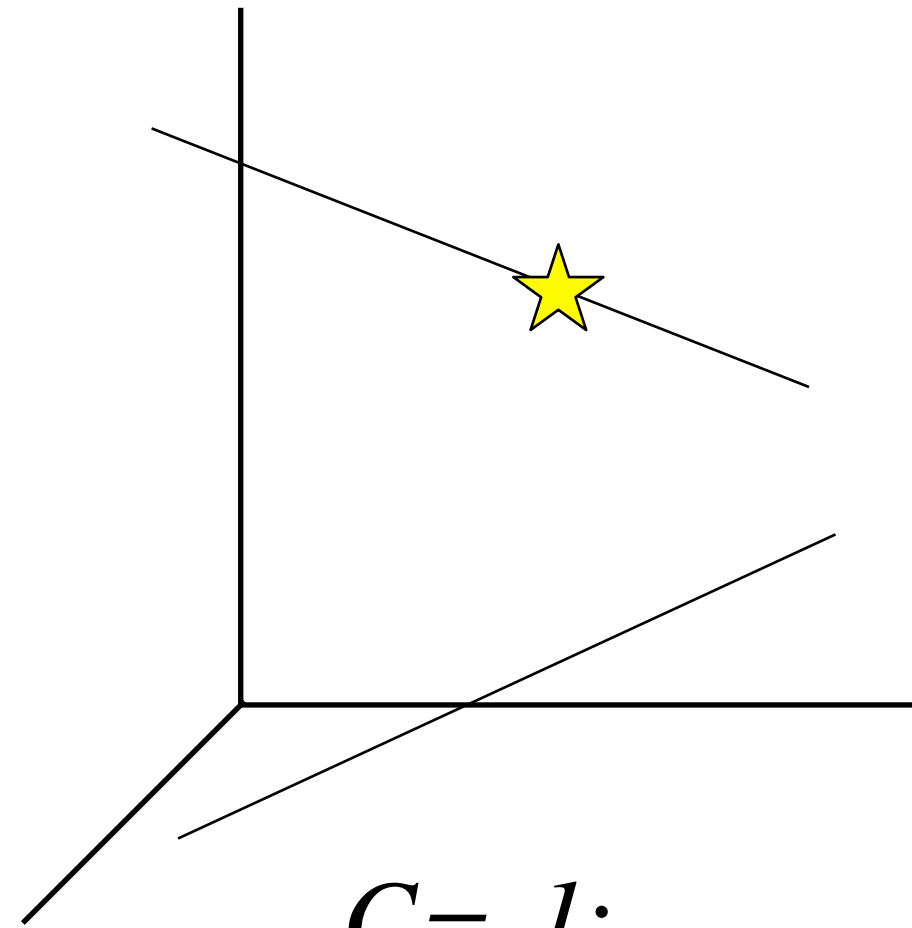
Co-Dimensions and Intersections



$C=0$:

Generic Intersection

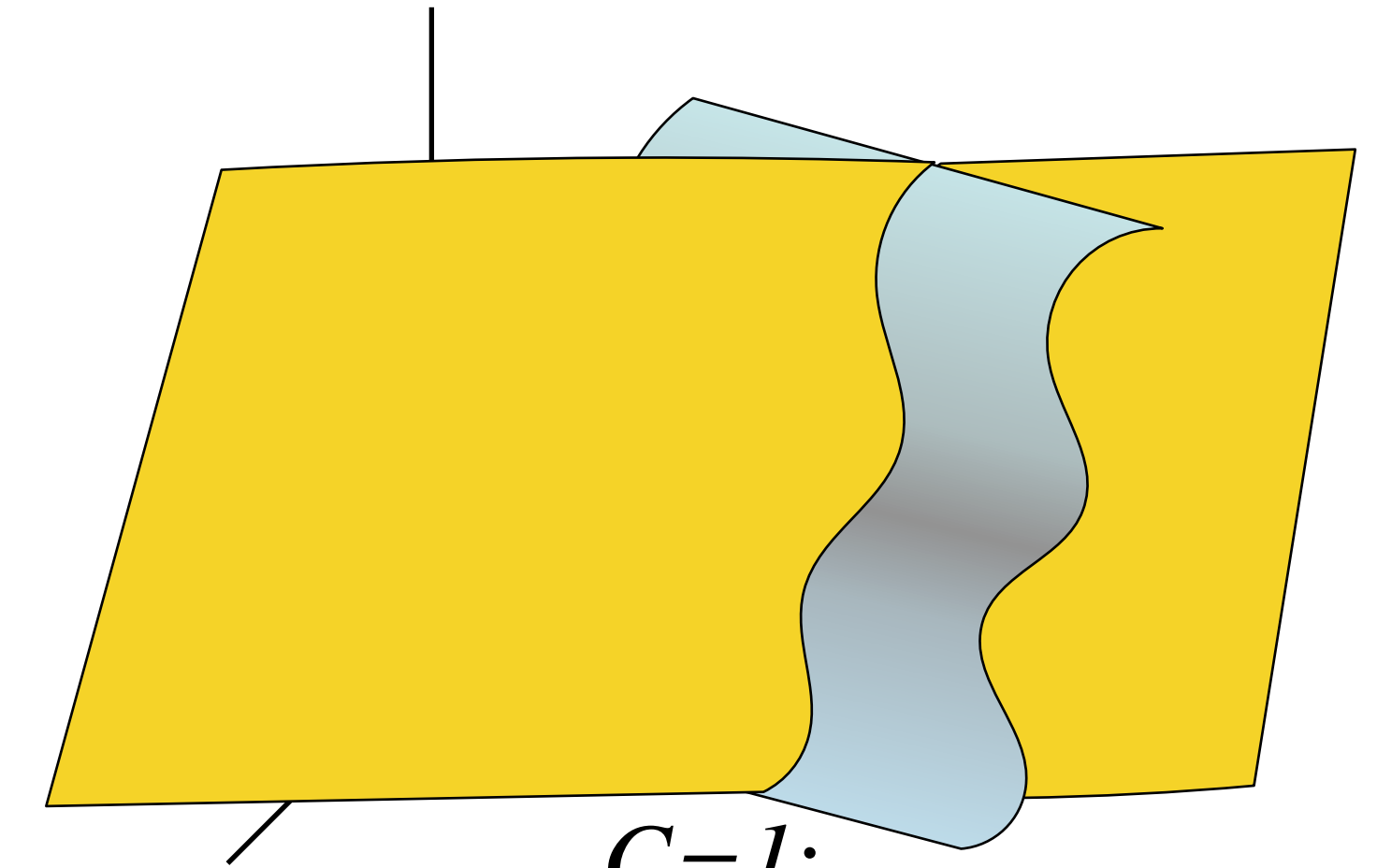
$$1 + 1 - 2 = 0$$



$C=-1$:

Need to search in 1-D

$$1 + 1 - 3 = -1$$



$C=1$:

Intersects in a 1-D line

$$2 + 2 - 3 = 1$$

Theory of General Position:

- The Co-Dimension of a k and l dimensional solution in an n dimensional space is $C = k + l - n$
- If $C \geq 0$ the intersection of these solutions is “generic”

Extends to arbitrary dimensions



The Ubiquity of Intersecting Solutions in Phase Space

- The existence of heteroclinic connections can be inferred from the co-dimension
 - The manifolds of 2-D QPO in phase space ($N=6$) will generically intersect
 - Manifolds of 1-D Periodic orbits in planar systems ($N=4$) will generically intersect
 - Heteroclinic orbits on a constant energy surface ($N = 5$ or 3) have a co-dimension of 1

Solution	Dimension	+ Stable/Unstable Manifold	General Co-Dimension $C = d+d-N$	Co-Dimension at a Fixed Energy
Planar Equilibrium	0	1	$1+1-4 = -2$	$1 + 1 - 3 = -1$
Planar Periodic Orbit	1	2	$2+2-4 = 0$	$2 + 2 - 3 = 1$
3-D Periodic Orbit	1	2	-2	$2 + 2 - 5 = -1$
2-D QPO in 3-D	2	3	0	$3 + 3 - 5 = 1$

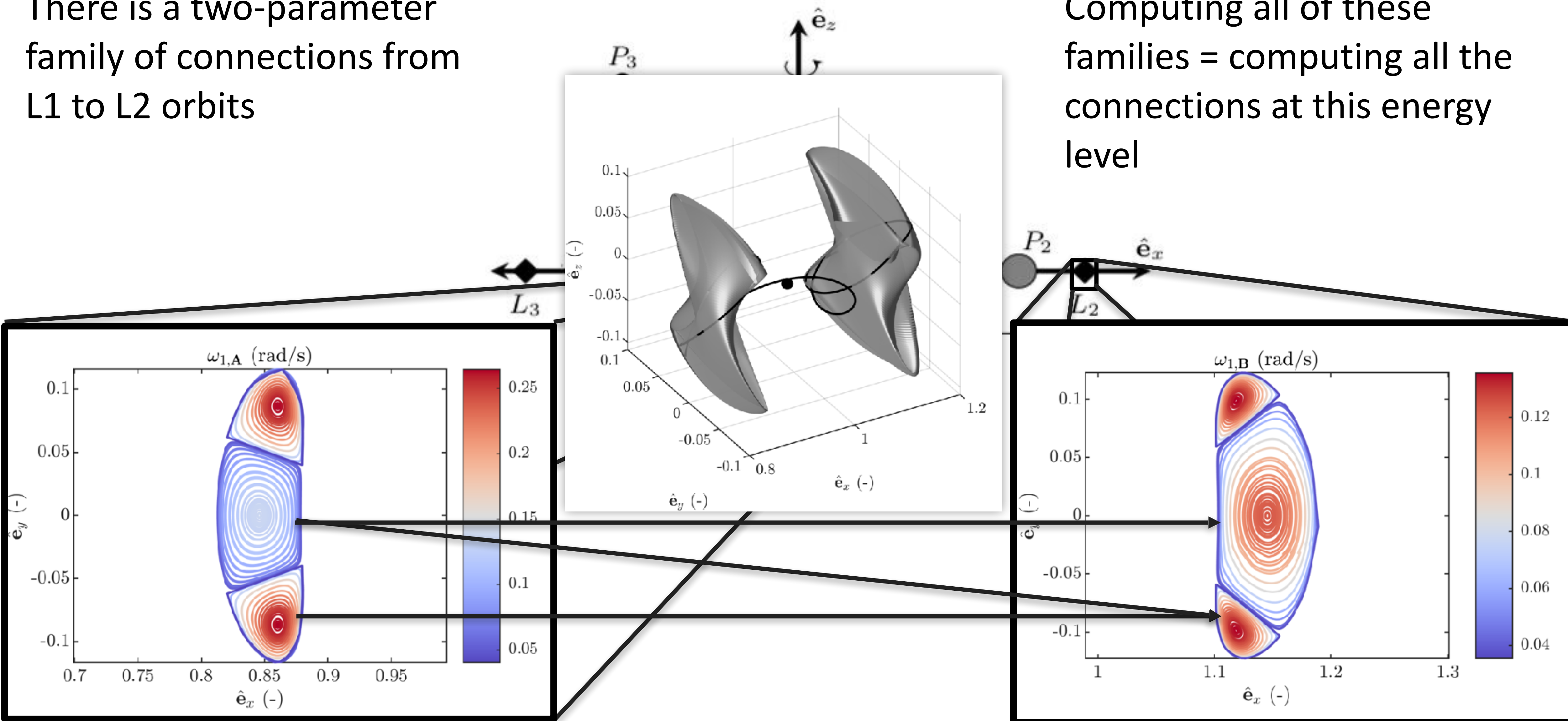


The Earth-Moon Connection Family



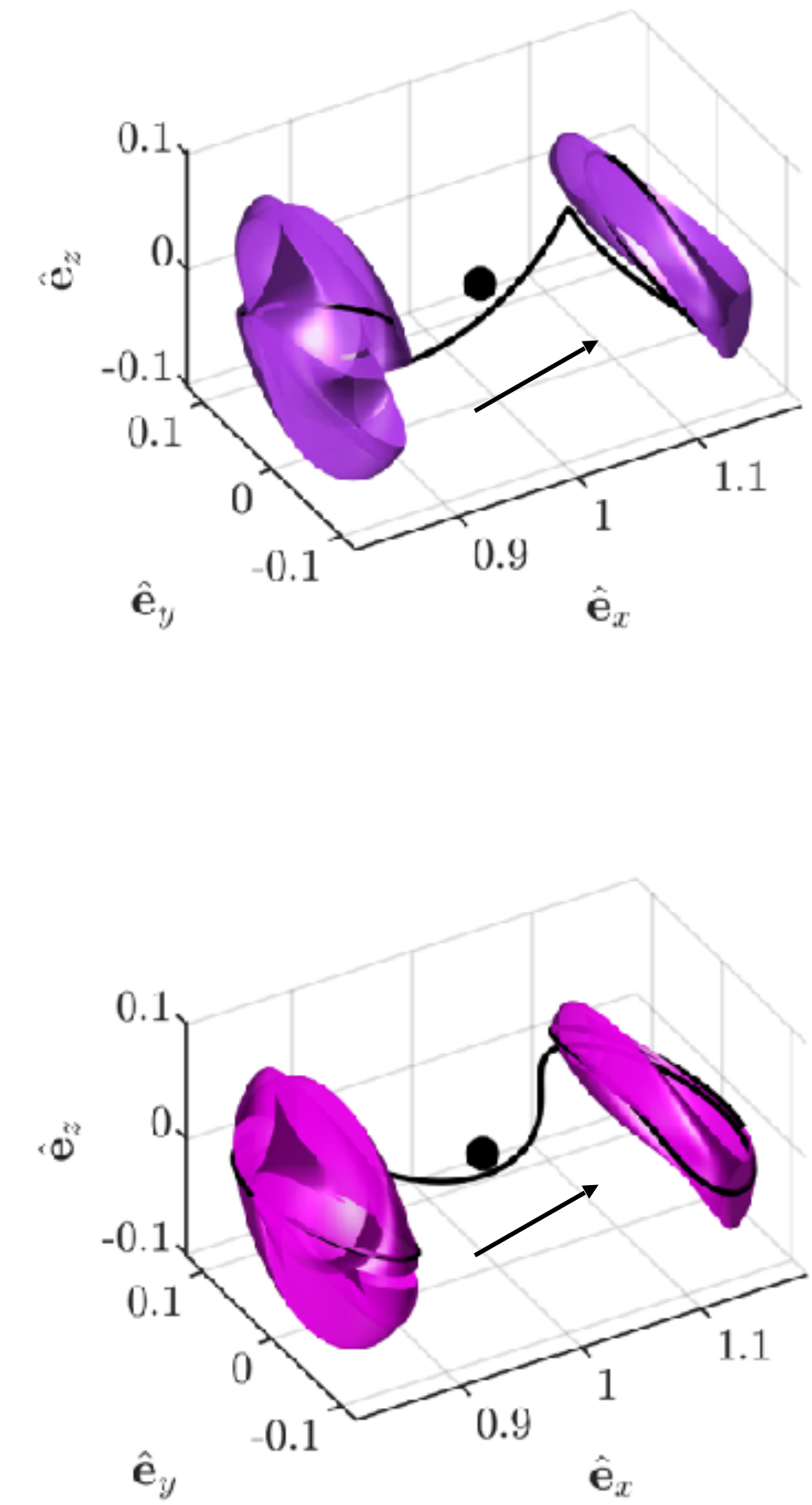
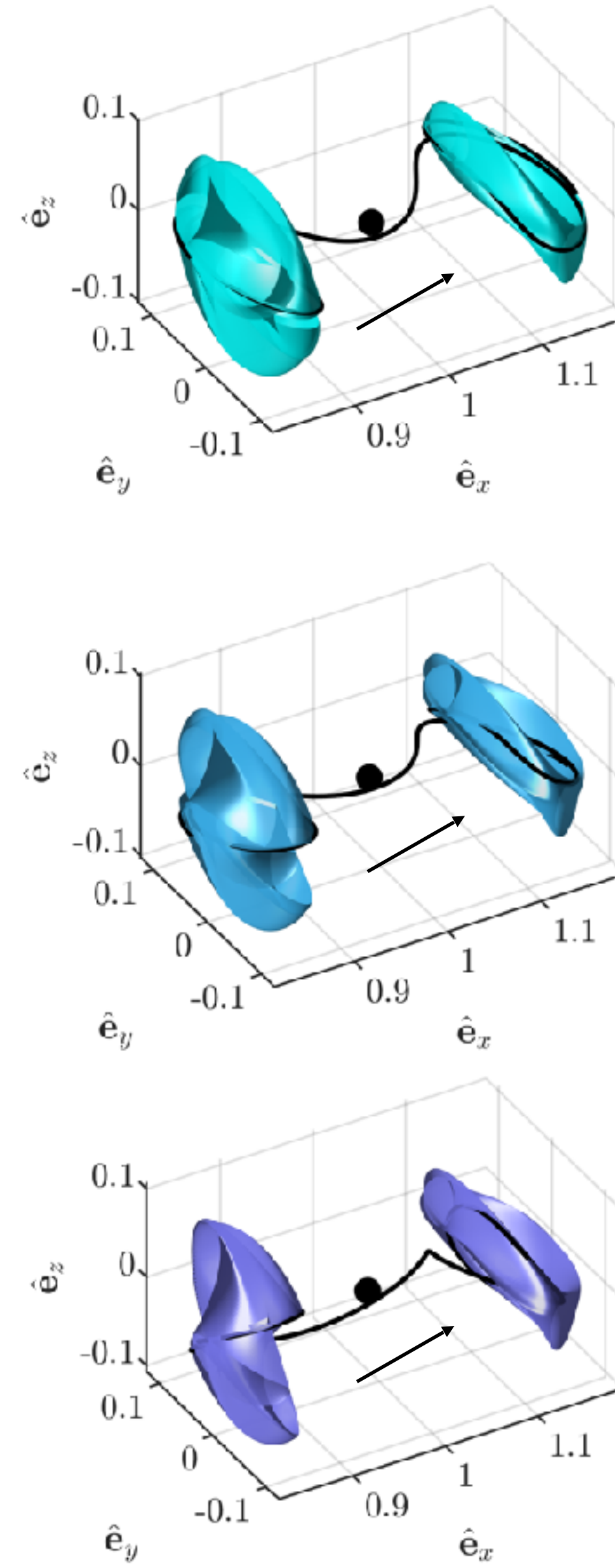
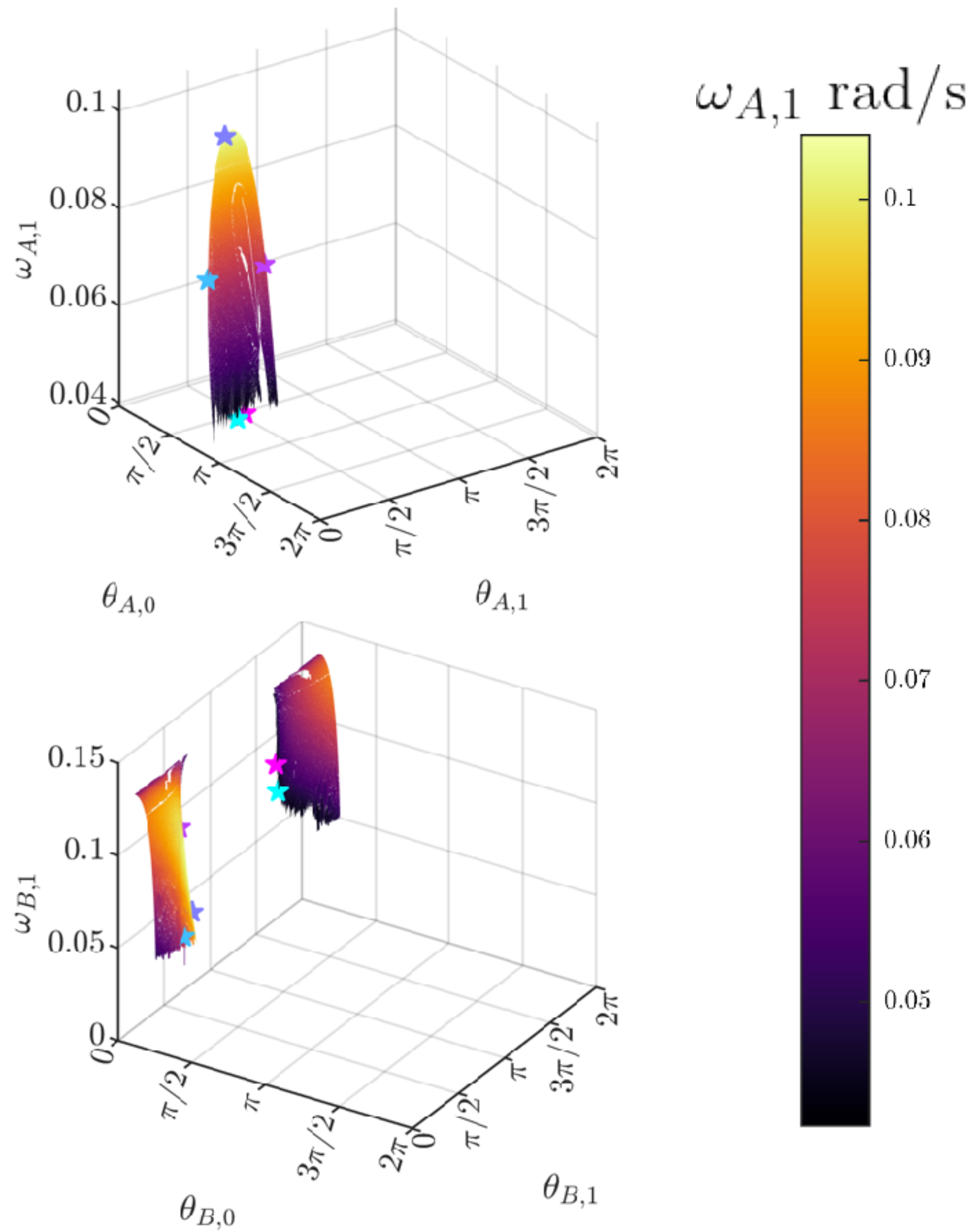
There is a two-parameter family of connections from L1 to L2 orbits

Computing all of these families = computing all the connections at this energy level



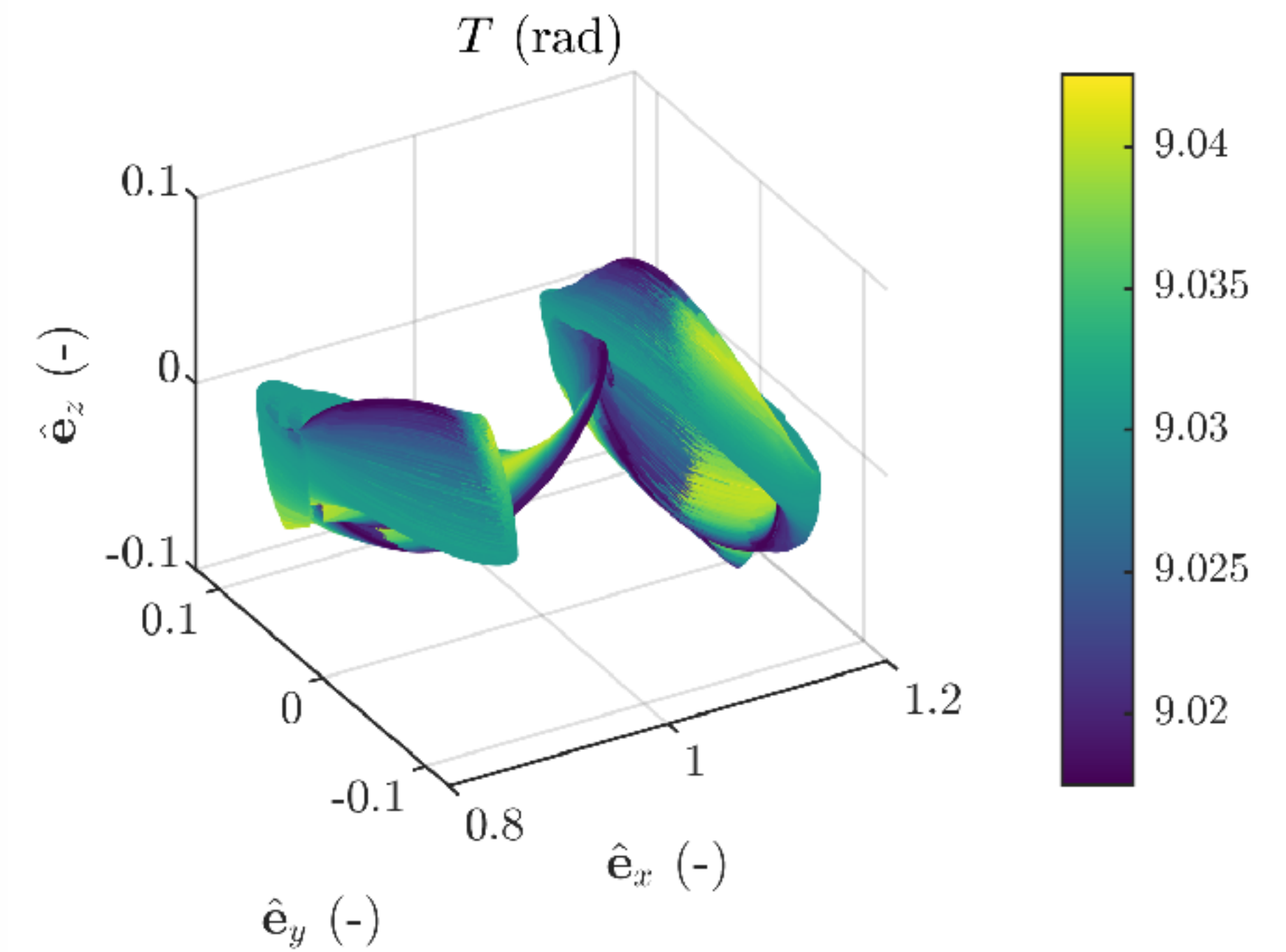
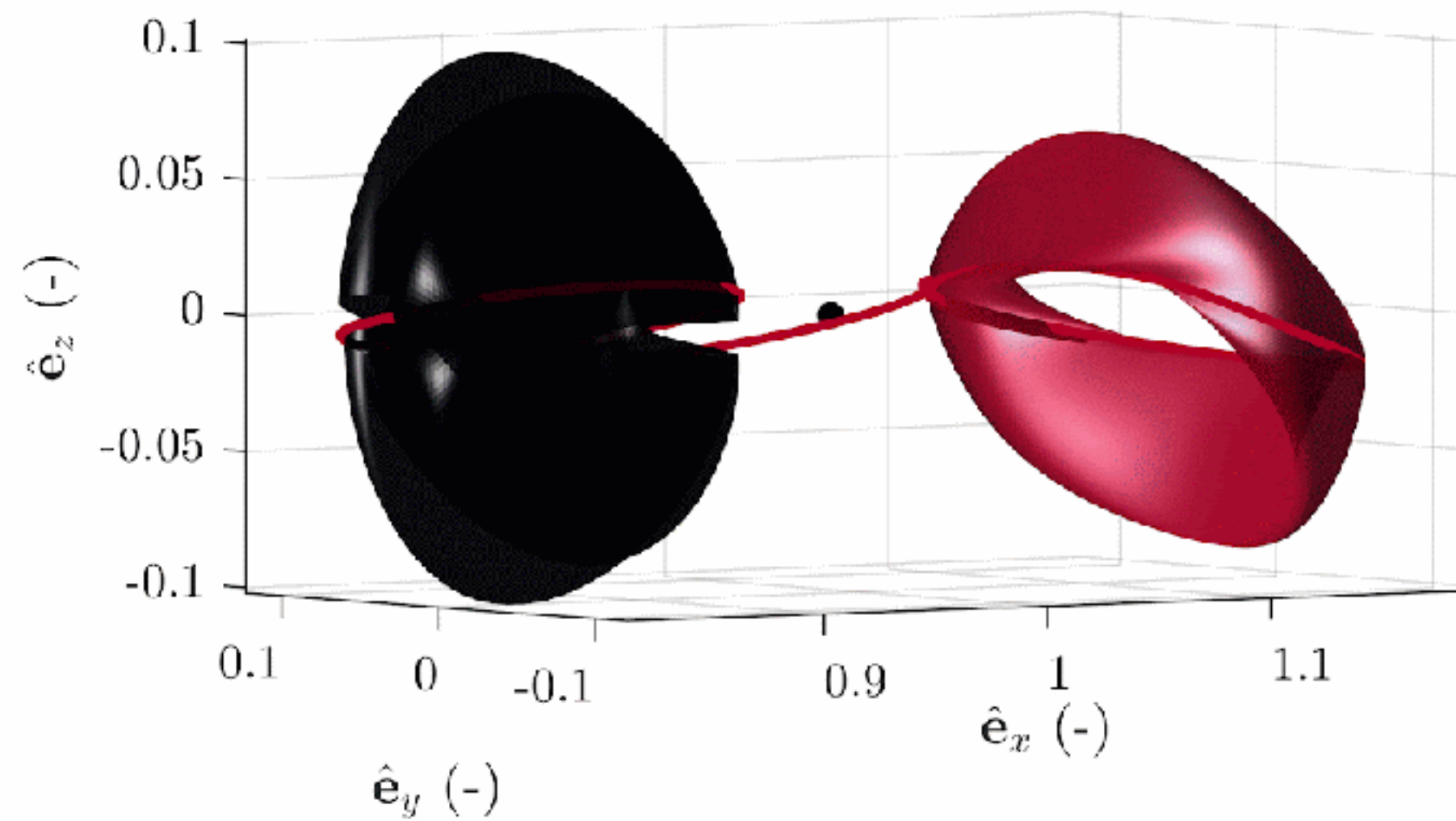
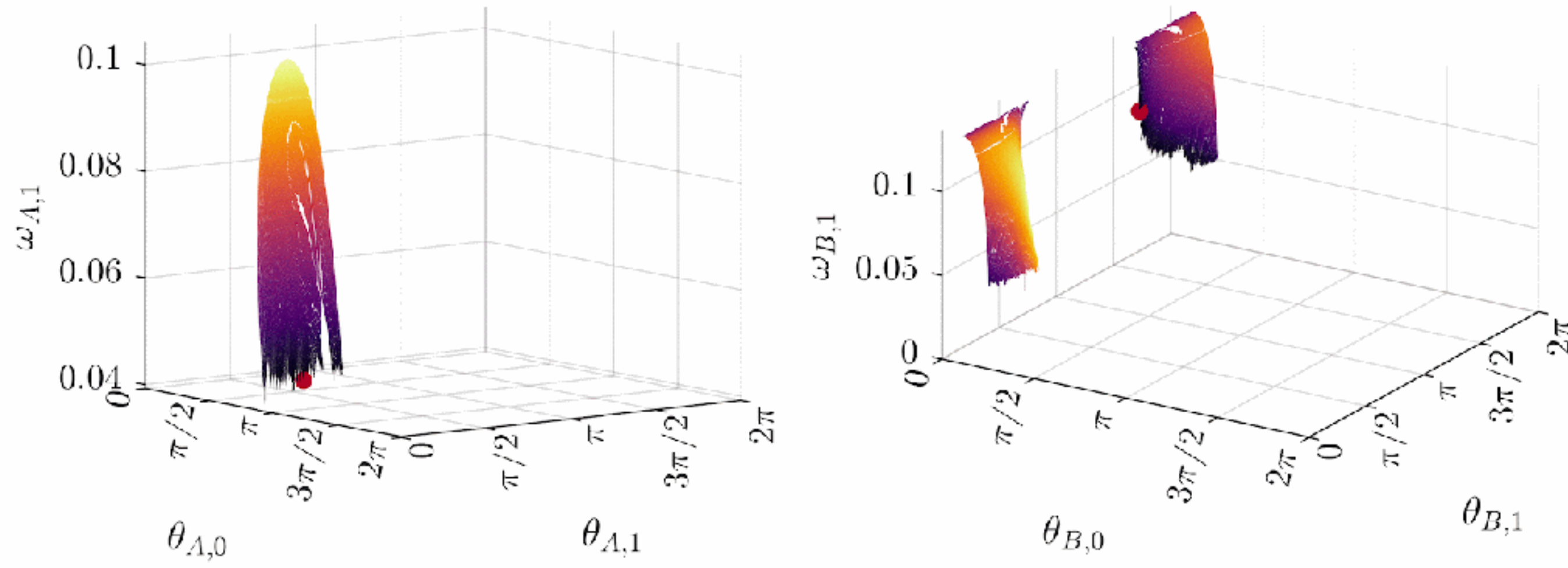


Summary: Lissajous to SQH Family



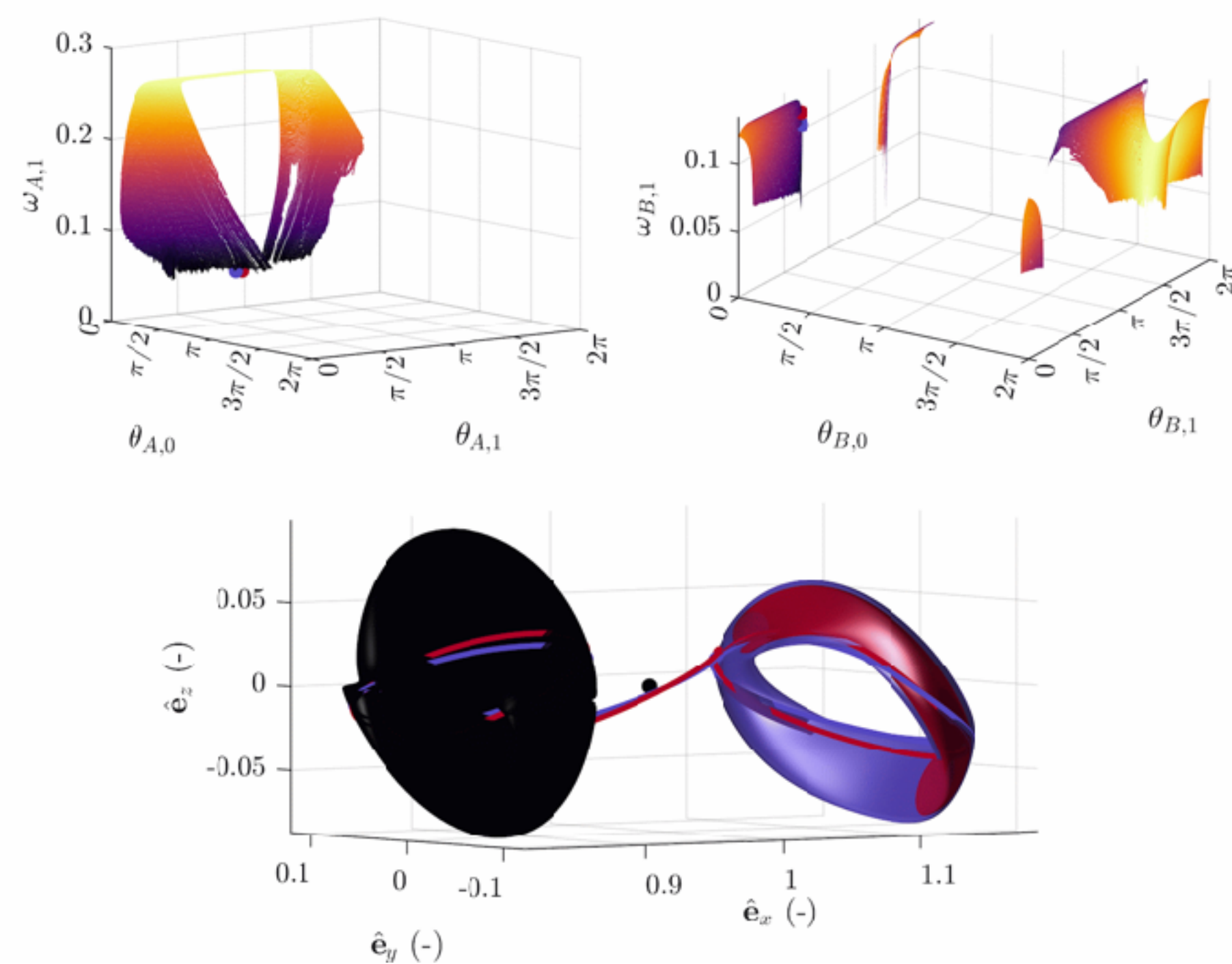


Lissajous to SQH Family

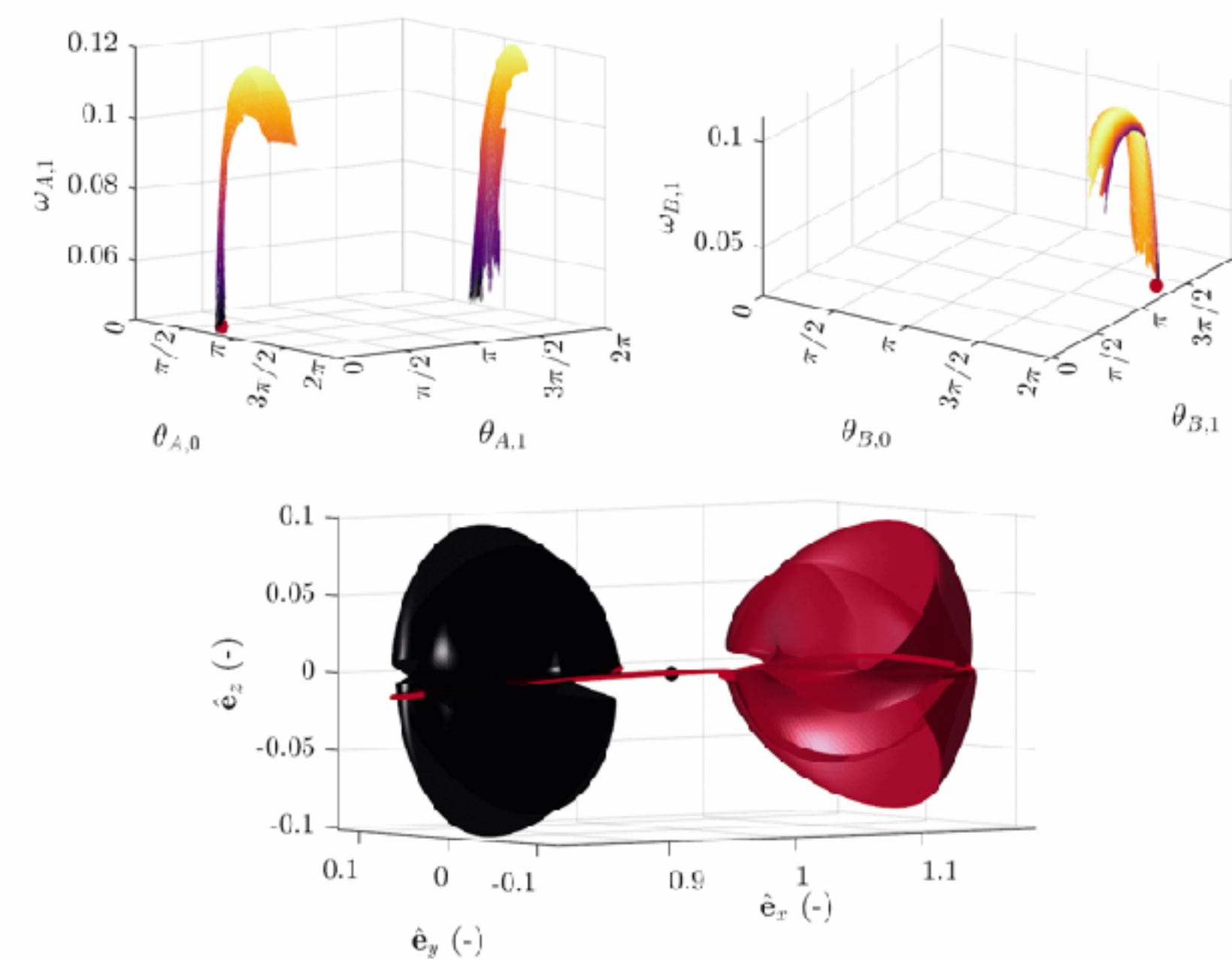




Northern to southern quasi-halo



Lissajous to Lissajous





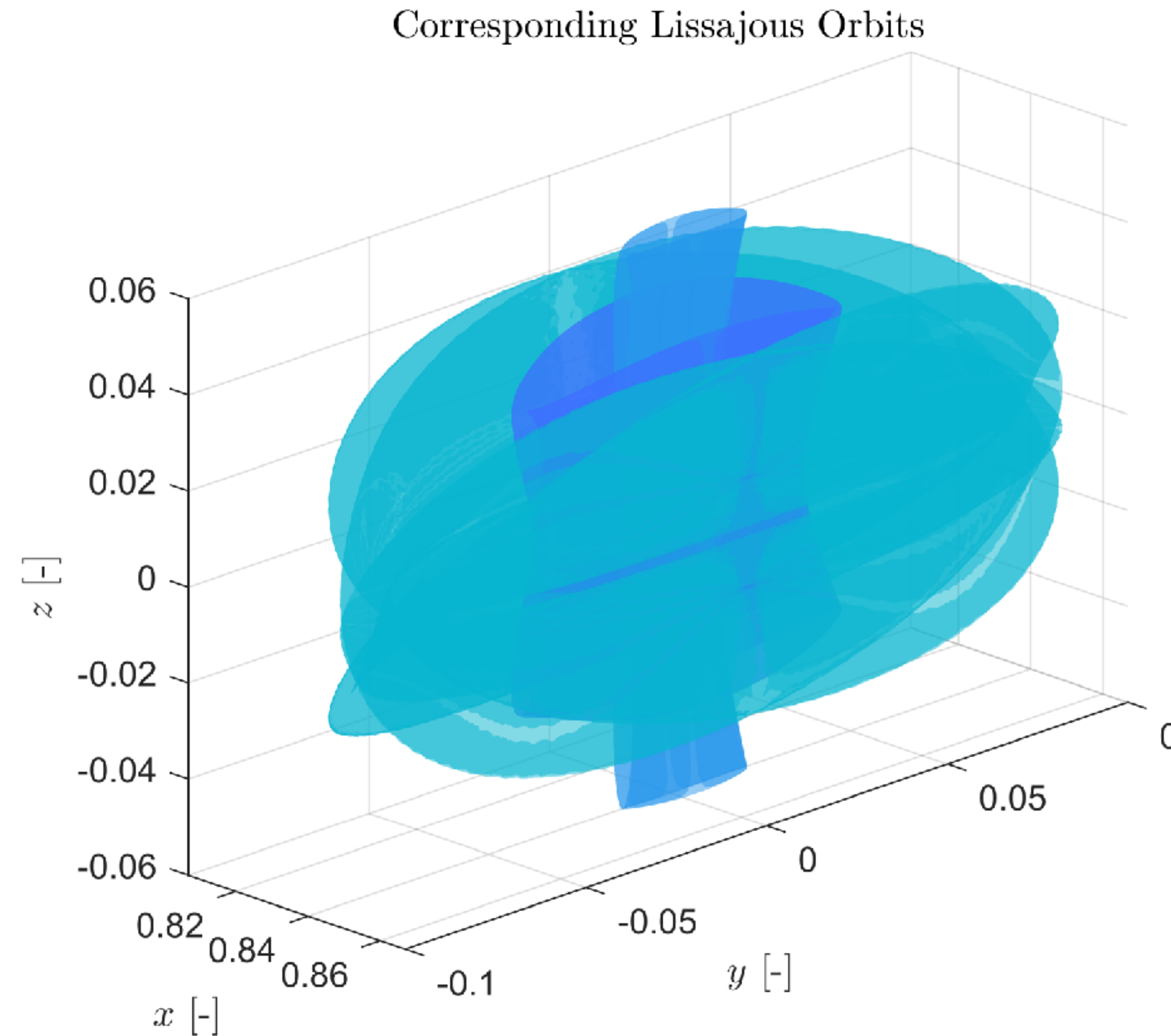
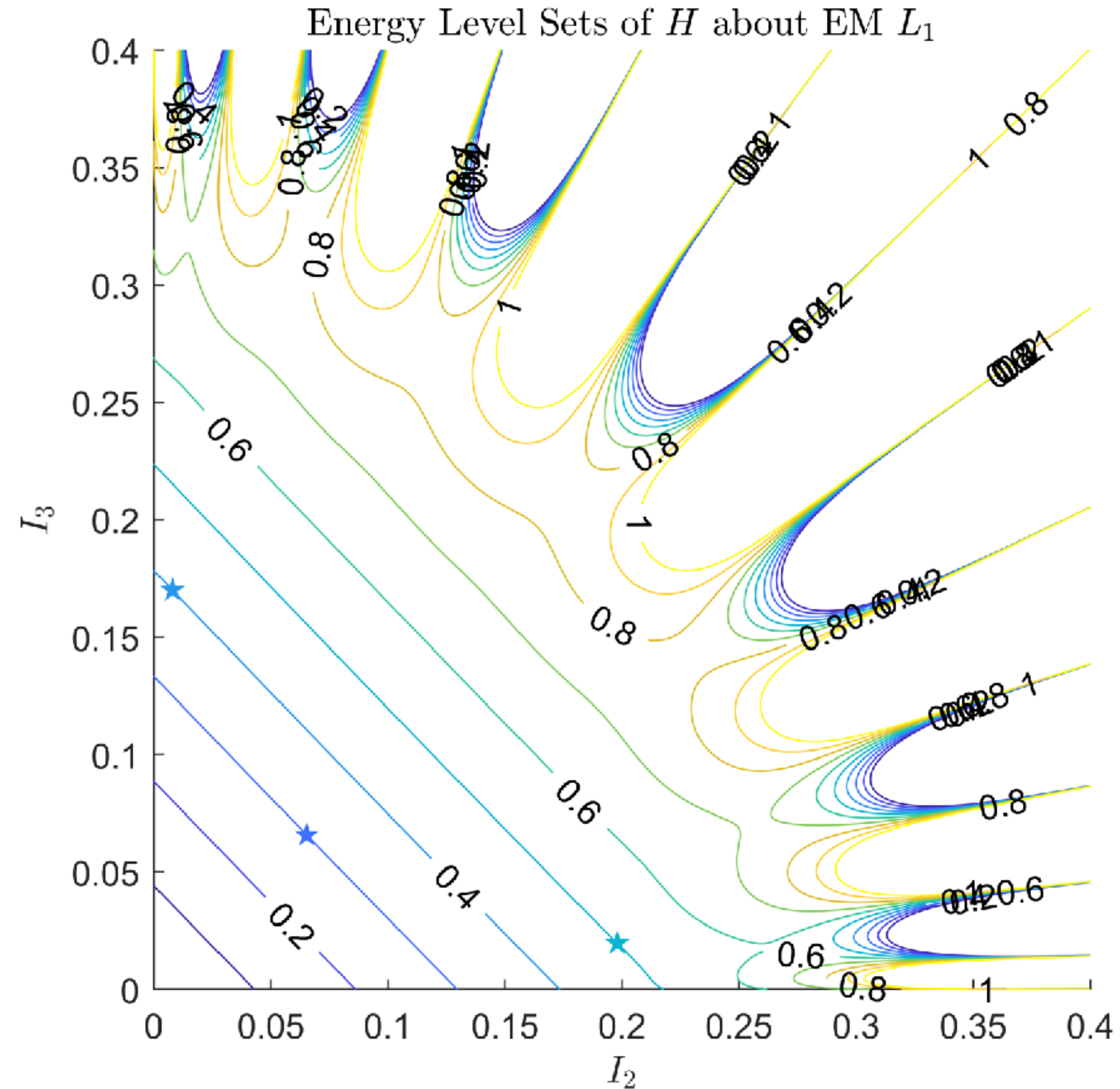
QPOs and “Orbit Elements”

Fundamental Decomposition of Phase Space

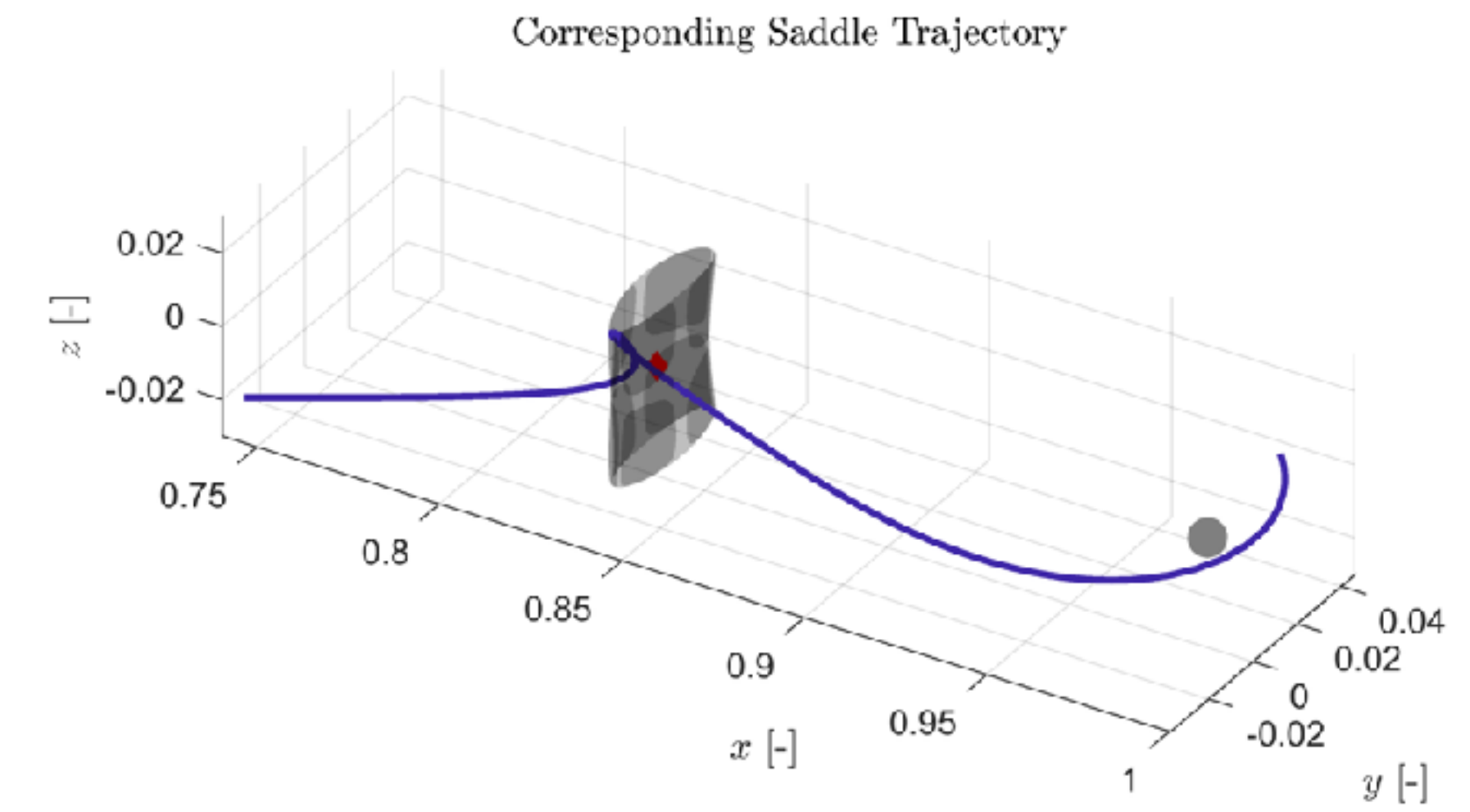
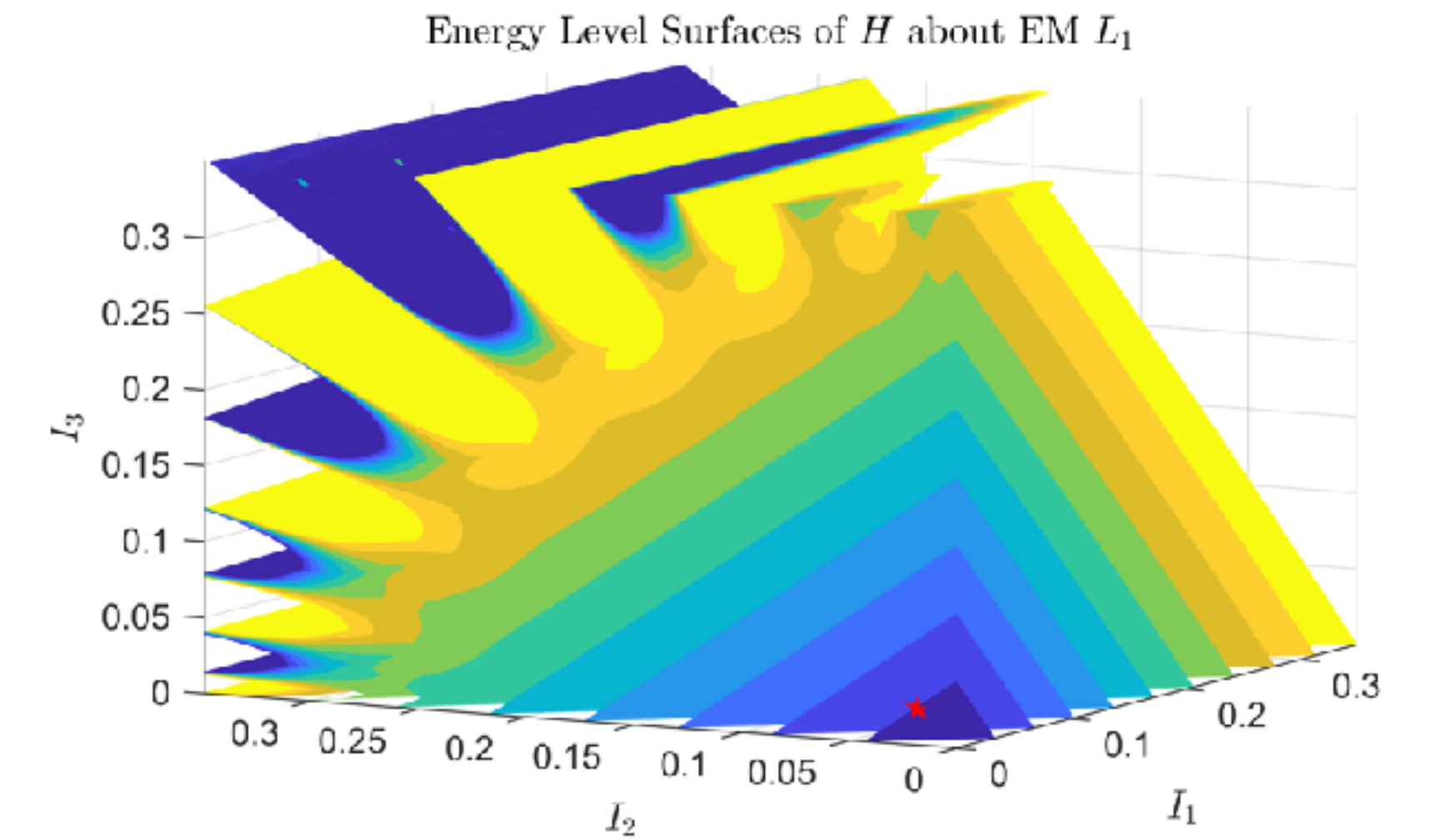
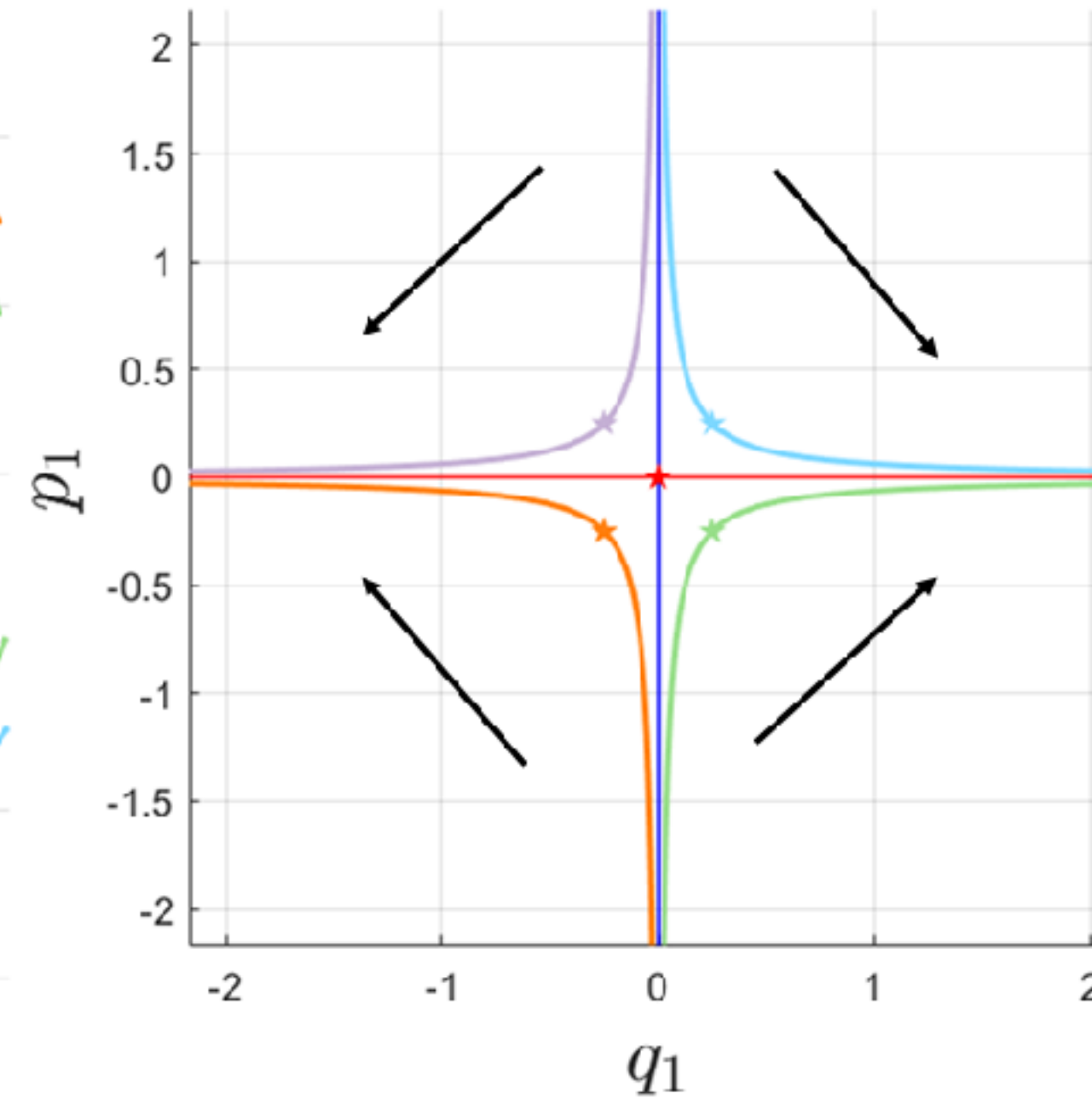
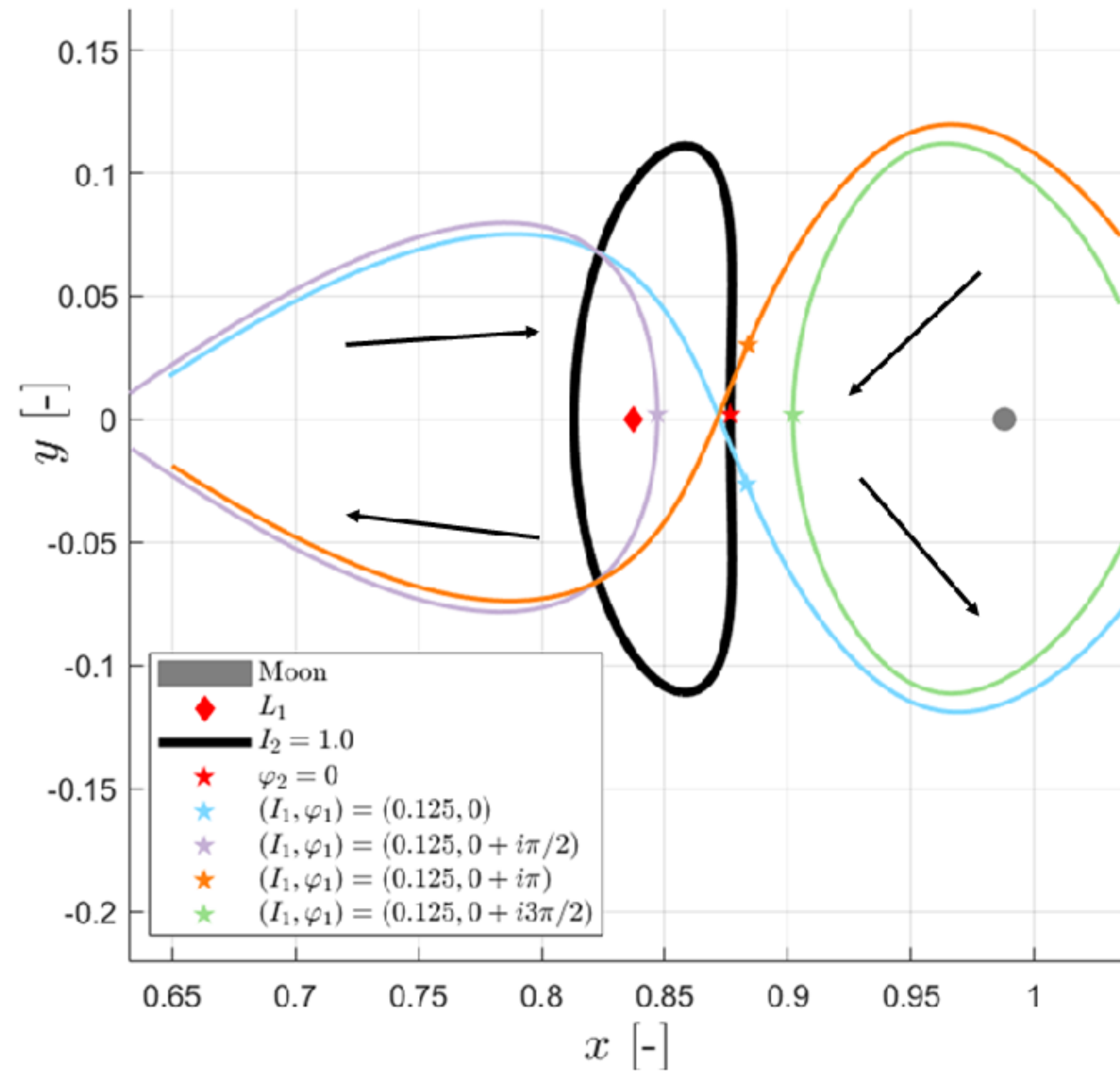


- Using analytical theories, the fundamental structures of quasi-periodic orbits and hyperbolic manifolds can be defined by convergent series
 - Develop series expansions that equal constant actions and angles that vary linearly in time
 - Our approach is following that developed by Jorba et al., and have limited convergence regions
- Result in functional relations between general Cartesian states and actions and angles associated with the phase flow
 - The actions and angles serve as “orbital elements” in the vicinity of the libration points
 - Can be used for a range of applications, ideally anything that orbit elements are used for
- Approach should be generalized to our numerically computed QPO and manifold structures as well
 - Limited by the analytic representation of these structures in phase space

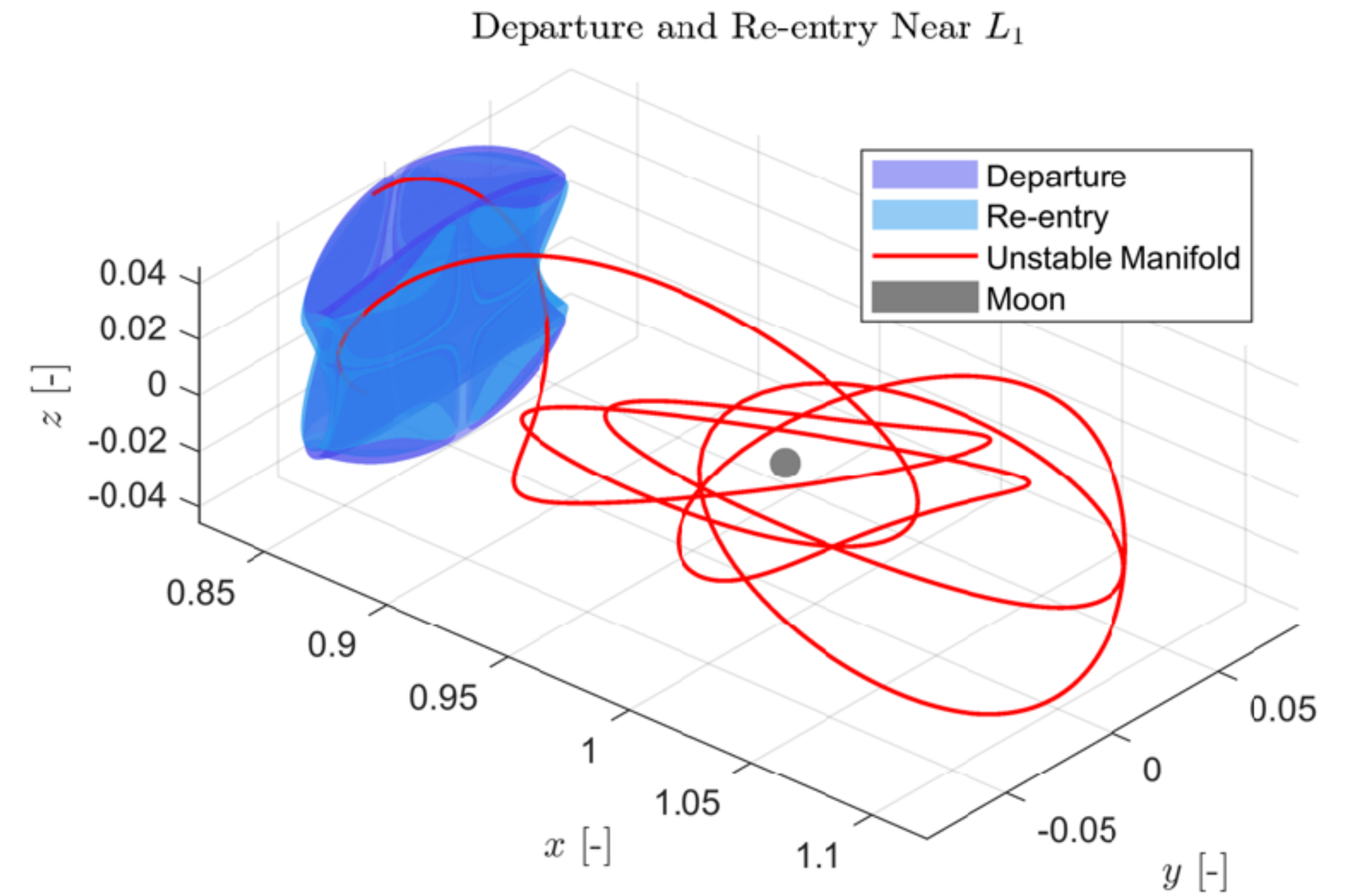
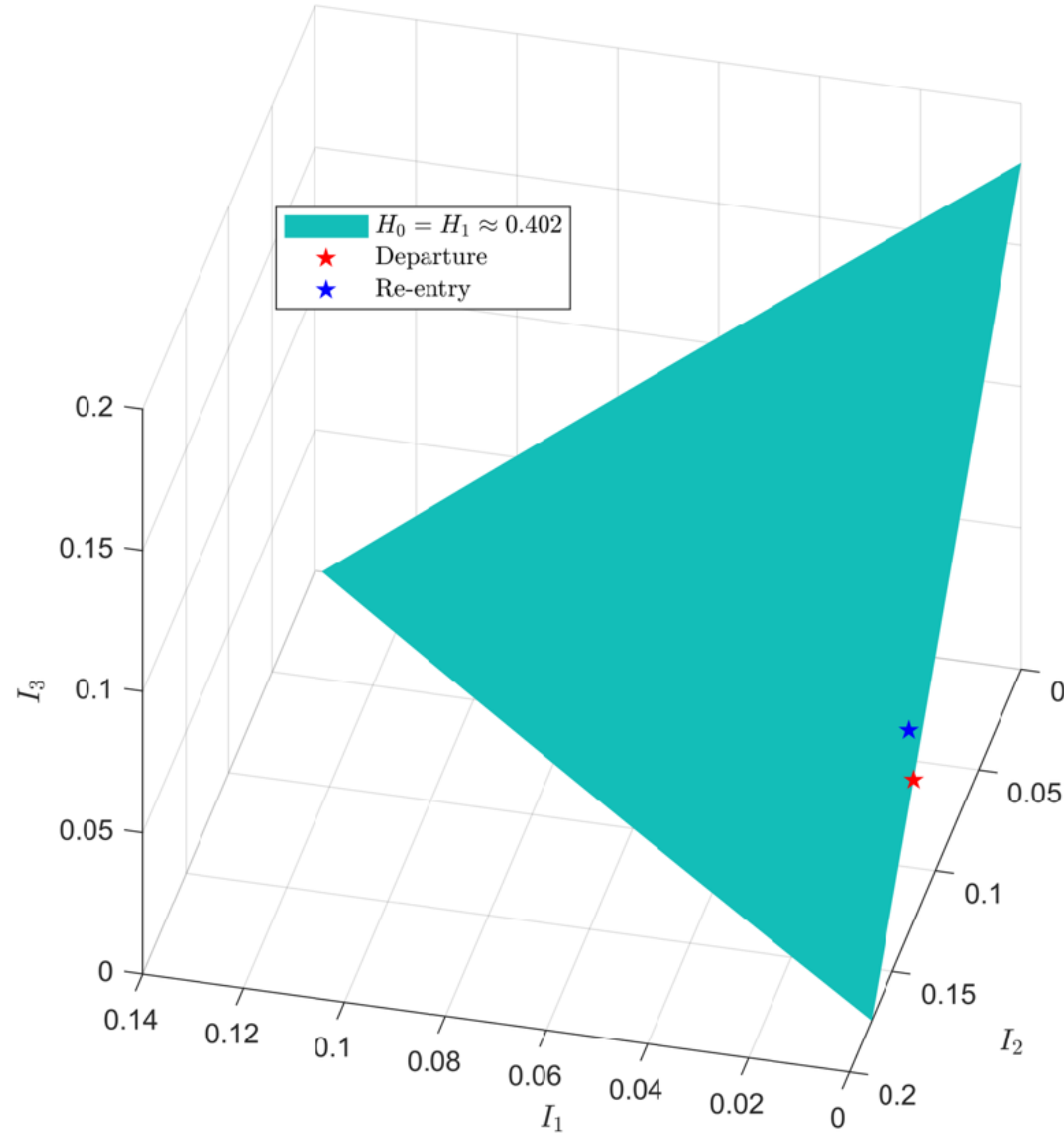
Analytical Representations of QPO



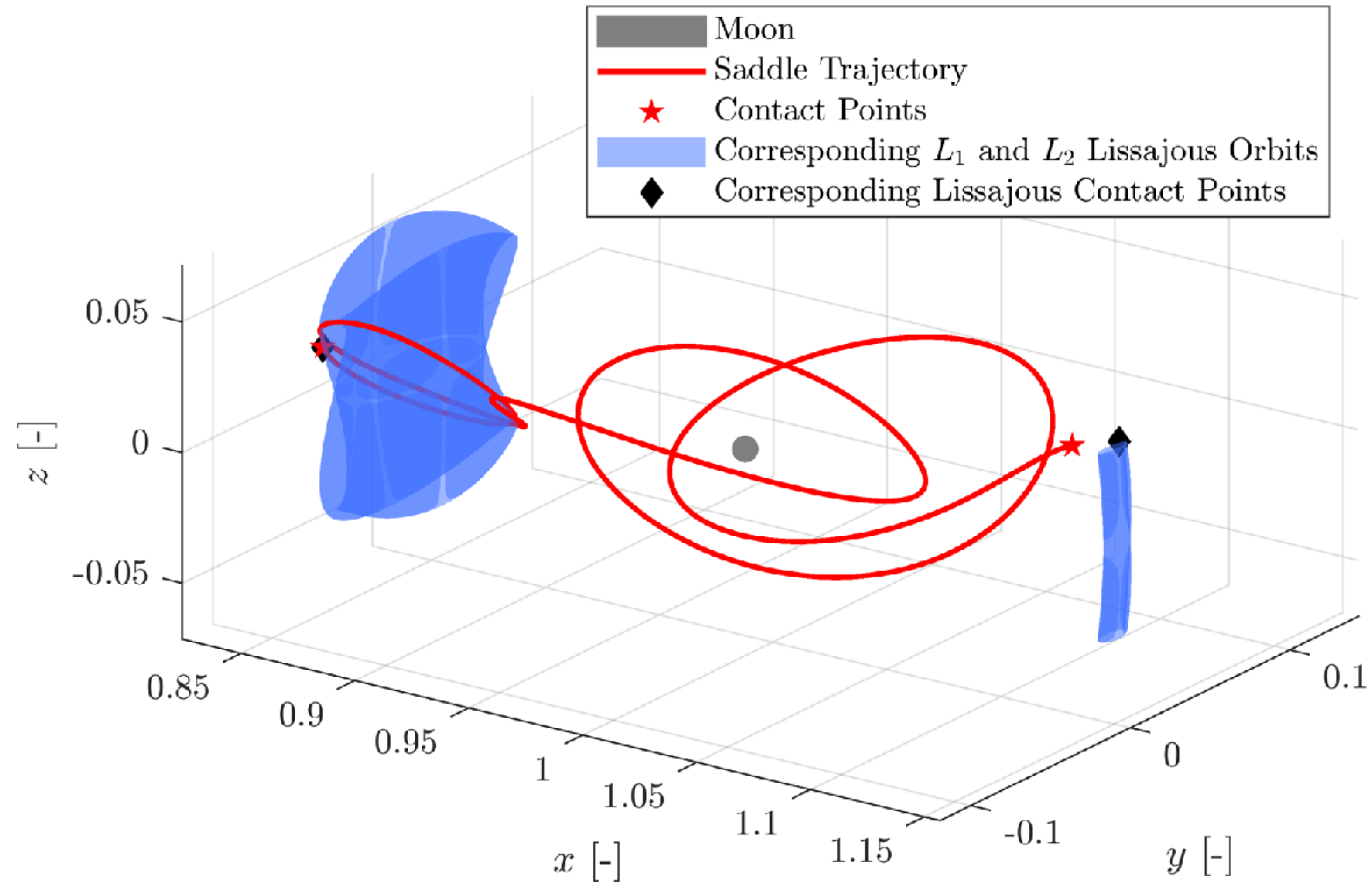
Action-Angle Orbit Elements



Non-Uniqueness of Elements



Possible Global Representations

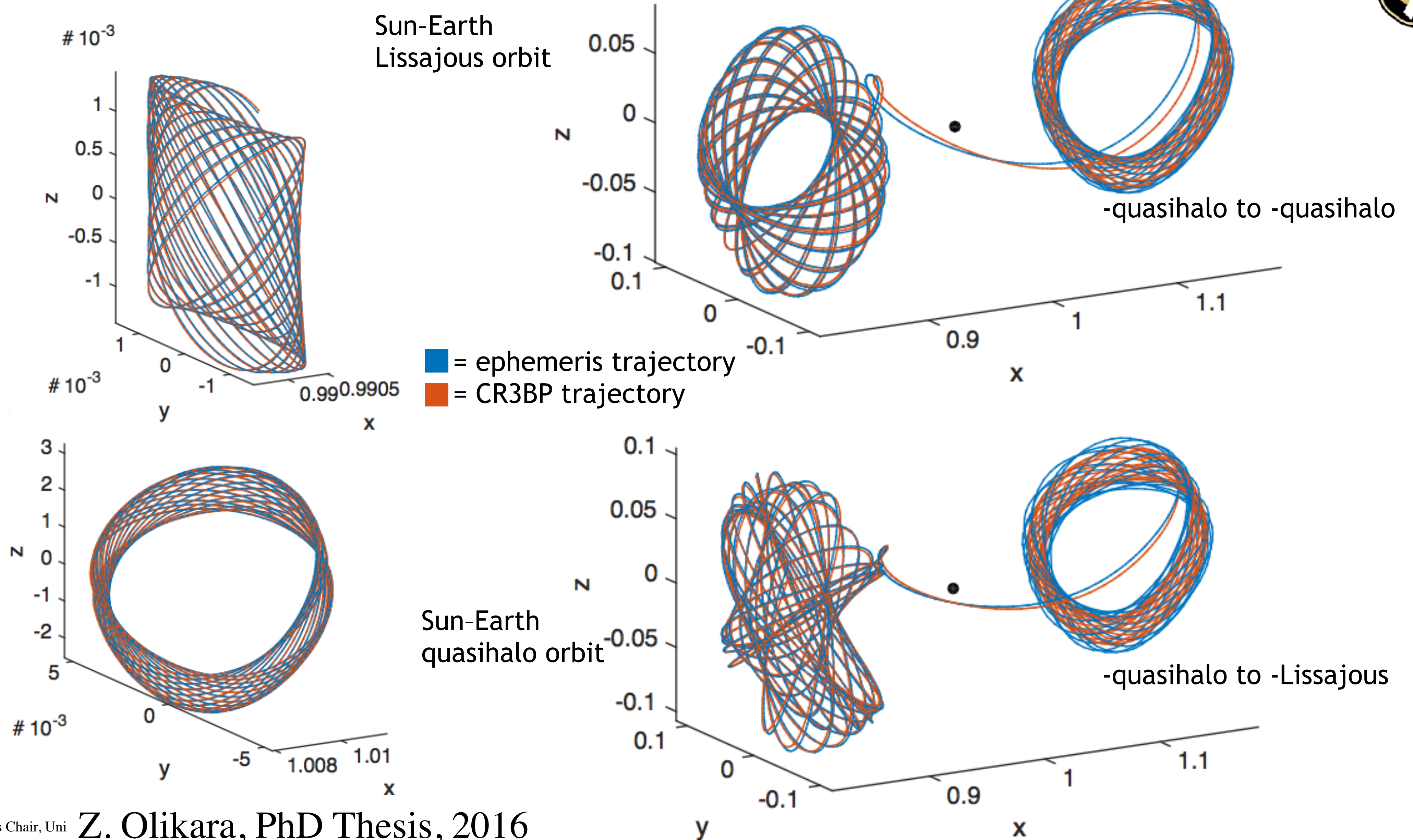




Do QPOs Persist in Real Models?

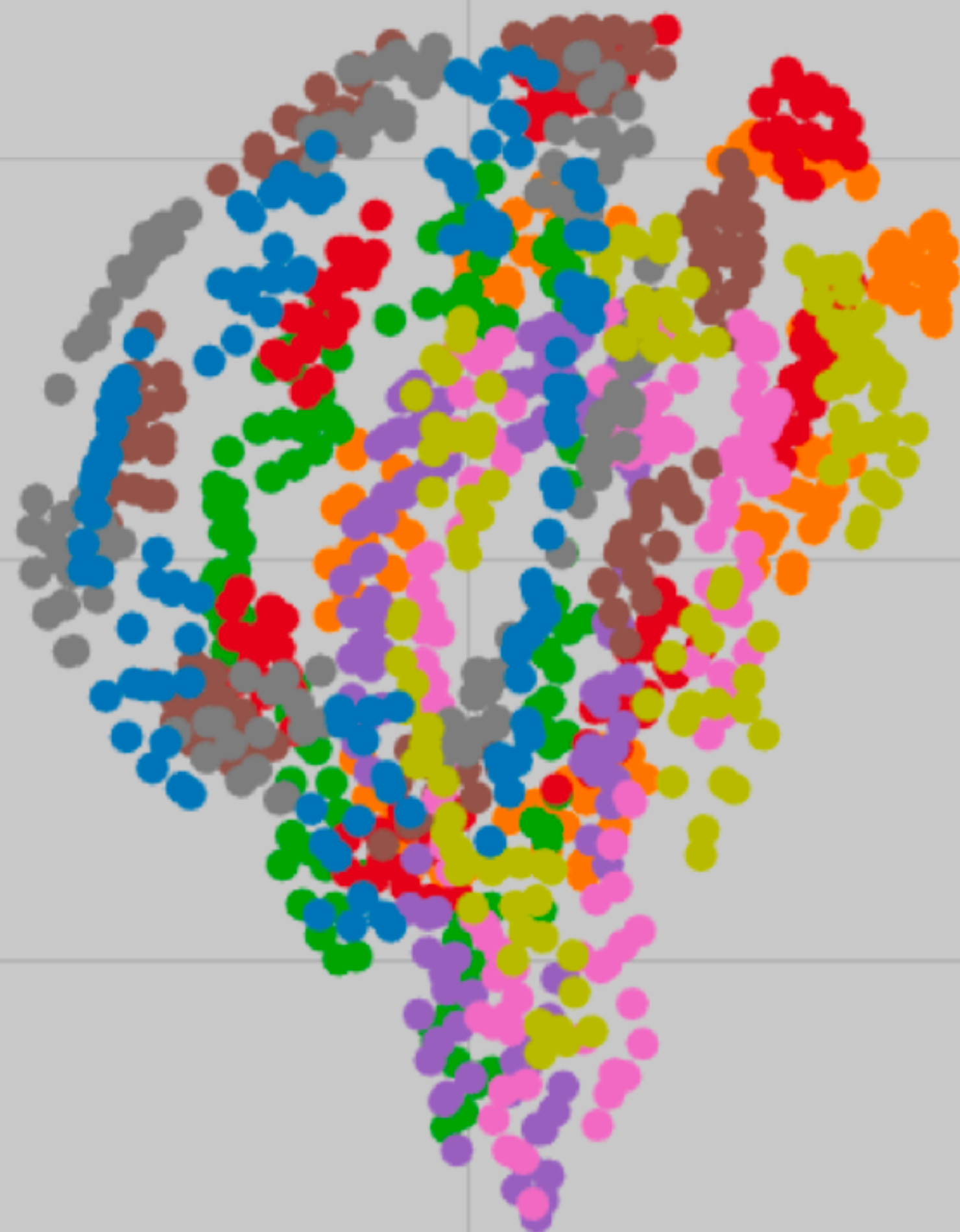


Do QPO persist in Real Models? *Yes*



Gateway's Orbit: L_2 9:2 NRHO

CR3BP (1D IEQ) CR4BP* (3D QHO) CR4BP (1D PO)



Park, Sanaga, & Howell, 2024

CR3BP to Ephemeris (Direct Transition) Results

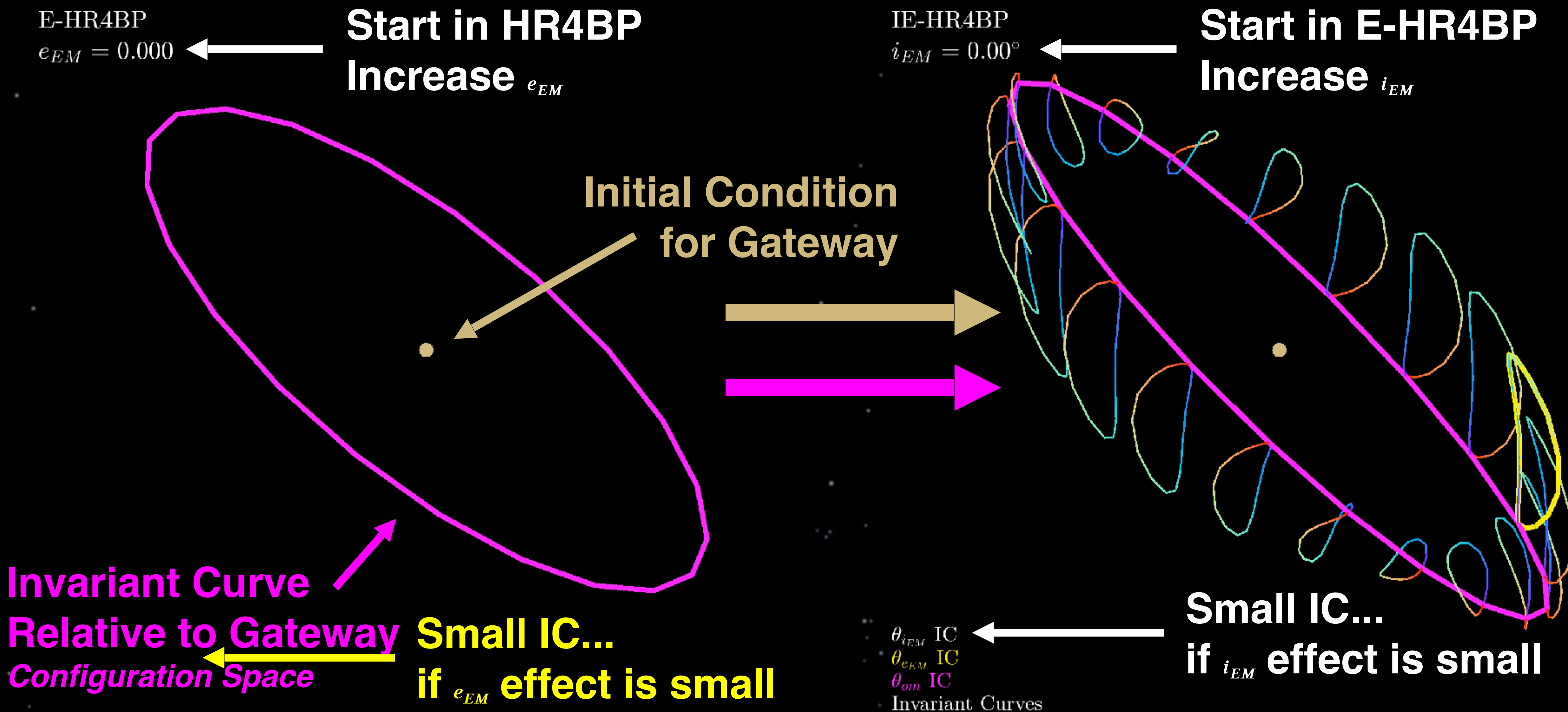
Surface of Section
 $z = 0, \dot{z} > 0$ (RP)



Park, Sanaga, & Howell, 2024

Zimovan-Spreen et al., 2023

Oscillatory Mode – Stability Analysis



IE-HR4BP Phase-Locked QPO Family

Proximity to Gateway:

Start: “Close” to GW

End: “Far” from GW

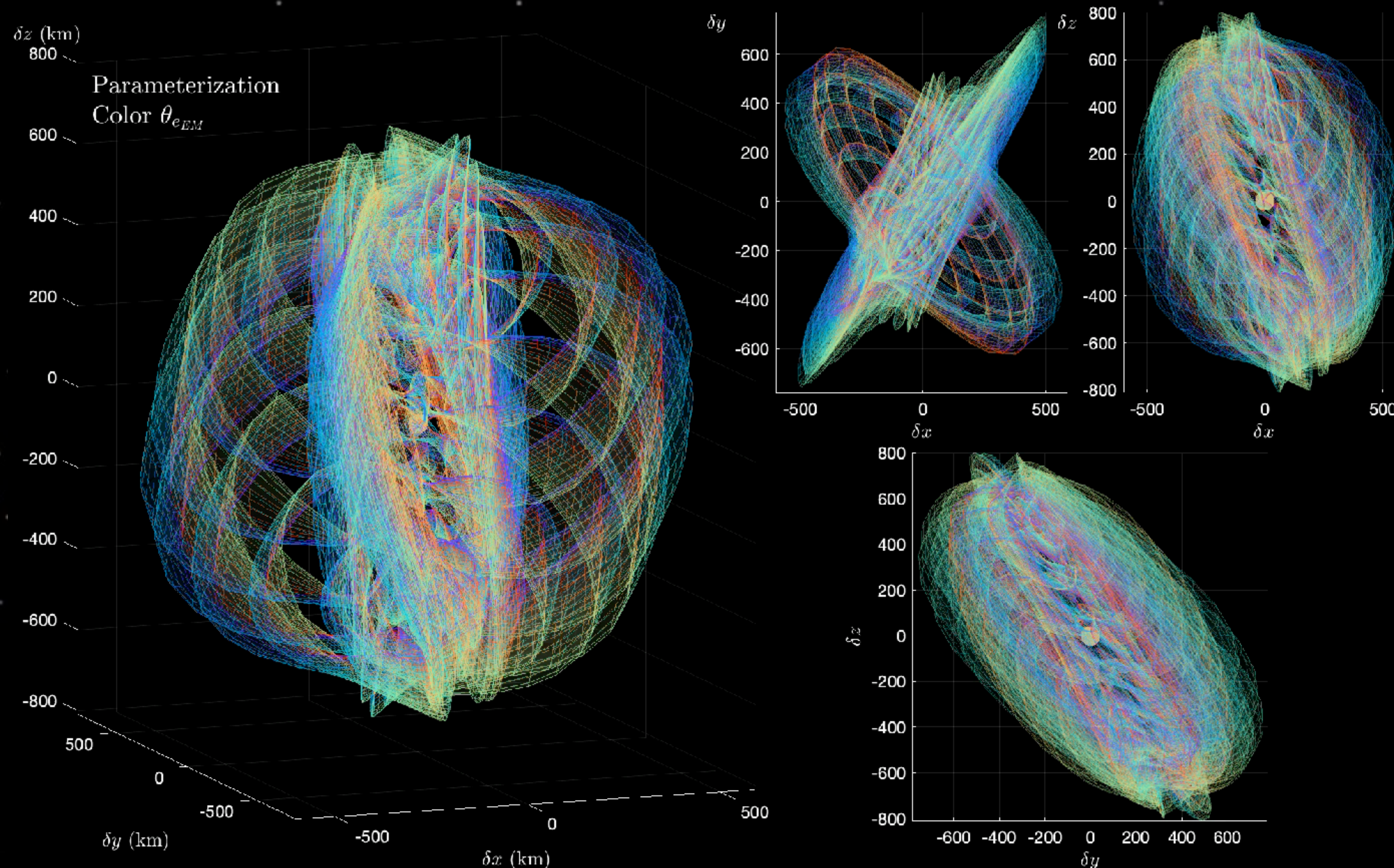
Oscillatory Mode:

The torus geometry is incredibly complex.

Ideally requires more nodes to compute it...

During Continuation:

Move further from resonance & recover expected geometry.



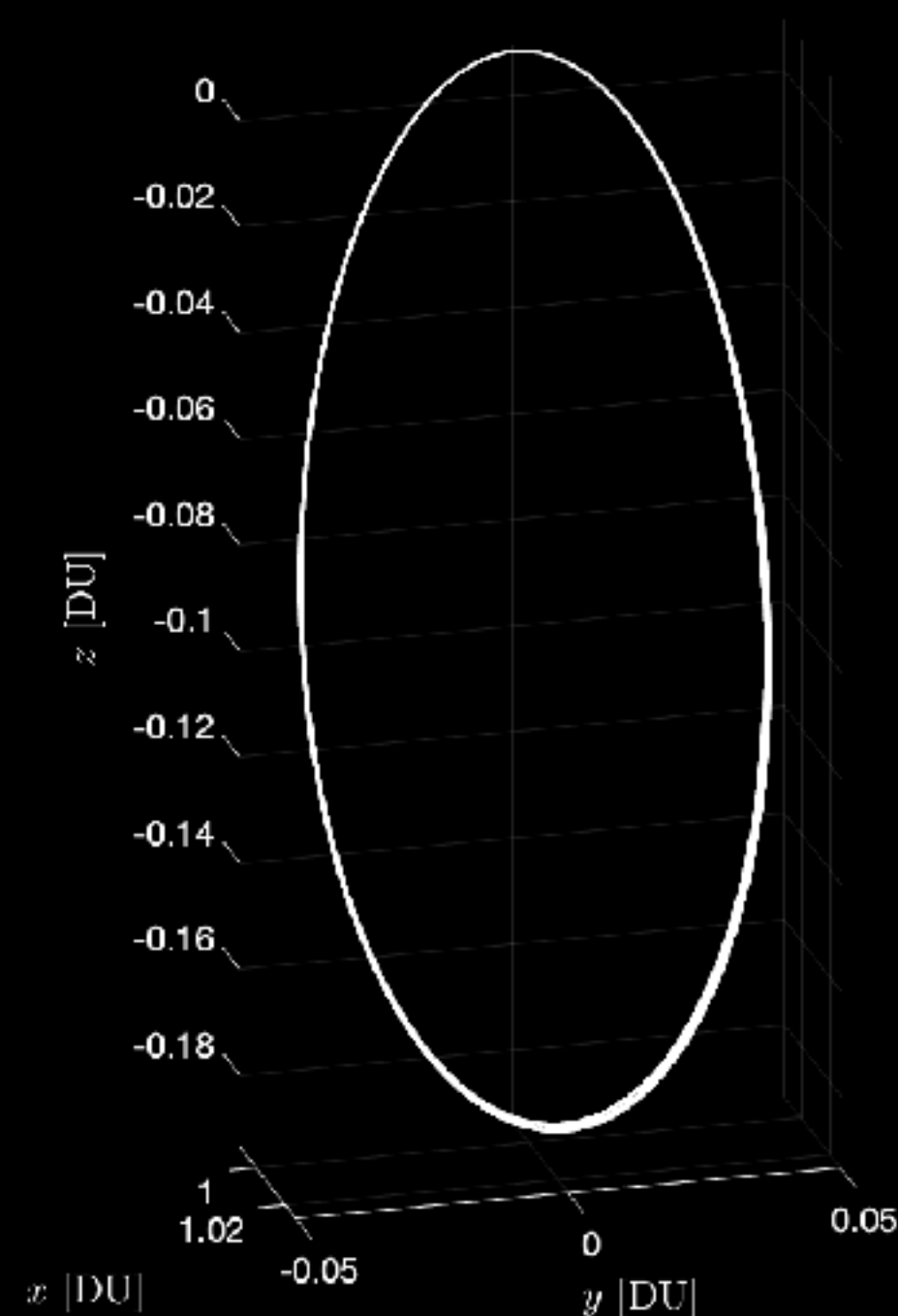
Intermediate Dynamical Models

Sun-Earth-Moon
HR4BP

Sun-Earth-Moon
E-HR4BP

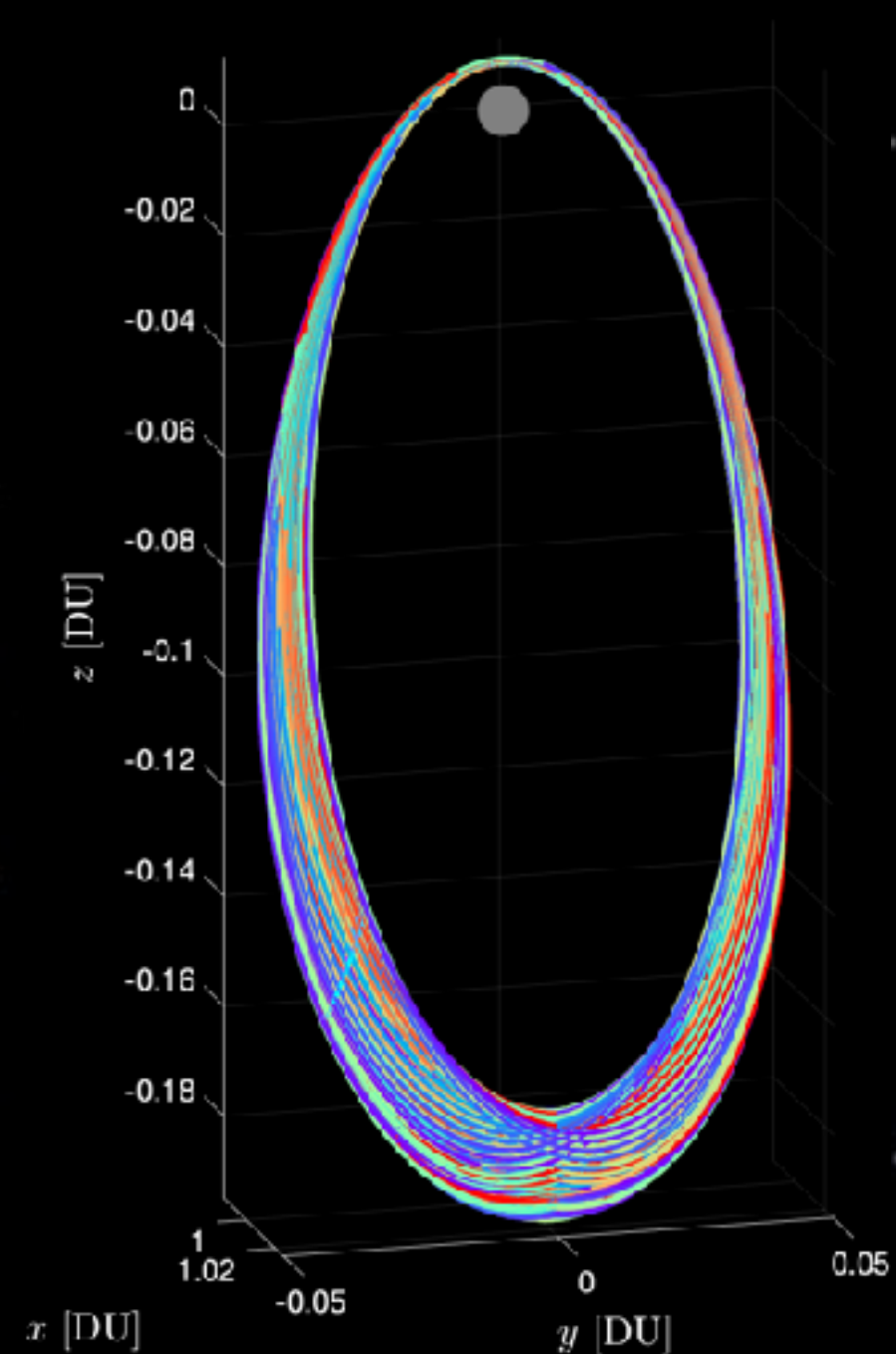
Sun-Earth-Moon
IE-HR4BP

Solar System /
Ephemeris Model



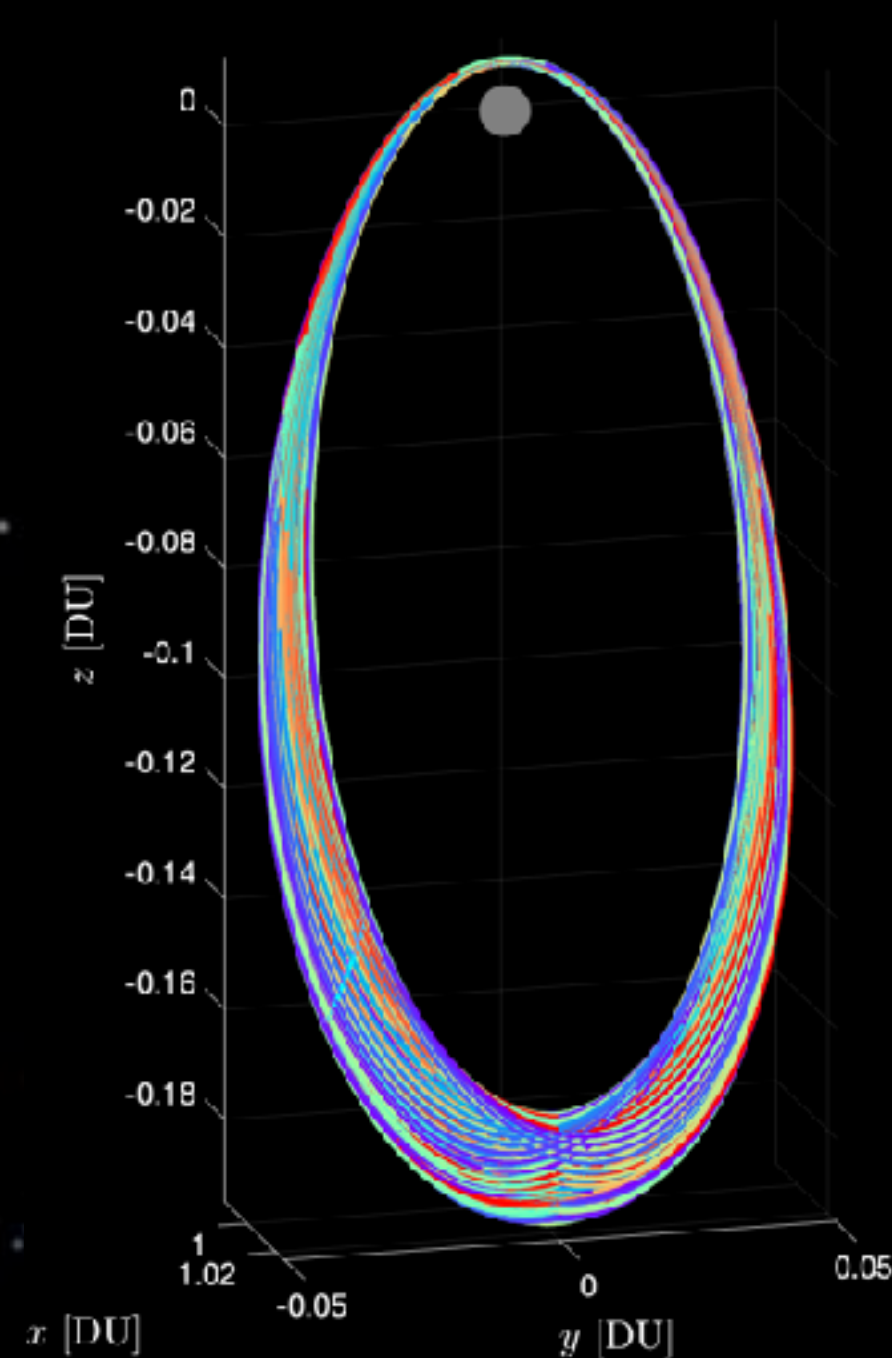
Antiperiodic ($n_e = 0$)
Simple Earth & Moon

Brown et al., 2024
Scheeres, 1998



Quasi-Periodic ($n_e = 2$)
+ EM Eccentricity

Park, Sanaga, & Howell, 2024



Quasi-Periodic ($n_e = 3$)
+ EM Inclination

Park, Sanaga, & Howell, 2024

...

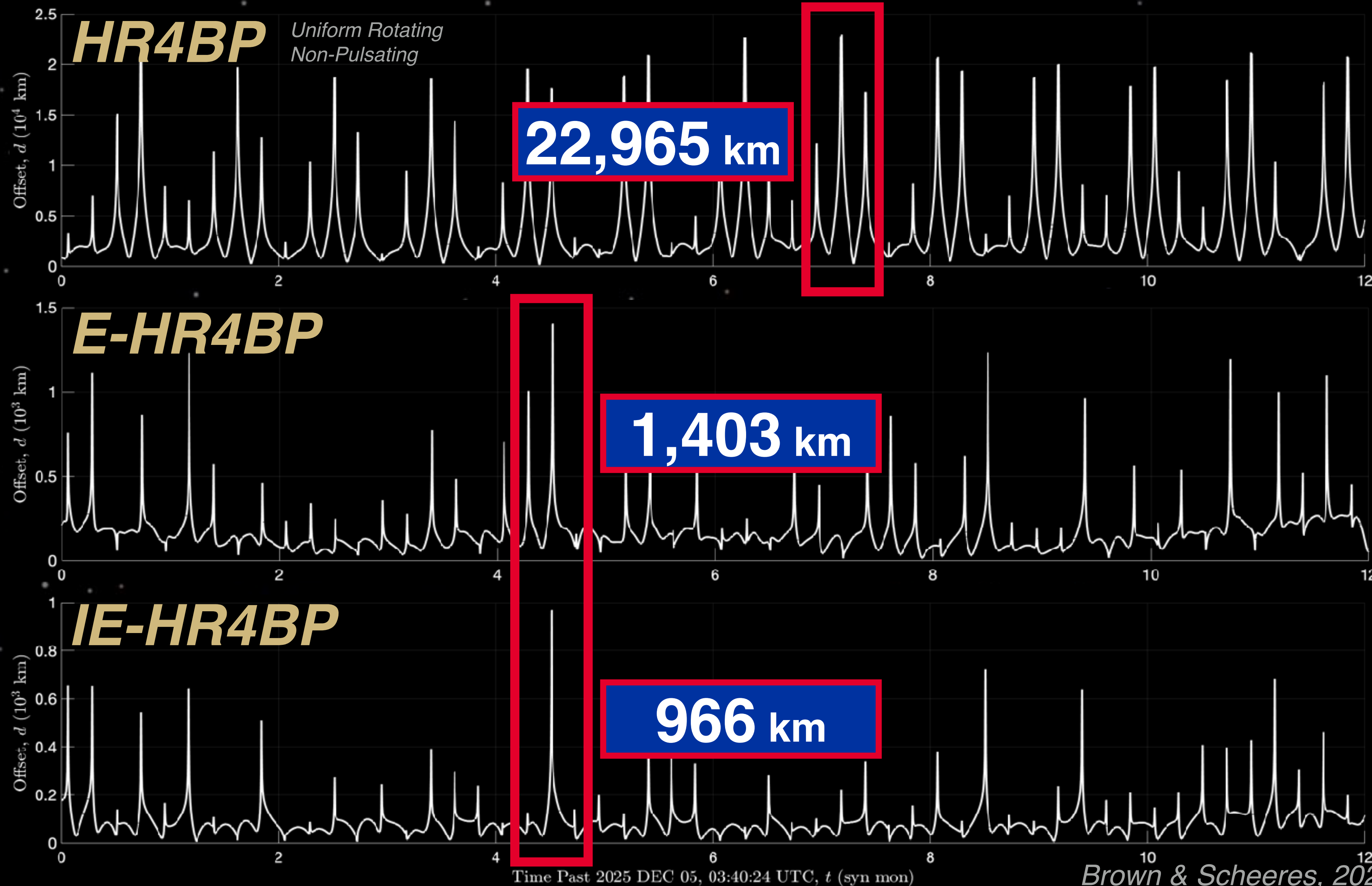
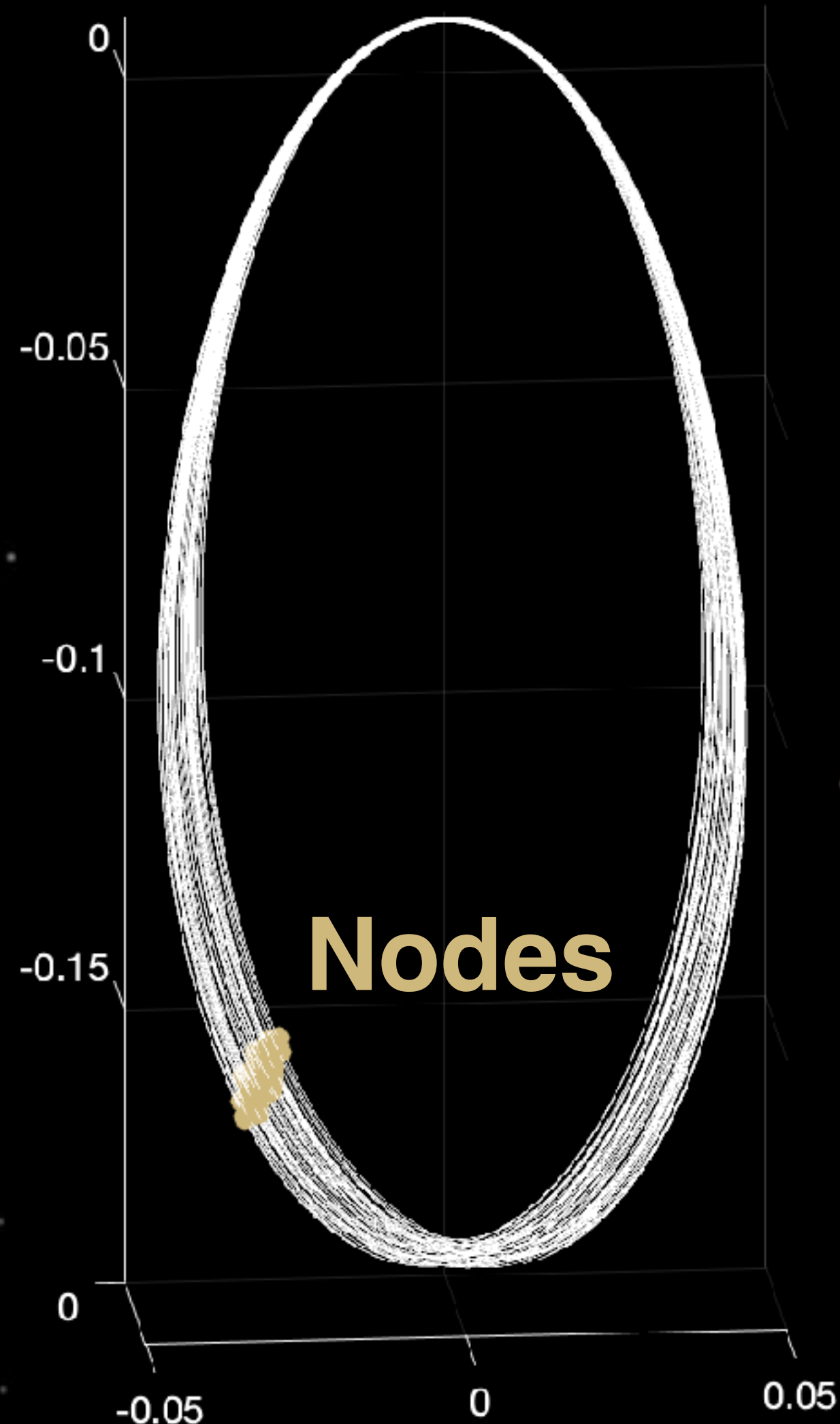


High Fidelity
 $n_e = 5$ / All Other Effects

NASA Gateway Mission Design Team
Gomez, Masdemont, & Mondelo, 2002

Model to Ephemeris Comparison

12 synodic months
SPICE Rotating Non-Pulsating

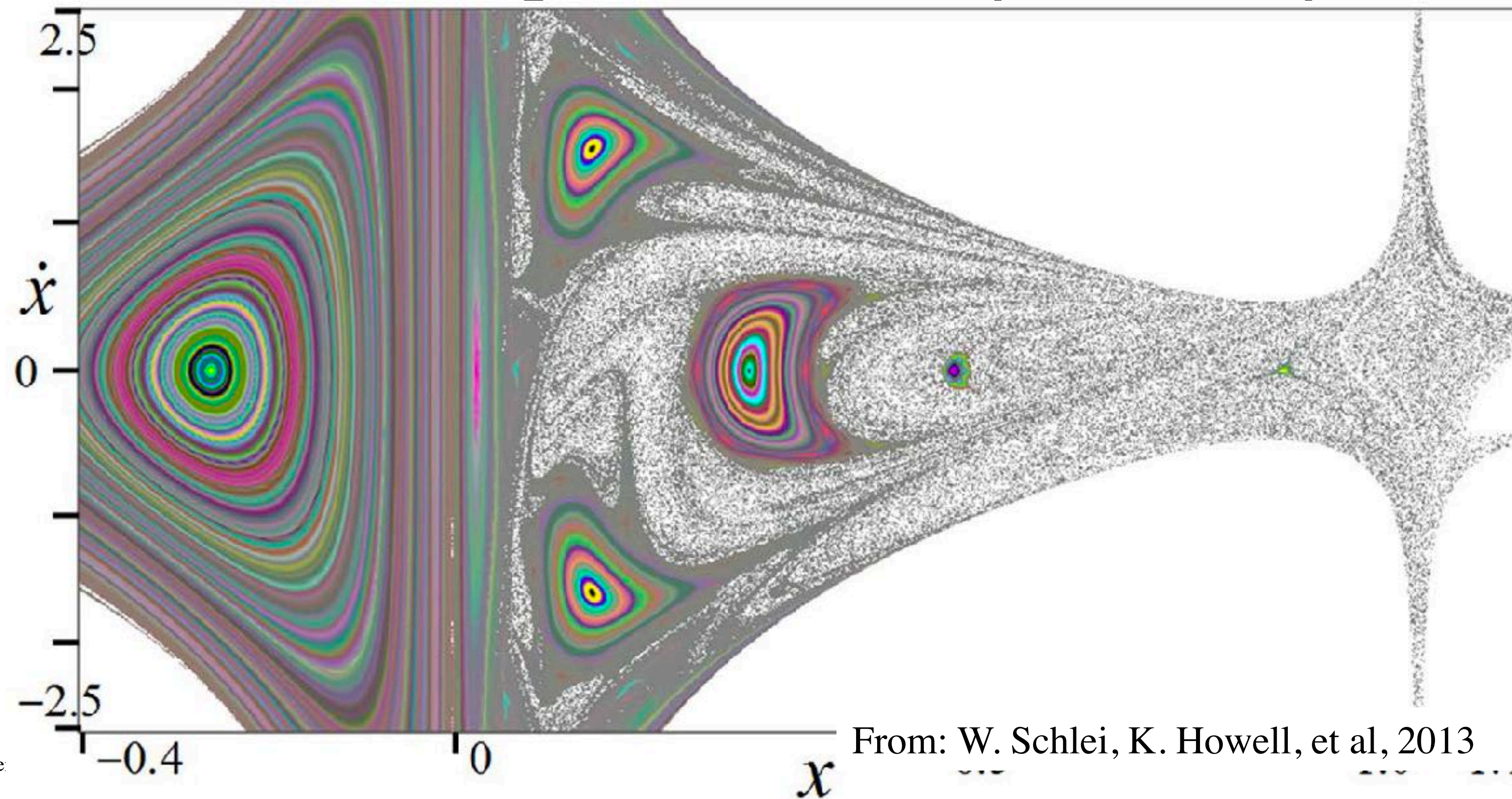


Brown & Scheeres, 2025c

Open Dynamical Systems Questions



- Where do QPO families end? Connection with KAM theory...
- What are the limits of integrable motion? Arnold diffusion...
- What are the effects of resonances & periodic orbits? Loss of dimension...
- Continuation methods through resonances, improved computational approaches

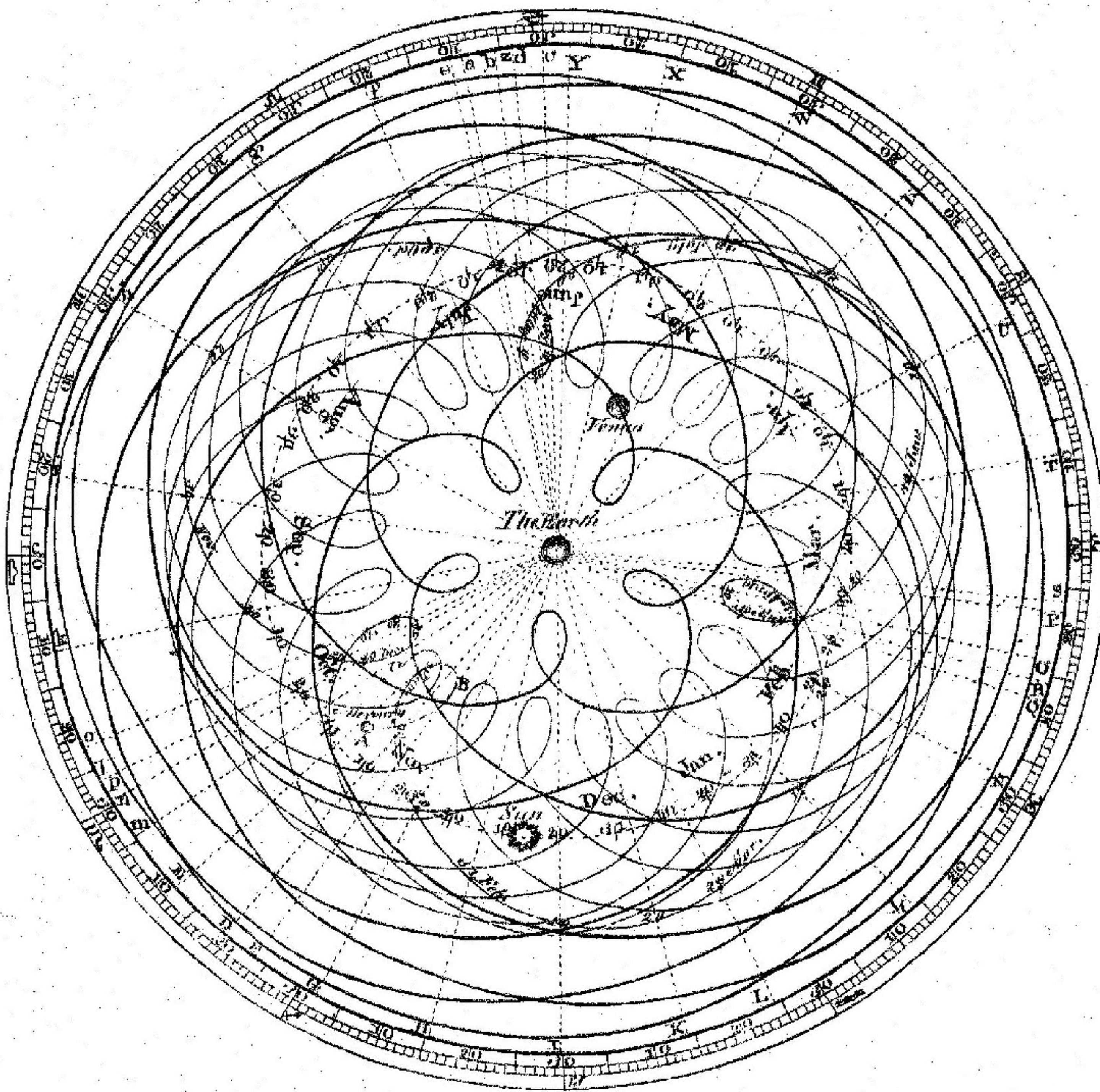


From: W. Schlei, K. Howell, et al, 2013

Higher Fidelity Models



- For analytical models, all perturbations are generally periodic, but add important additional details
 - Examples include the Elliptic-Restricted 3-Body Problem, the Earth-Sun-Moon-Spacecraft Restricted 4-Body Problem, motion about Jupiter (with the Laplace Resonance), etc...
 - For a Hamiltonian Dynamical System every new periodic perturbation is equivalent to an added degree of freedom
 - This increases the dimension of existing tori by 1, and can cause previously stable tori to become unstable
- Pathway that enables higher fidelity models to be developed... take a stable halo orbit in the vicinity of the Lunar Gateway orbit for example
 - CR3BP $\Rightarrow$ 3-D torus
 - + Sun perturbation $\Rightarrow$ Hill Restricted 4-BP $\Rightarrow$ 4-D torus
 - + Earth/Moon node circulation (Saros) $\Rightarrow$ 5-D torus, ...
 - Each new dimension can destabilize earlier tori through resonances
 - Each new torus brings system “closer” to an ephemeris system
- Motivates definition of the “Ptolemaic Restricted N Body Problem”



Ptolemaic N -Body Problem

$$\ddot{\mathbf{r}} = \mathbf{F}(\mathbf{r}, \dot{\mathbf{r}}, \vec{\theta}) \in \mathbb{R}^3$$

$$\vec{\theta} = \vec{\theta}_o + \vec{\omega}(t - t_o) \in \mathbb{R}^M$$

$\vec{\theta}$ models M perturbing bodies, each periodic with different frequencies, chosen to model the solar system just as the Ptolemaic system did.

QPOs of dimension: M , $1 + M$, $2 + M$ exist with Stable/Unstable Manifolds

Stable motions are QPOs with dimension $3 + M$

Mission Design

Motivation for Studying QPOs



Models

2BP

CR3BP

HR4BP

PtolemaicRBNP

Ephemeris

Fundamental Solutions

Conic Sections

EP, PO, QPO, and $\pm$ Manifolds

PO, QPO and $\pm$ Manifolds

QPO and $\pm$ Manifolds

None

*Transfer Designs: ΔV + **Heteroclinic***

ΔV only

ΔV only: EP, PO, + Heteroclinic: **QPO**

ΔV only: PO, + Heteroclinic: **QPO**

ΔV + Heteroclinic: **QPO**

“Shadow” trajectories

QPOs are the only solutions that persist in higher-fidelity models and are the only ones that robustly exhibit heteroclinic (i.e., cost free) transfers. Allows us to design trajectories with dynamical systems techniques that can be made arbitrarily “close” to ephemeris models.

Summary



- Quasi-Periodic Orbits are the dominant class of motion in complex dynamical systems
- Their precise computation is now feasible with efficient algorithms
 - Single shooting / Multiple shooting / Colocation / PDE formulations
 - Enables computation of multi-parameter families
 - Enables manifold computations and heteroclinic / homoclinic connections
- Applications are numerous and range far beyond mission design
- Our future research:
 - Formation and proximity flight and control of S/C
 - Computation of high-dimension QPO and their application
 - Use as “orbital elements” for describing generalized motion in non-integrable problems
 - Extension to N-body systems and Full body problems

Some Relevant CSML Papers:



- Zubin Olikara PhD Thesis: “Computation Of Quasi-Periodic Tori And Heteroclinic Connections In Astrodynamics Using Collocation Techniques,”
- N. Baresi, Z. Olikara and D.J. Scheeres. 2018. “Fully numerical methods for continuing families of quasi-periodic invariant tori in astrodynamics,” *Journal of the Astronautical Sciences* 65: 157-182.
- D.B. Henry and D.J. Scheeres. 2021. “Expansion Maps: Designing Relative Trajectories on Quasi-periodic Orbits,” *Journal of Guidance, Dynamics and Control* 44(3): 457- 468.
- D. Lujan and D.J. Scheeres. 2022. “The Earth-Moon L2 Quasi-Halo Orbit Family: Characteristics and Manifold Applications,” *Journal of Guidance, Control and Dynamics* 45(11): 2029-2045.
- D. Lujan and D.J. Scheeres. 2023. “Dynamics in the Vicinity of the Stable Halo Orbits,” *Journal of the Astronautical Sciences* 70:20.
- D.B. Henry and D.J. Scheeres. 2023. “Quasi-periodic Orbit Transfer Design via Whisker Intersection Sets,” *Journal of Guidance, Control and Dynamics* 46(10): 1929-1944.
- L.T. Peterson and D.J. Scheeres. 2023. “Local Orbital Elements for the Circular Restricted Three-Body Problem,” *Journal of Guidance, Control and Dynamics* 46(12): 2275-2289.
- D.B. Henry and D.J. Scheeres. 2024. “Fully numerical computation of heteroclinic connection families in the spatial three-body problem,” *Communications in Nonlinear Science and Numerical Simulation* 130: 107780.
- G.M. Brown, D.B. Henry, L.T. Peterson, and D.J. Scheeres. 2025. “Structure of Periodic Orbit Families in the Hill Restricted 4-Body Problem,” *SIAM Journal on Applied Dynamical Systems* 24(1): 346-375. DOI 10.1137/24M1637301
- G.M. Brown, L.T. Peterson, D.B. Henry and D.J. Scheeres. 2025. “Behavior of Symmetric Periodic Orbits with High-Order Resonance in the Sun-Earth-Moon System,” *Journal of the Astronautical Sciences* 72:43 (2025). <https://doi.org/10.1007/s40295-025-00517-3>
- G.M. Brown and D.J. Scheeres. 2026. “Orbit Capacity of Higher-Dimensional Tori around Gateway,” *Journal of Guidance, Control and Dynamics* 49(3): 879-893. <https://doi.org/10.2514/1.G009219>

